Fall 2024 CS582: Machine Learning for Bioinformatics

Ge Liu

Lecture 5: Protein sequence model

Feb 5, 2024



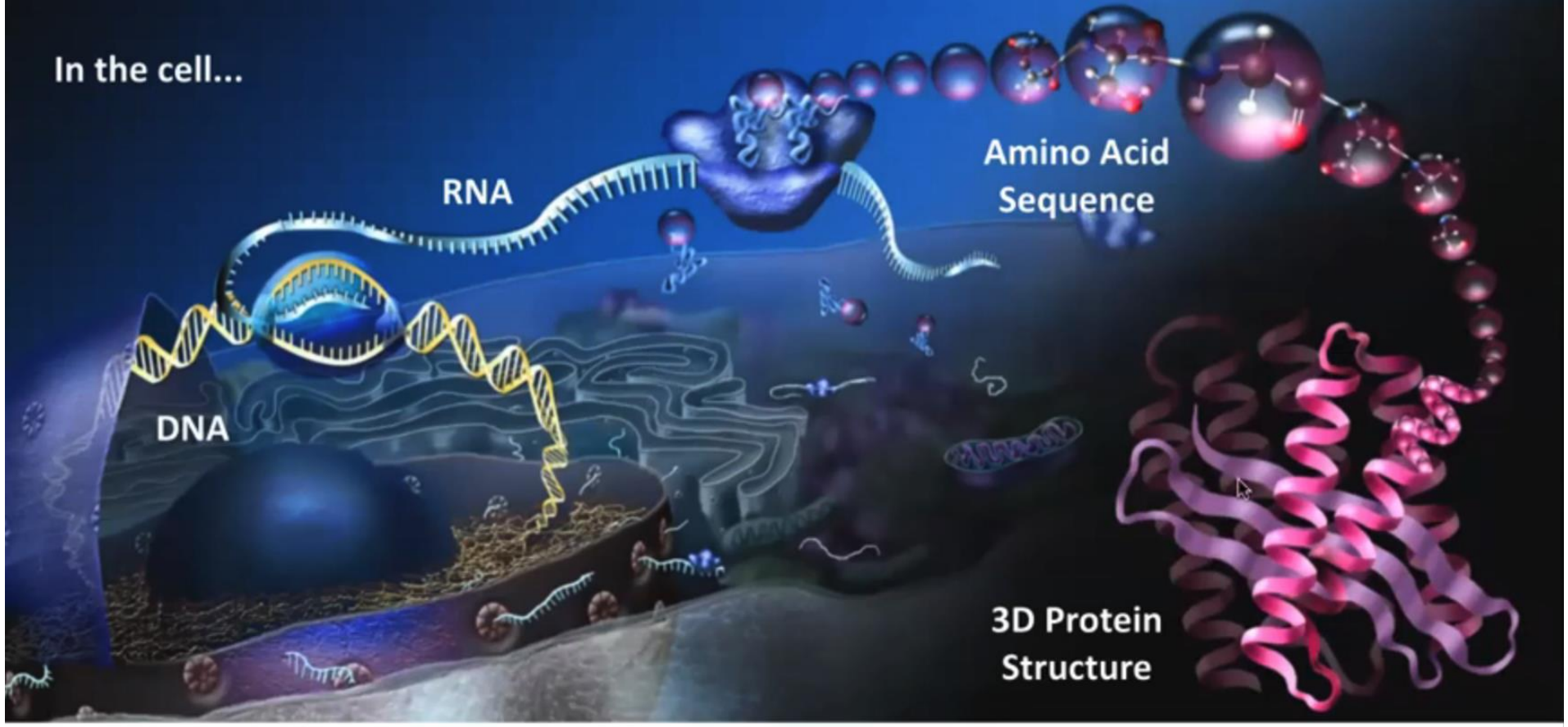
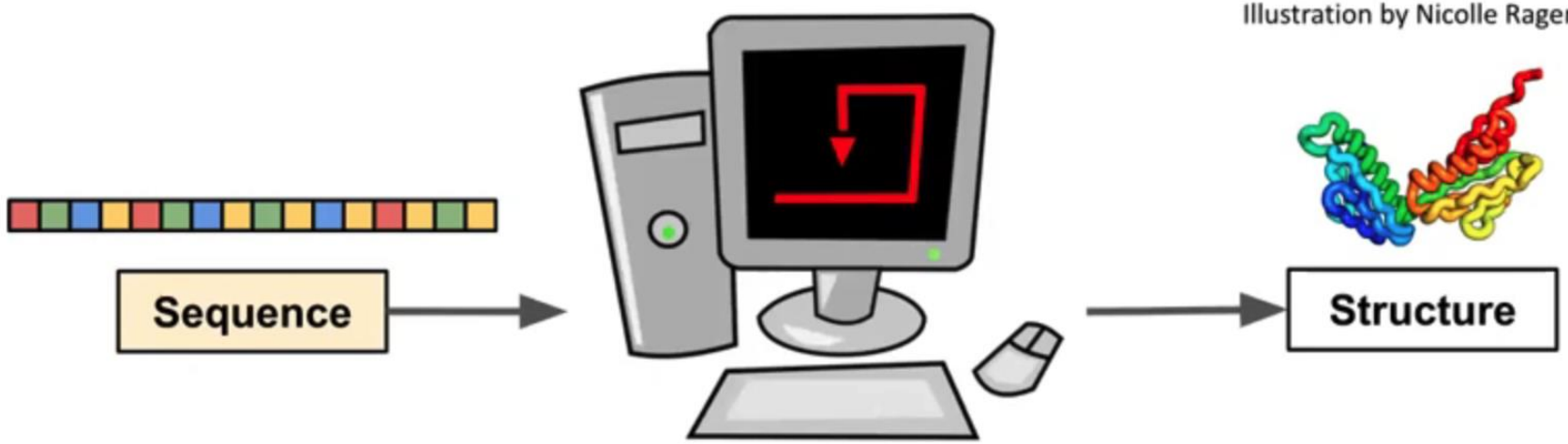
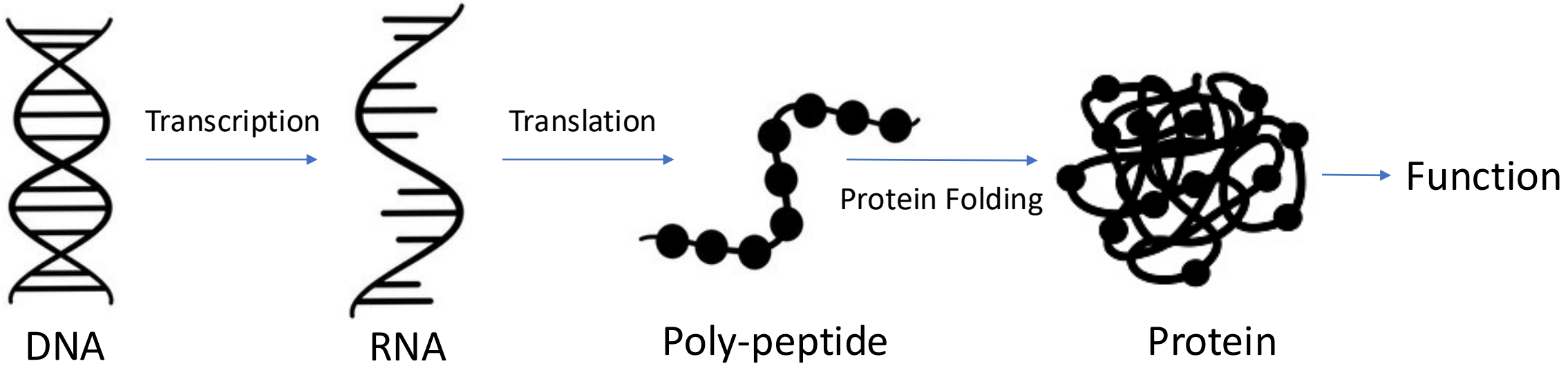


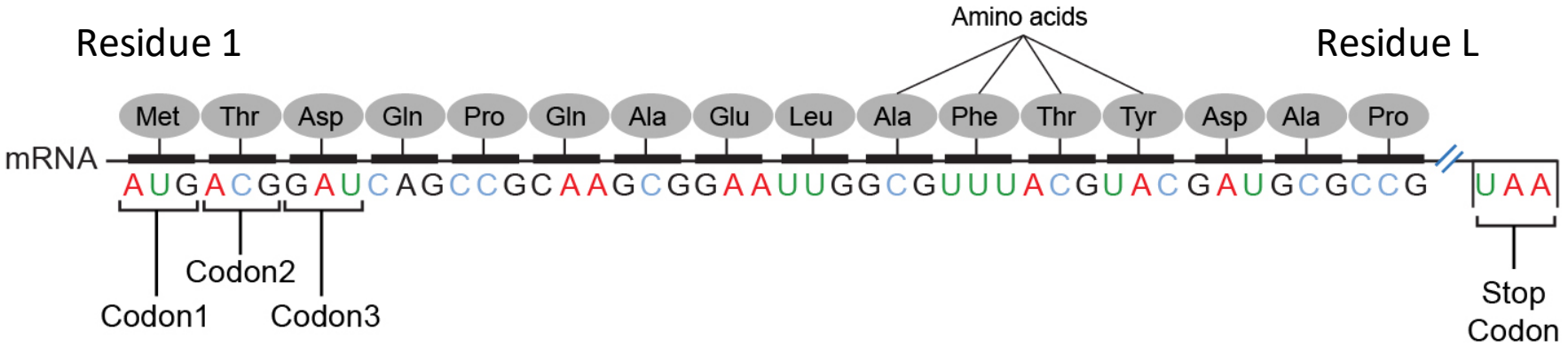
Illustration by Nicolle Rager Fuller



The central dogma of molecular biology

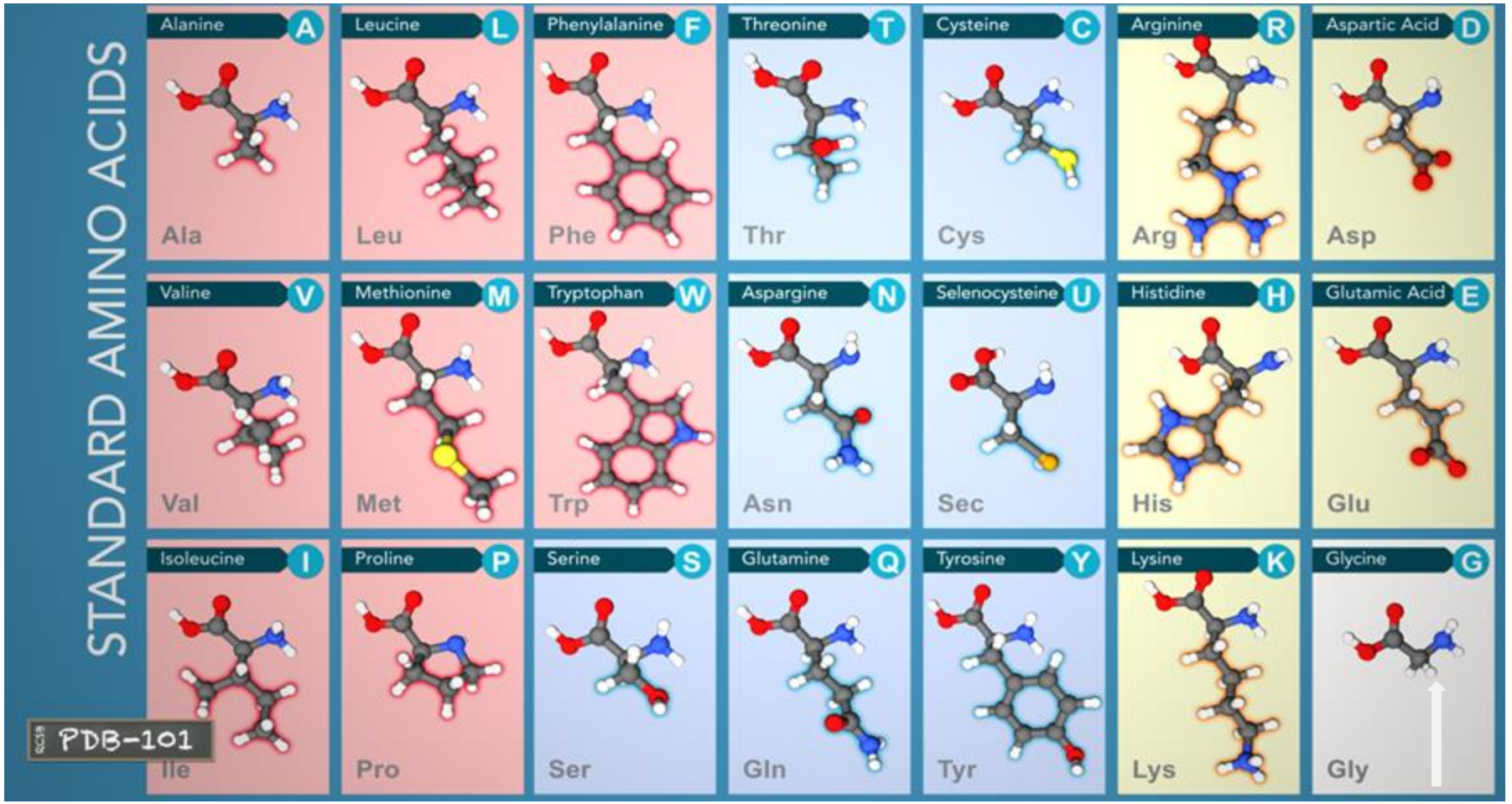
Major focus of the course: **deep learning** for modeling, understanding, and designing the sequence, structure and function of biological molecules.





All proteins are made out of **21 building blocks**, called **amino acids**.

The side chain is the only part that varies from amino acid to amino acid and determines its properties.



Hydrophobic | Hydrophilic or polar | Charged | No side chain

Protein vs genome sequence

	Natural language	Proteins	Genomics
length	~10k	hundreds	**1k - millions**
resolution (tokenizer)	word-level	character-level 21+X	**character-level** 4+X

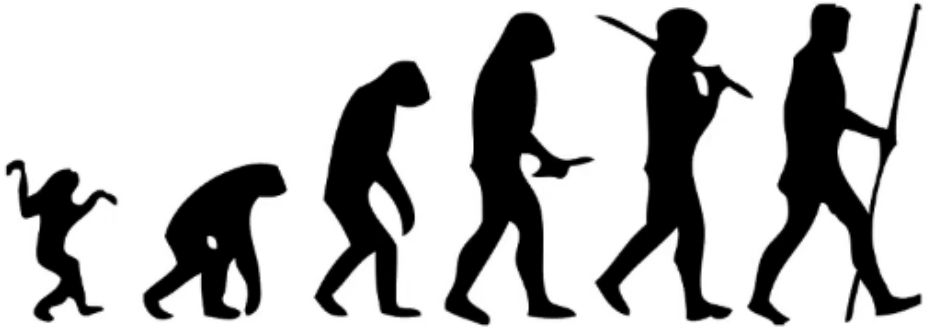
Ultralong sequences, and **character-level** modeling important in genomics

Two schools of thoughts for solving folding

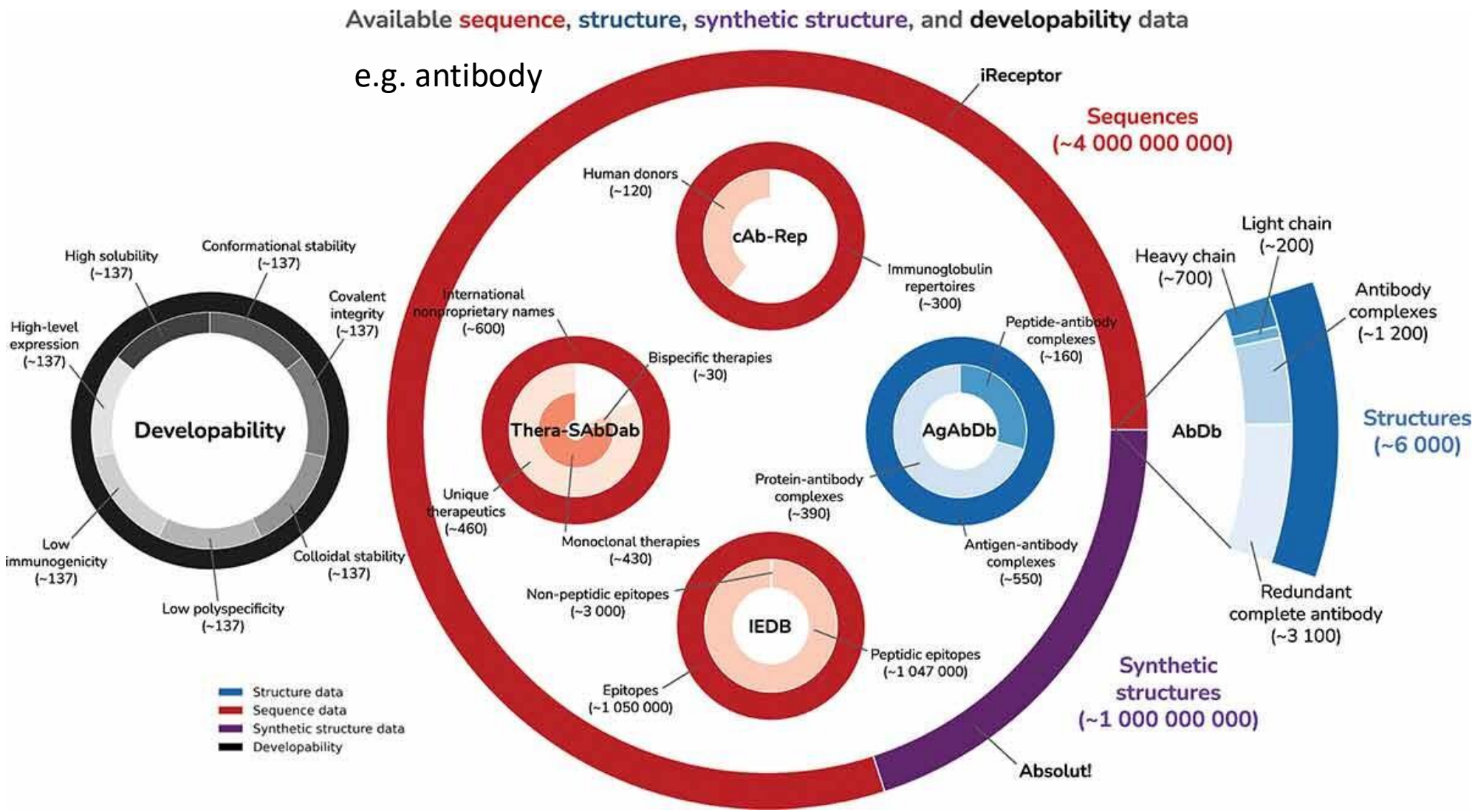
1. Physics-based approaches: Anfinsen's Dogma
   - Rosetta, MD Simulations, etc. don't scale well
   - Levinthal's Paradox



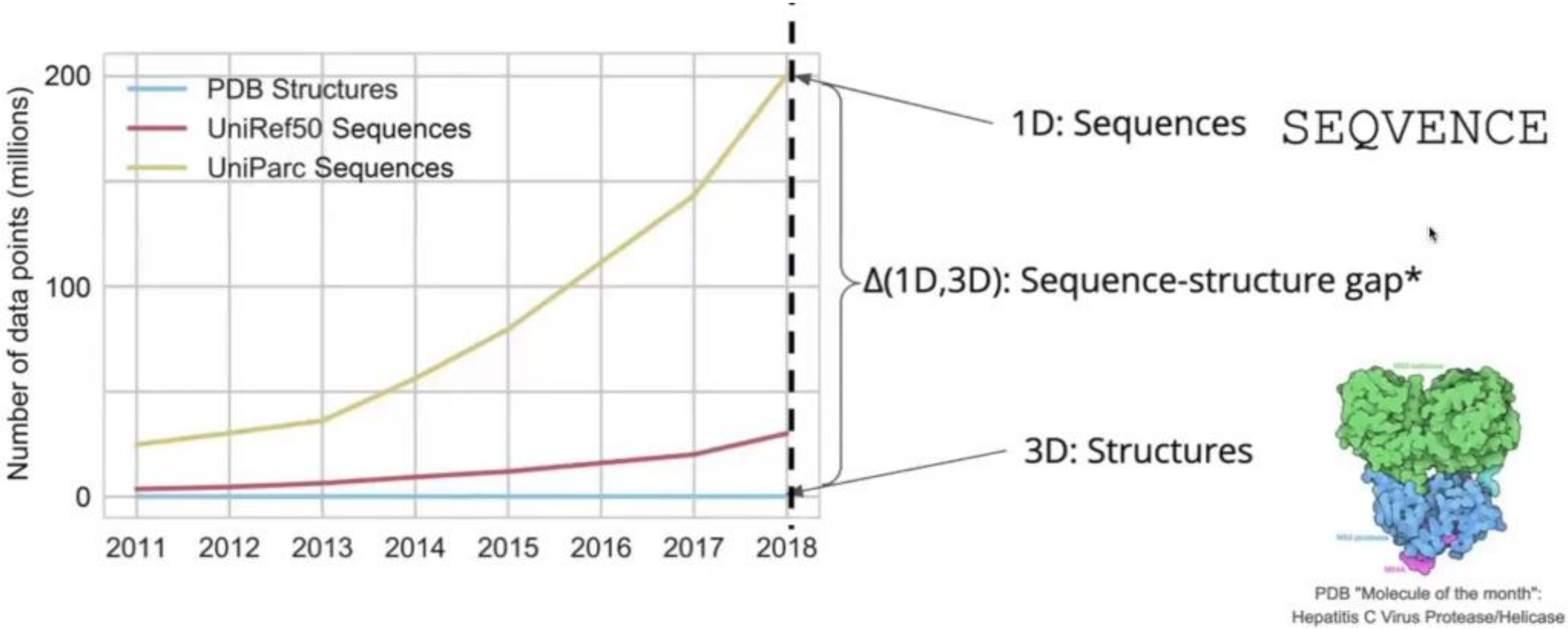
2. Evolutionary Information
   - Homologous Structures
   - Multiple Sequence Alignment (MSA)



Protein sequence data is more available



Data landscape



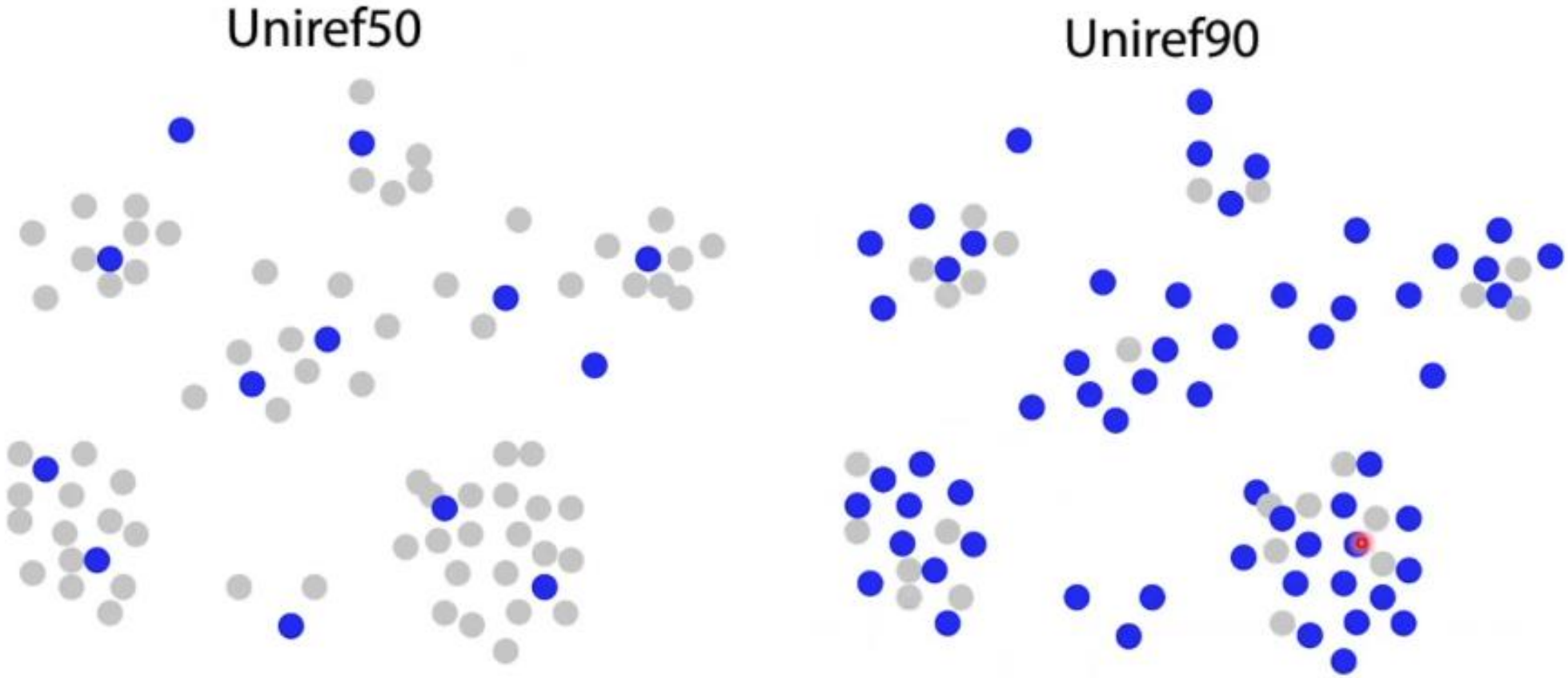
Dataset

- Protein sequences are not i.i.d.
- Models trained on all data might overfit to large clusters

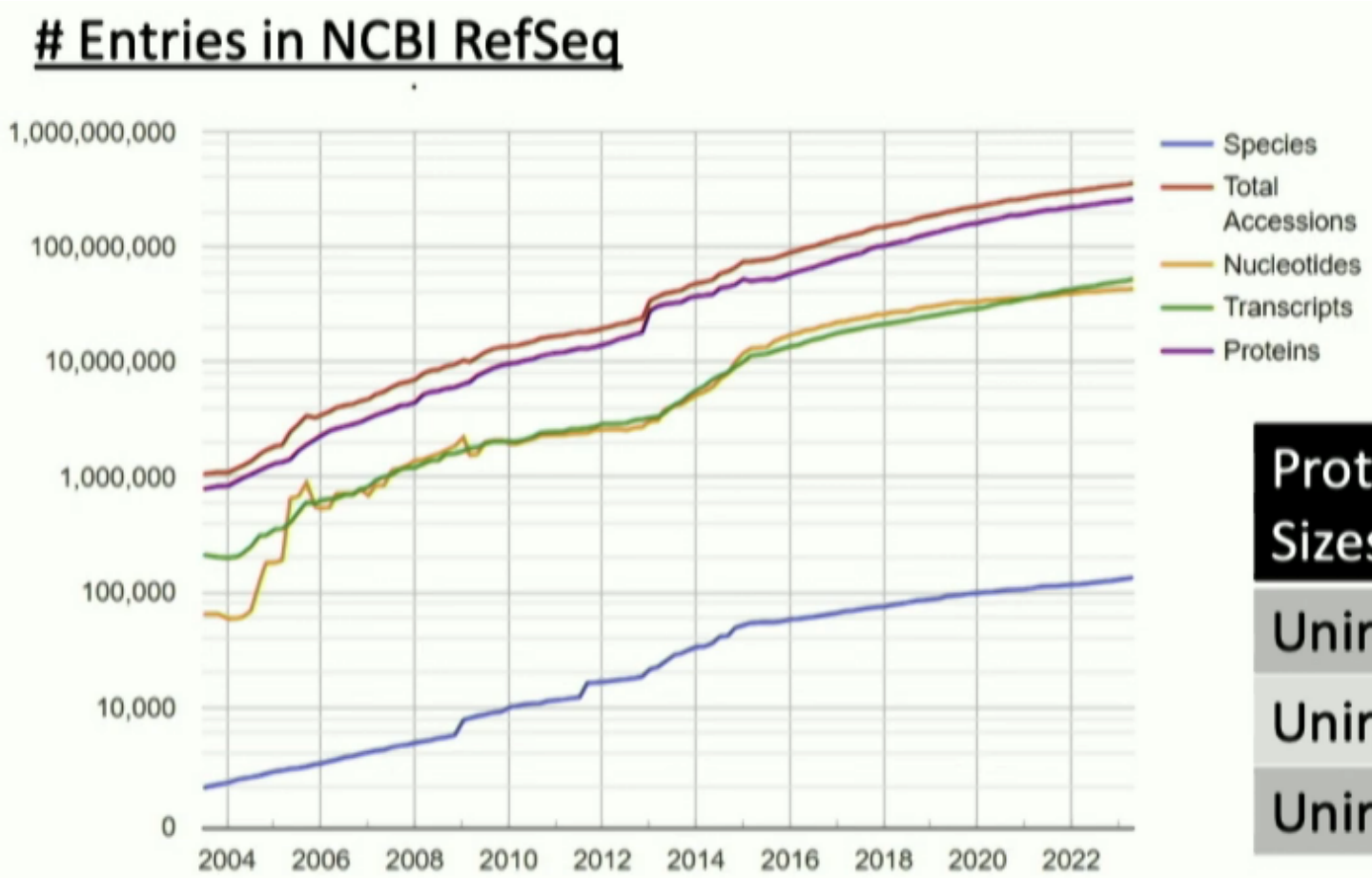


Dataset

- Control Cluster by % of sequence identify
- sequence of a representative (best-annotated) protein

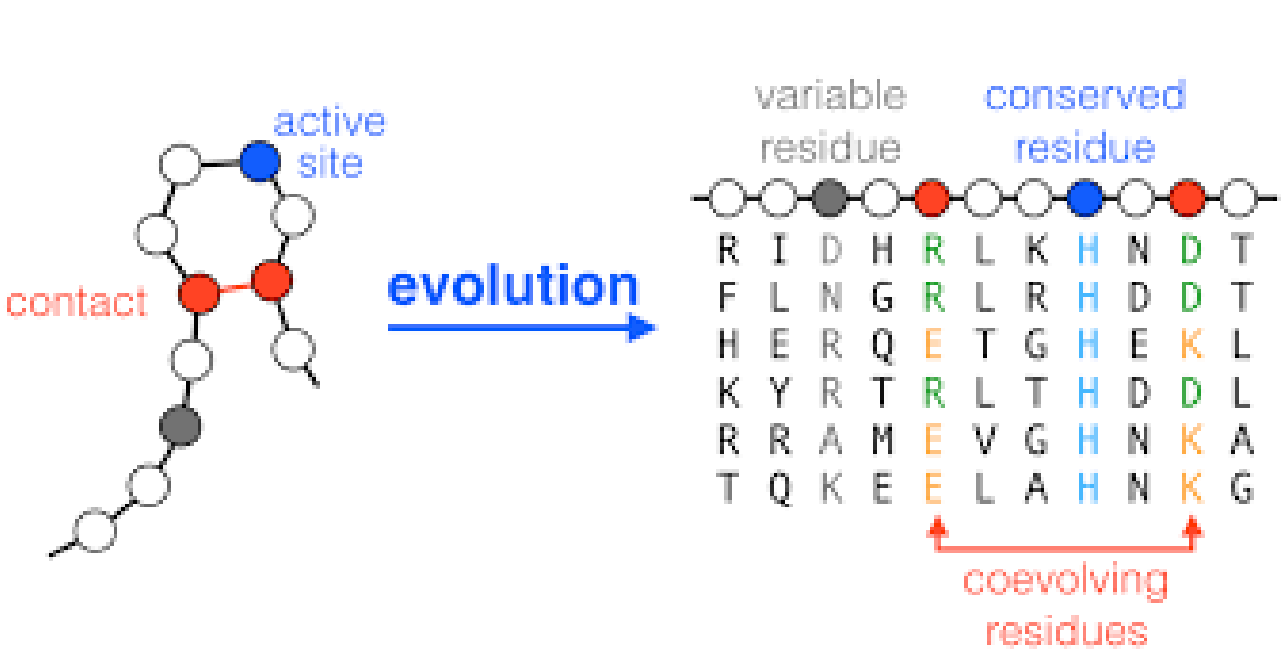


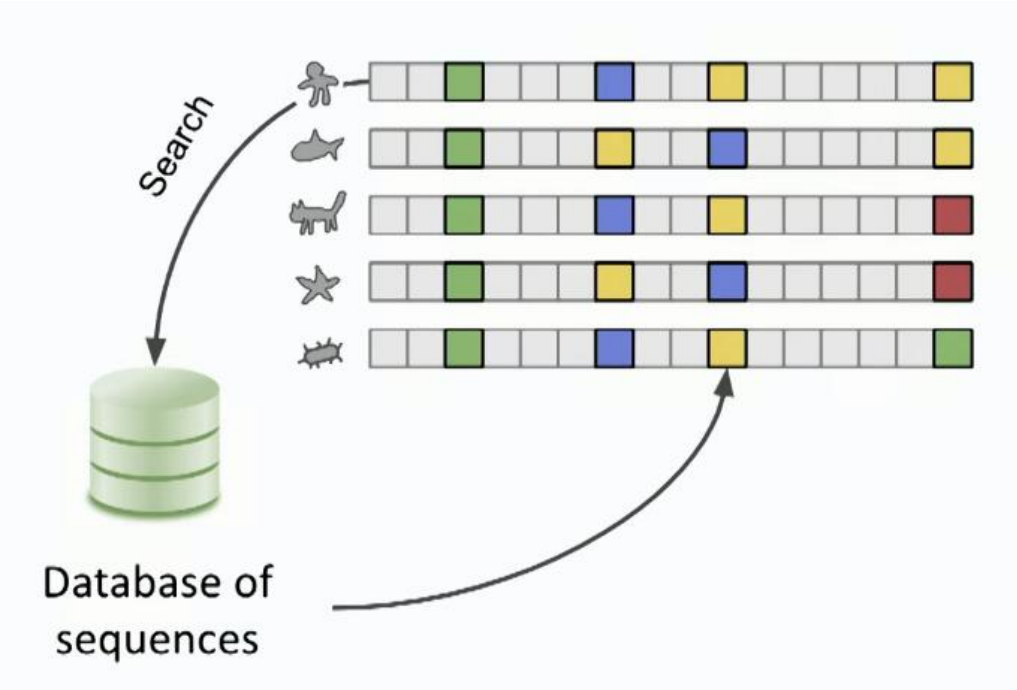
Uniref database



Protein Dataset Sizes	Count (in millions)
Uniref 100 (i.e. all)	351
Uniref 90	171
Uniref 50	60.5

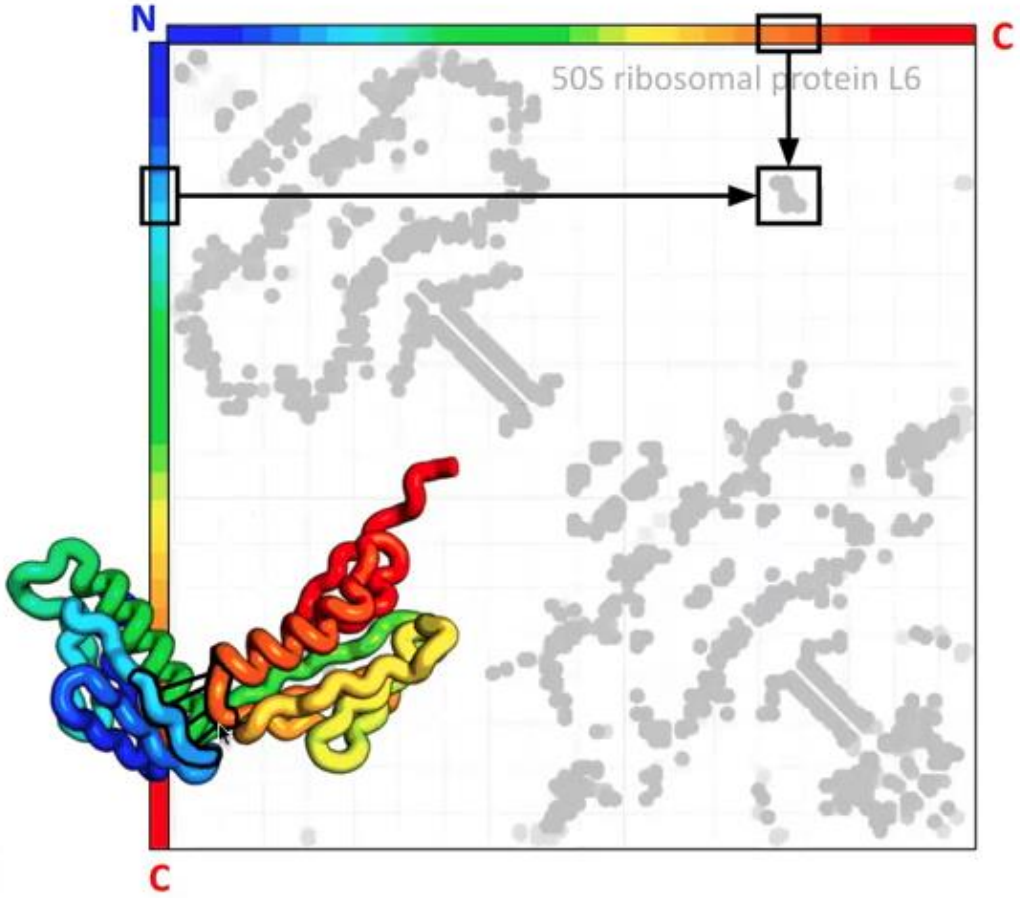
Sequence data contains rich information about evolution (and structure / function)

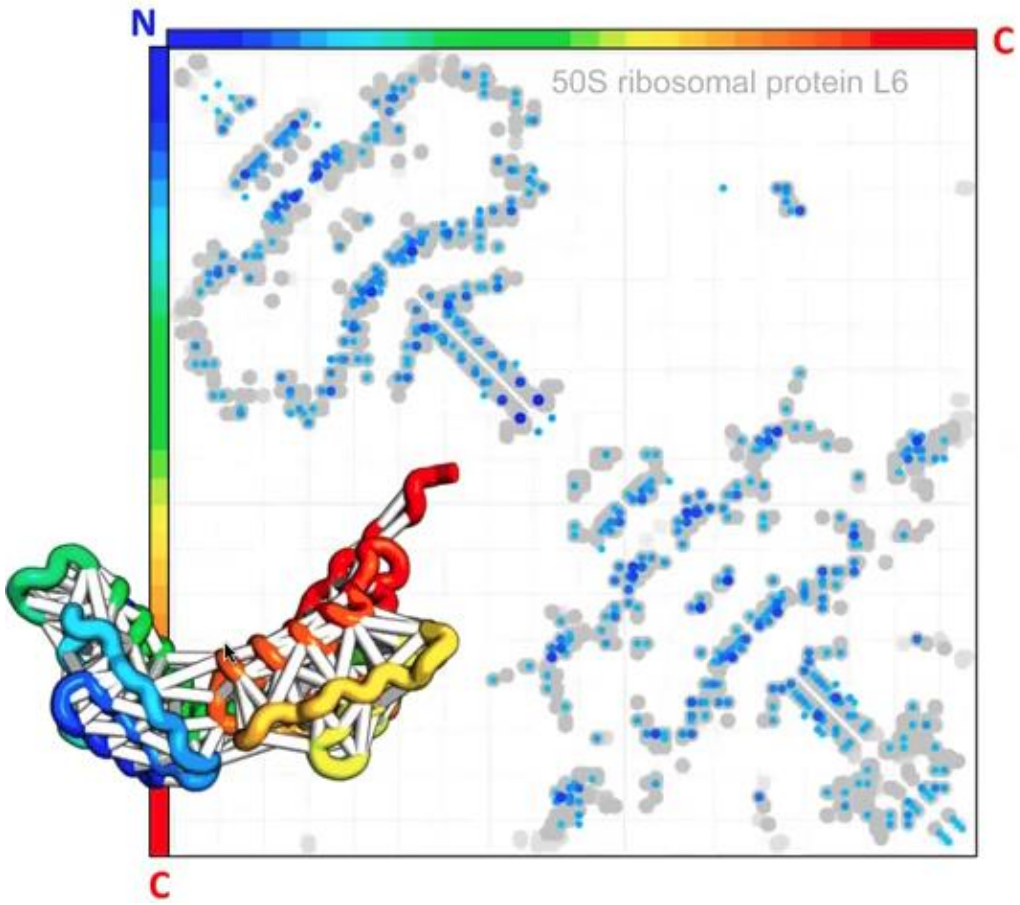




By measuring **co-evolution**, we can infer contacts in proteins!

Contact map of protein

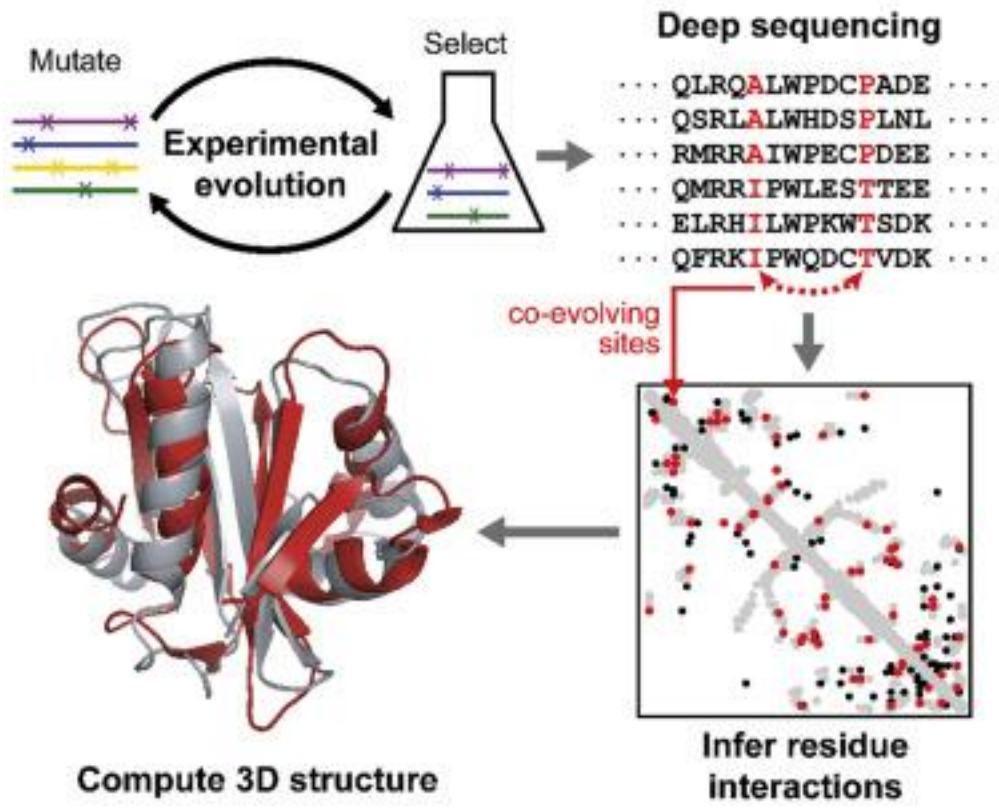


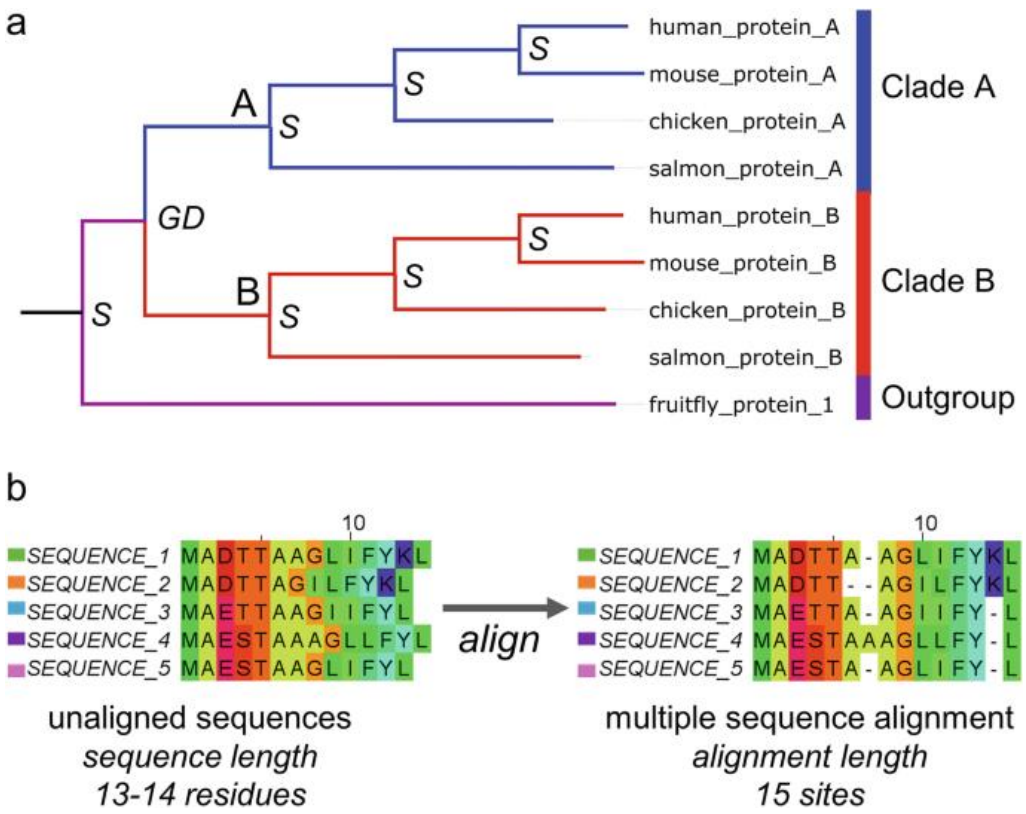


XRAY Contacts

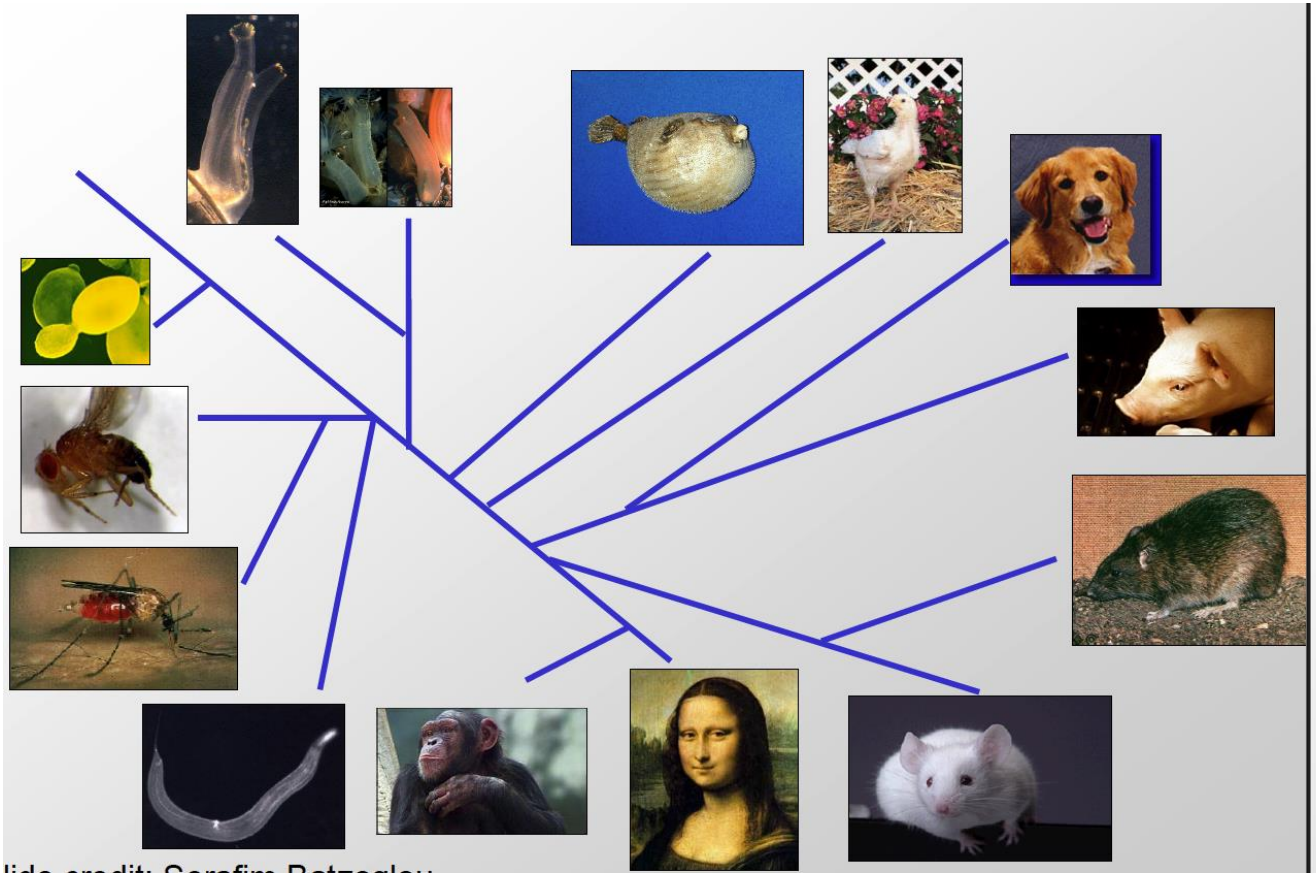
Predicted Contacts

Multiple sequence alignment (MSA)

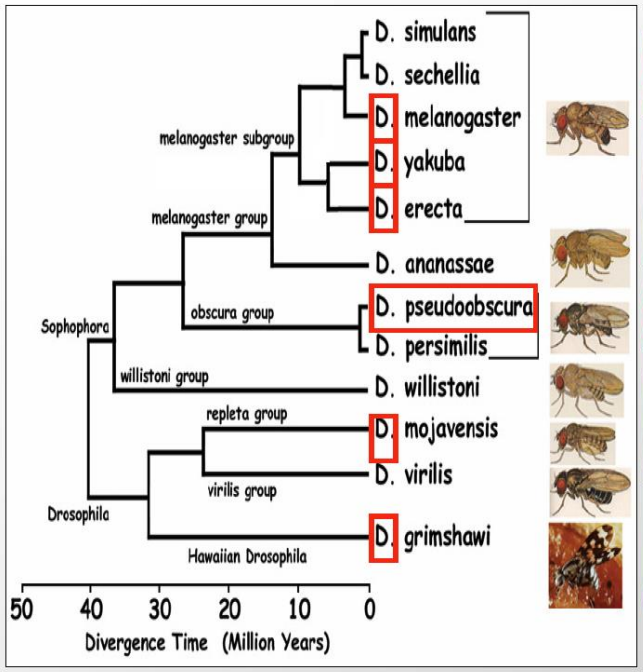


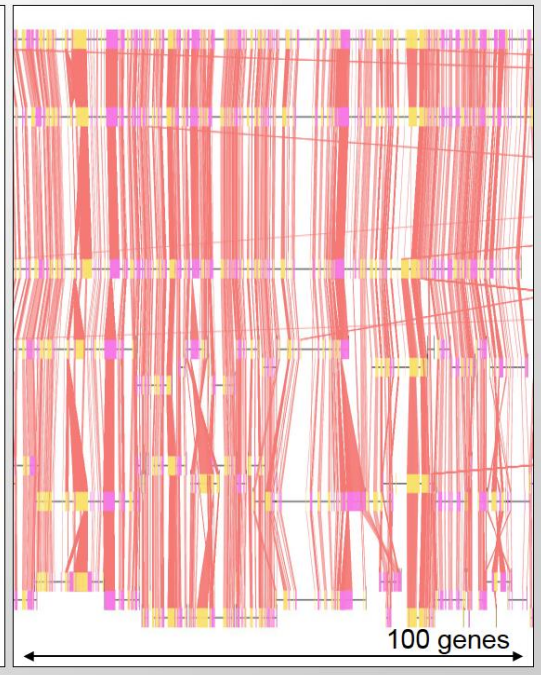


Alignment: all species/genes share common ancestry



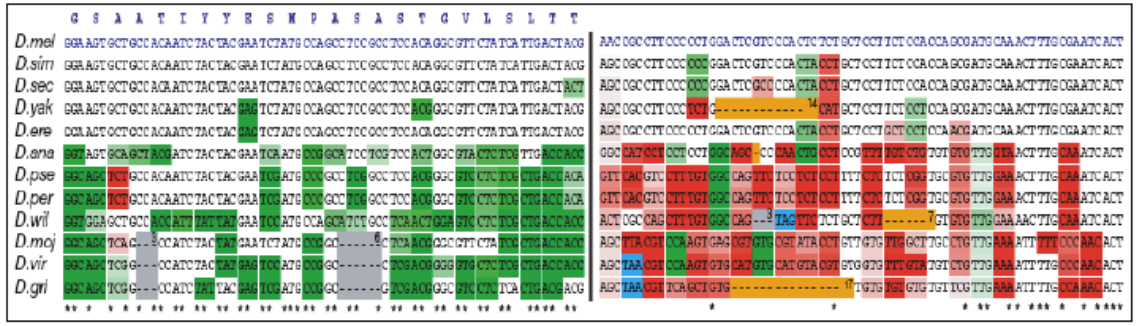
Genome-wide alignments reveal orthologous segments





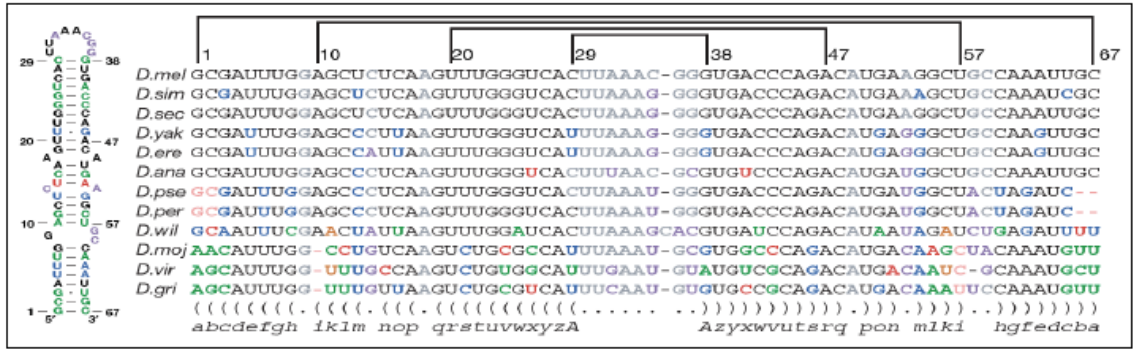
- **Genome-wide alignments span entire genome**
- **Comparative identification of functional elements**

Evolutionary signatures for diverse functions



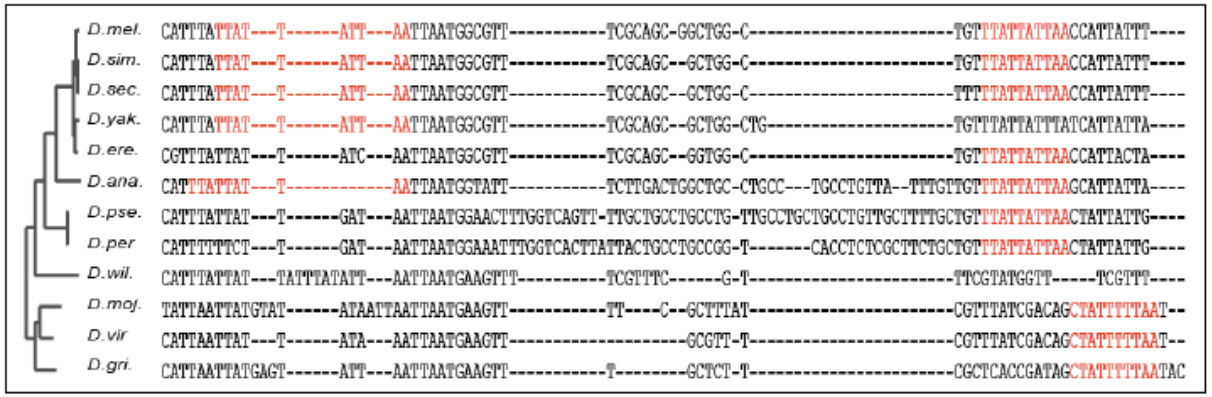
Protein-coding genes

- Codon Substitution Frequencies
- Reading Frame Conservation



RNA structures

- Compensatory changes
- Silent G-U substitutions

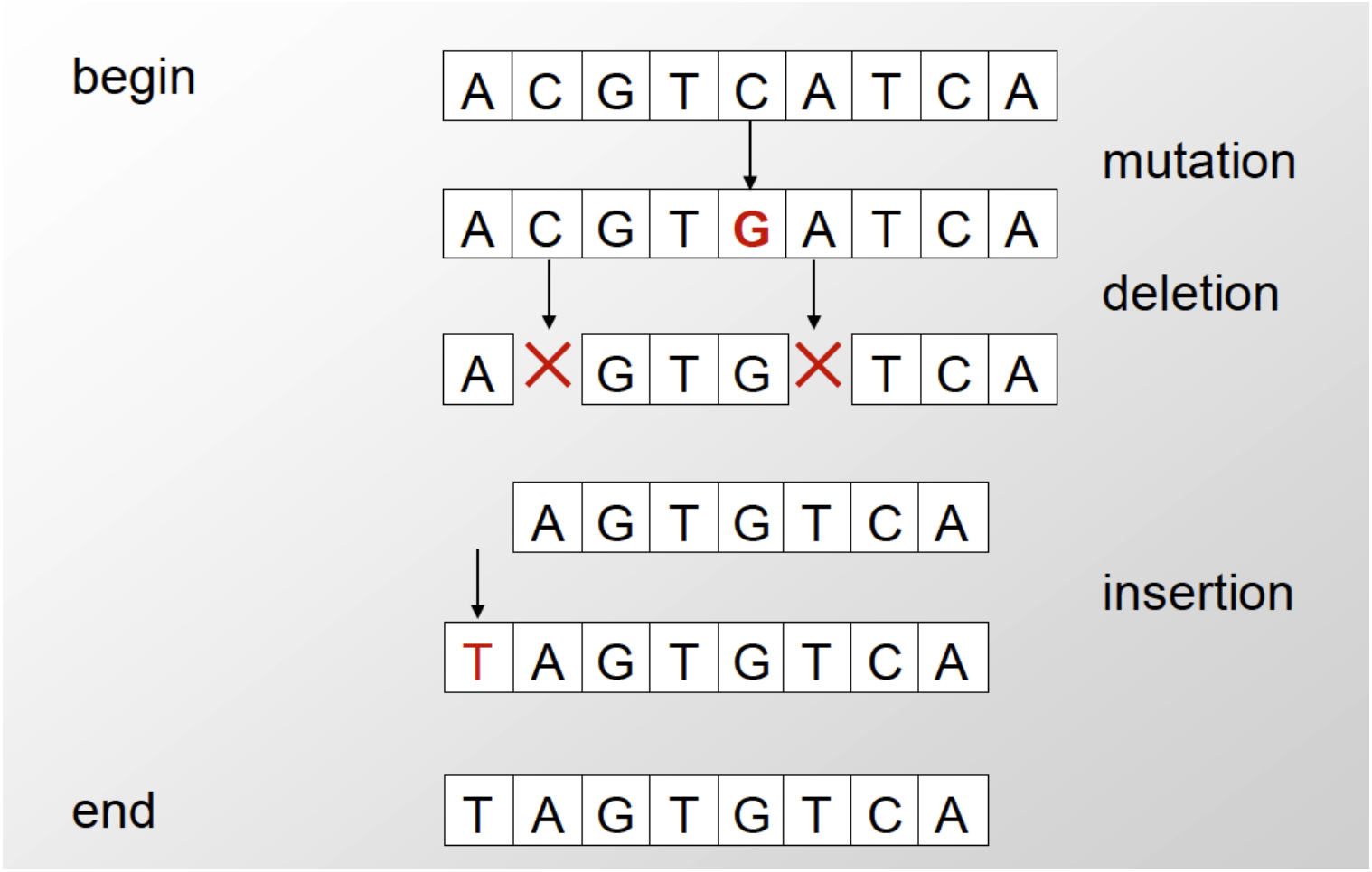


Regulatory motifs

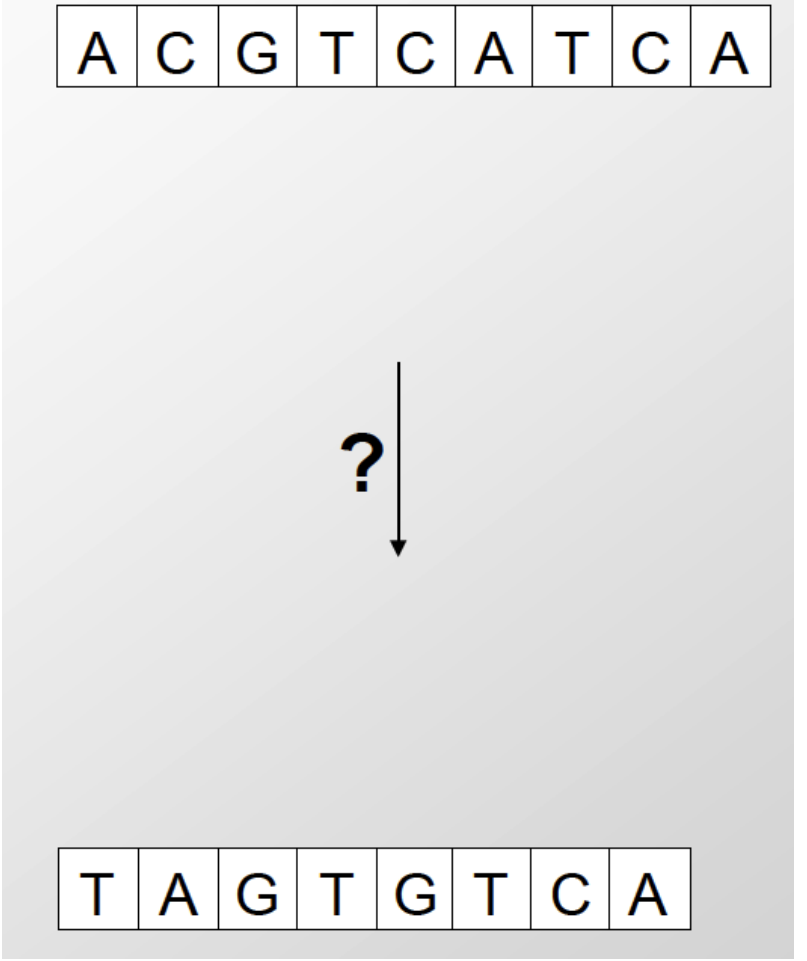
- Mutations preserve consensus
- Increased Branch Length Score
- Genome-wide conservation

Stark et al, Nature 2007

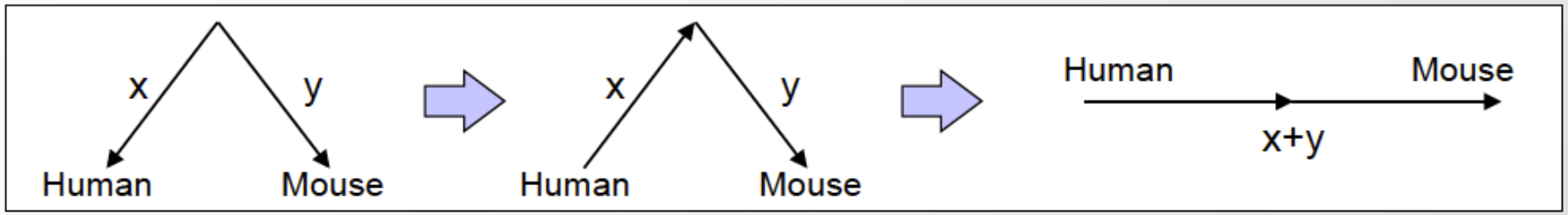
Genomes change over time



Alignment: infer edit operations

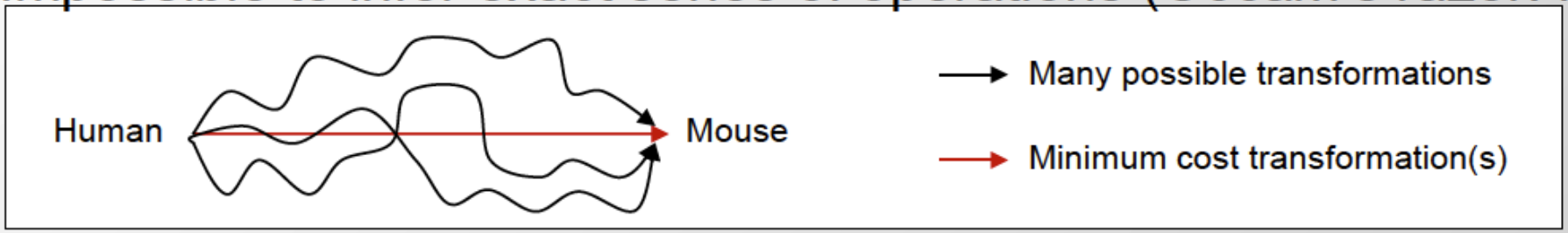


- Define set of evolutionary operations (insertion, deletion, mutation)
  - Symmetric operations allow time reversibility (part of design choice)

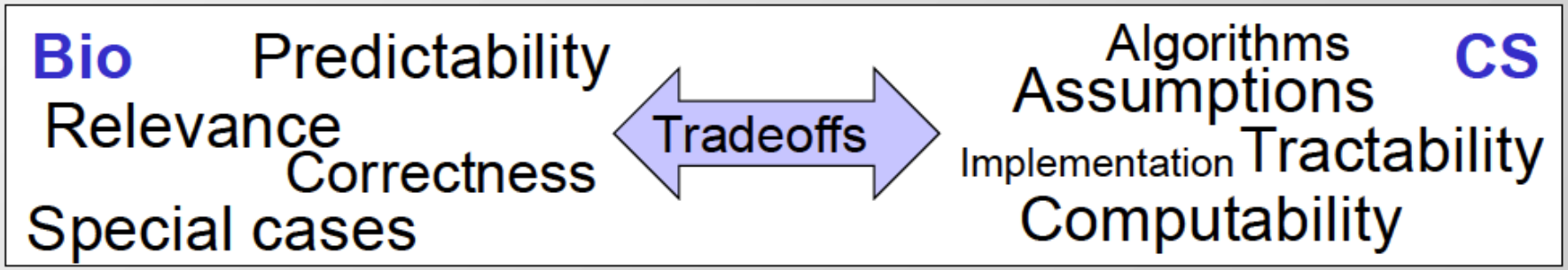


(Exception: methylated CpG dinucleotides → TpG/CpA non-symmetric)

- Define optimality criterion (min number, min cost)
  - Impossible to infer exact series of operations (Occam's razor: find min)

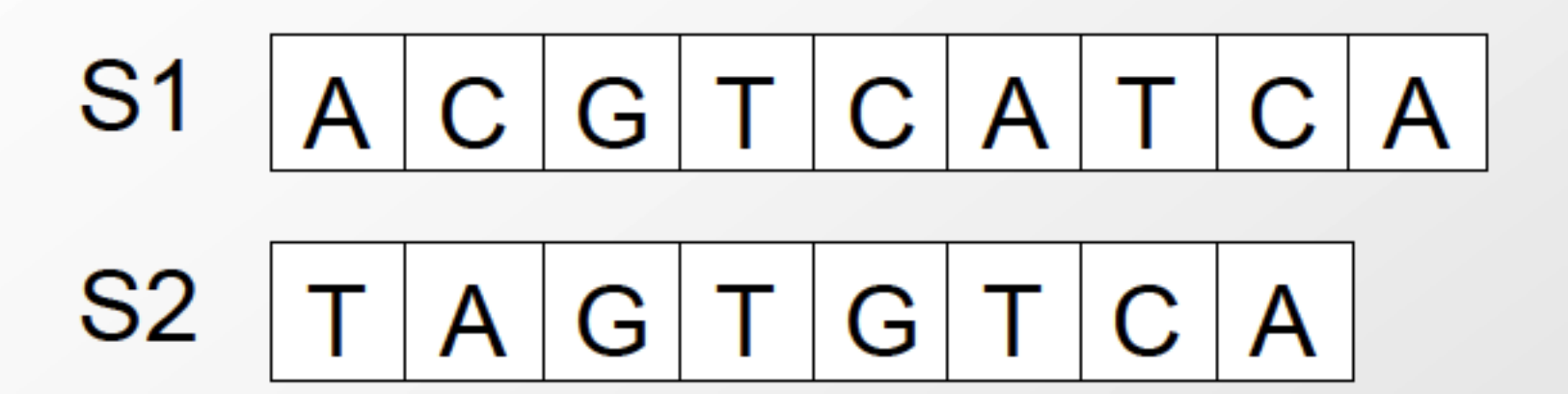


- Design algorithm that achieves that optimality (or approximates it)
  - Tractability of solution depends on assumptions in the formulation



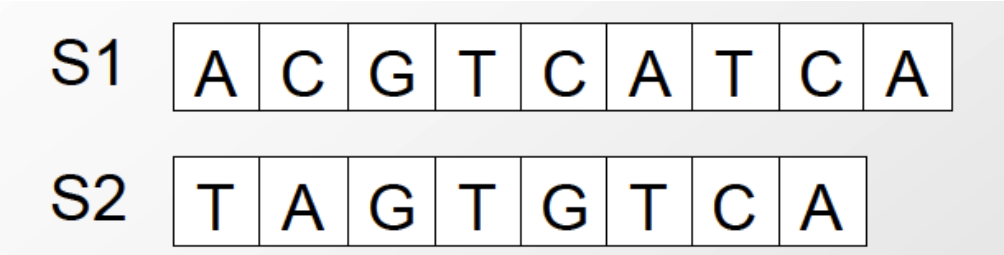
Naïve solution: longest common substring

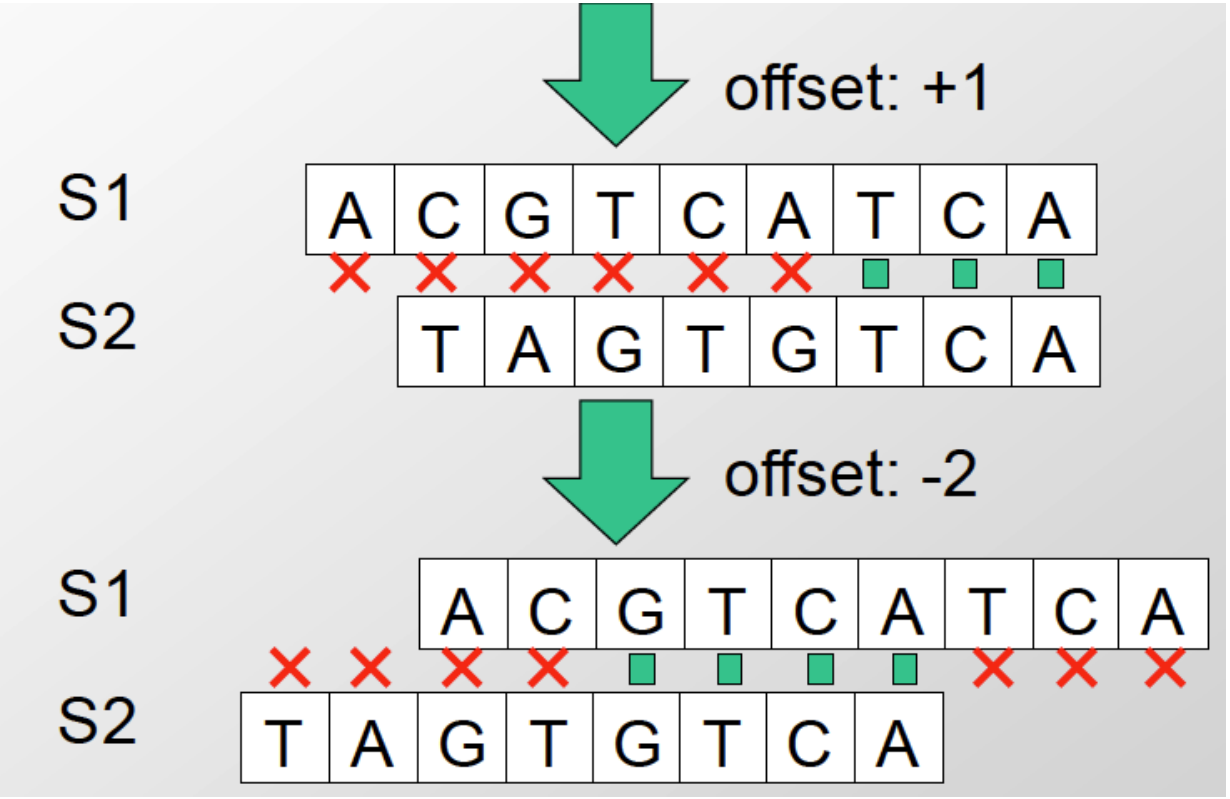
Given two possibly related strings S1 and S2What is the longest common substring? (no gaps)



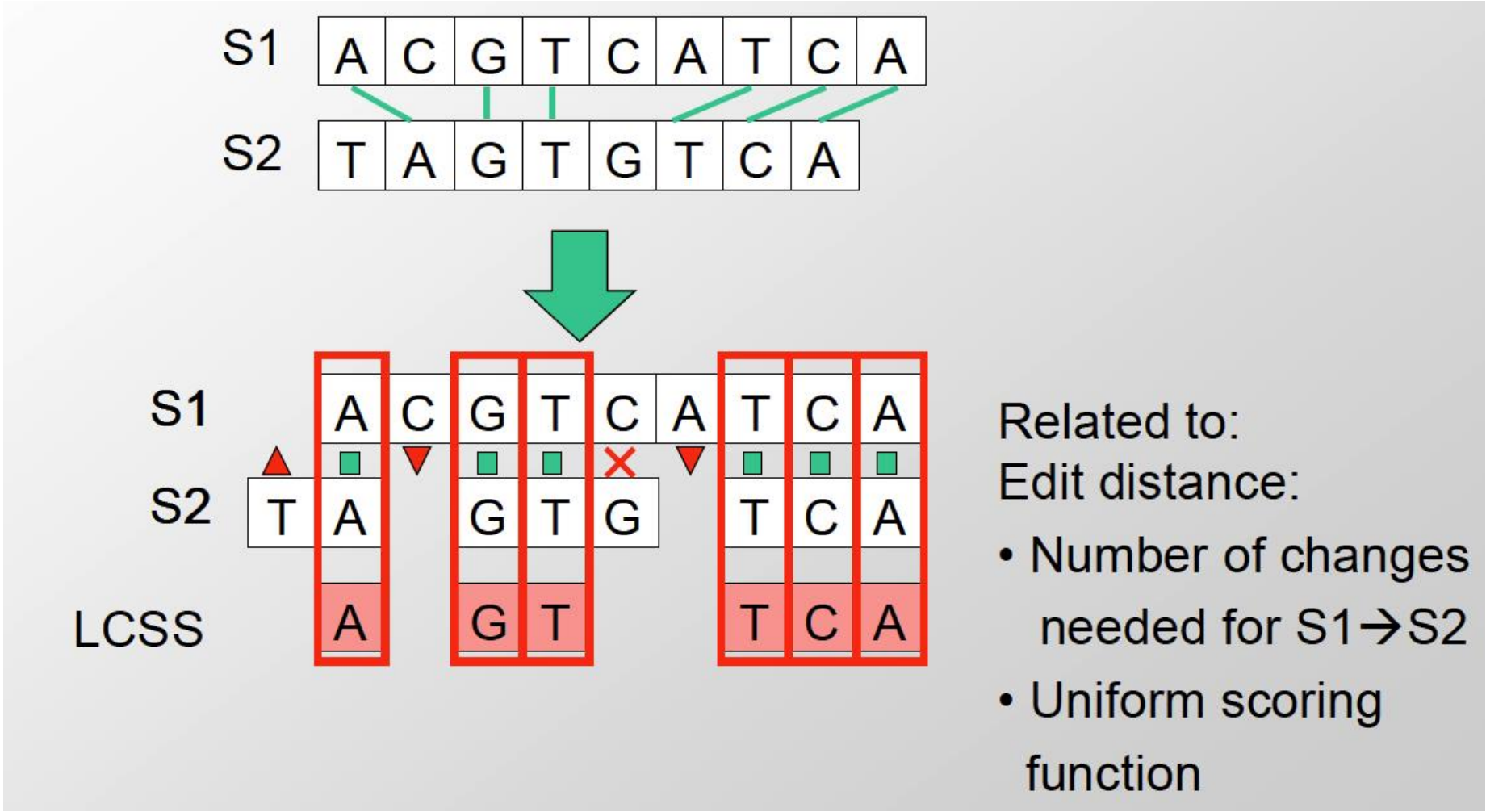
Naïve solution: longest common substring

Given two possibly related strings S1 and S2What is the longest common substring? (no gaps)





Better formulation: Longest common subsequence (allow gaps)



Final formulation: sequence alignment

- Allow gaps (fixed penalty) -> advanced: varying gap costs
  - Insertion & deletion operations
  - Unit cost for each character inserted or deleted
- Varying penalties for edit operations
  - Transitions (Pyrimidine<->Pyrimidine, Purine<->Purine)
  - Transversions (Purine <-> Pyrimidine changes)
  - Polymerase confuses Aw/G and Cw/T more often

Scoring function:
Match(x,x) = +1
Mismatch(A,G)= -½
Mismatch(C,T)= -½
Mismatch(x,y) = -1

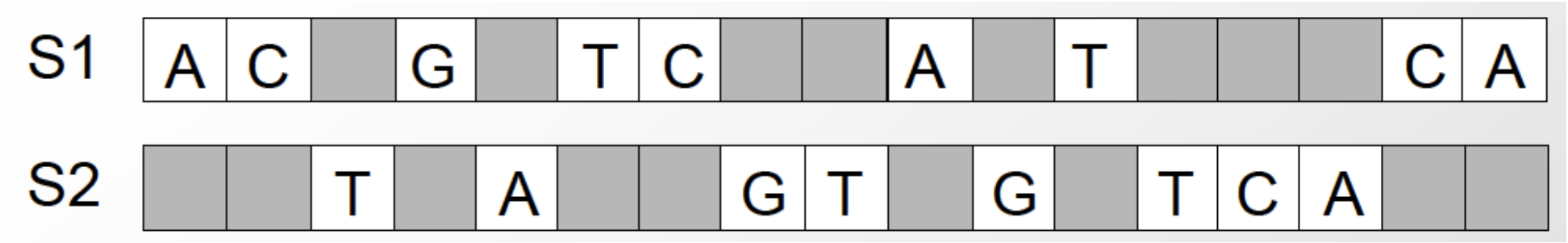
	A	G	T	C
A	+1	-½	-1	-1
G	-½	+1	-1	-1
T	-1	-1	+1	-½
C	-1	-1	-½	+1

purine pyrimid.

Transitions:
A⇔G, C⇔T common (lower penalty)

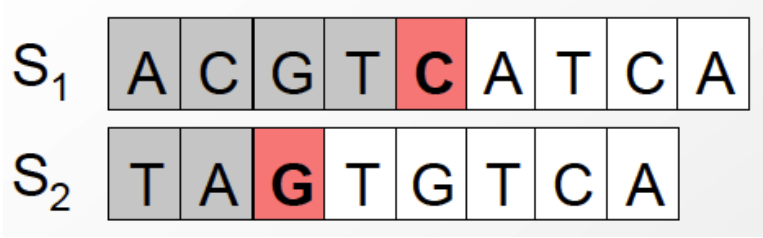
Transversions:
All other operations

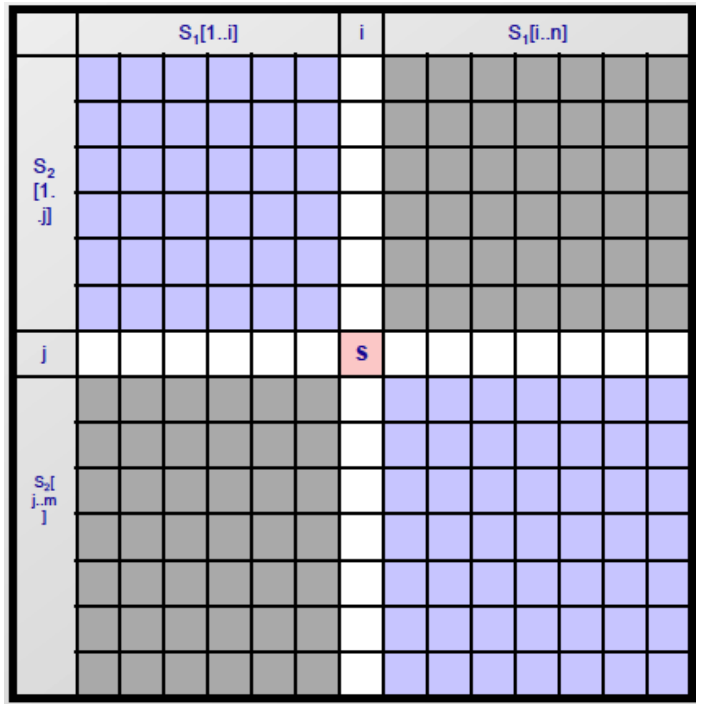
Complexity of pair-sequence alignment

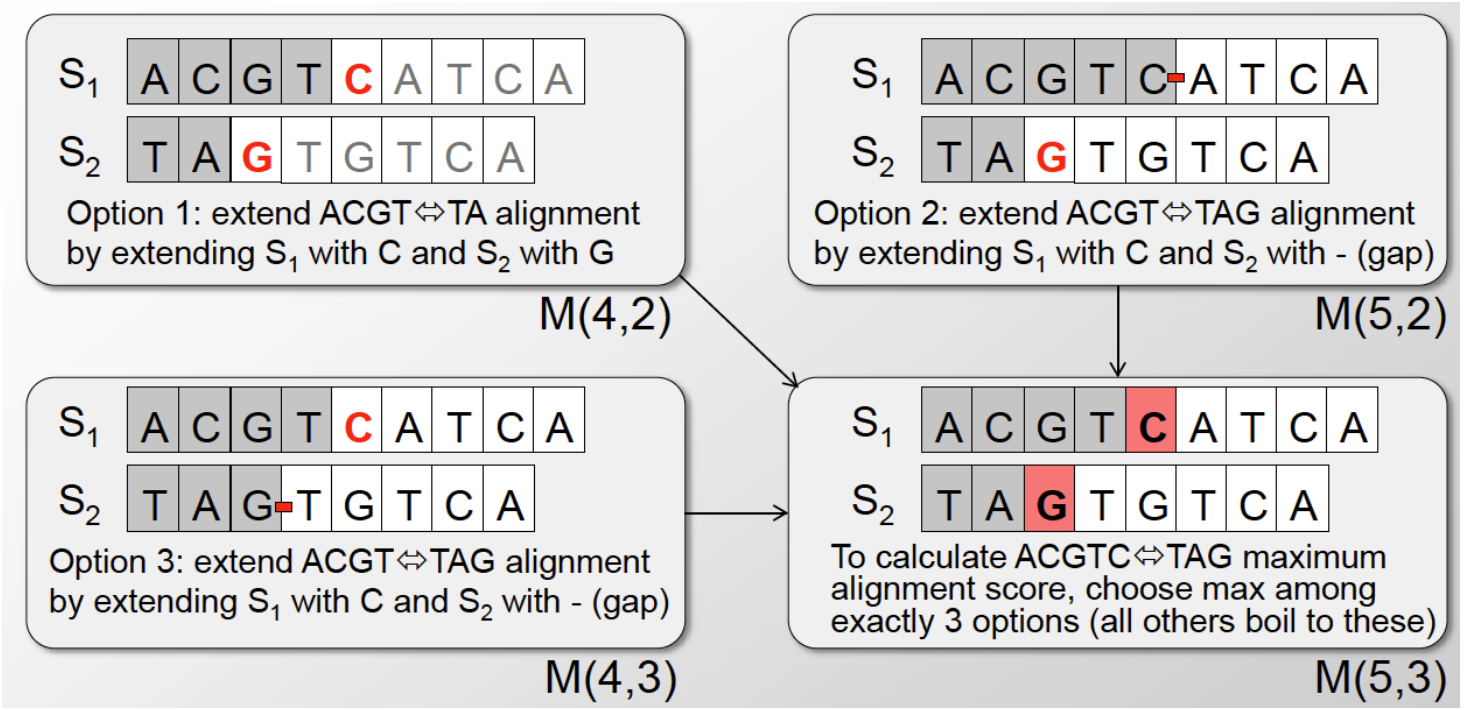


- Longest ‘non-boring’ alignment: n+m entries
  - Otherwise a gap will be aligned to a gap -> condense
- Alignment is equivalent to gap placement
  - (n+m choose n) ways to choose S1 placement
  - At each position yes/no answer of placing character
  - Exponential number of possible placements
- Need faster solution for finding best alignment: Dynamic Programming!

Compute best alignment recursively: small -> big

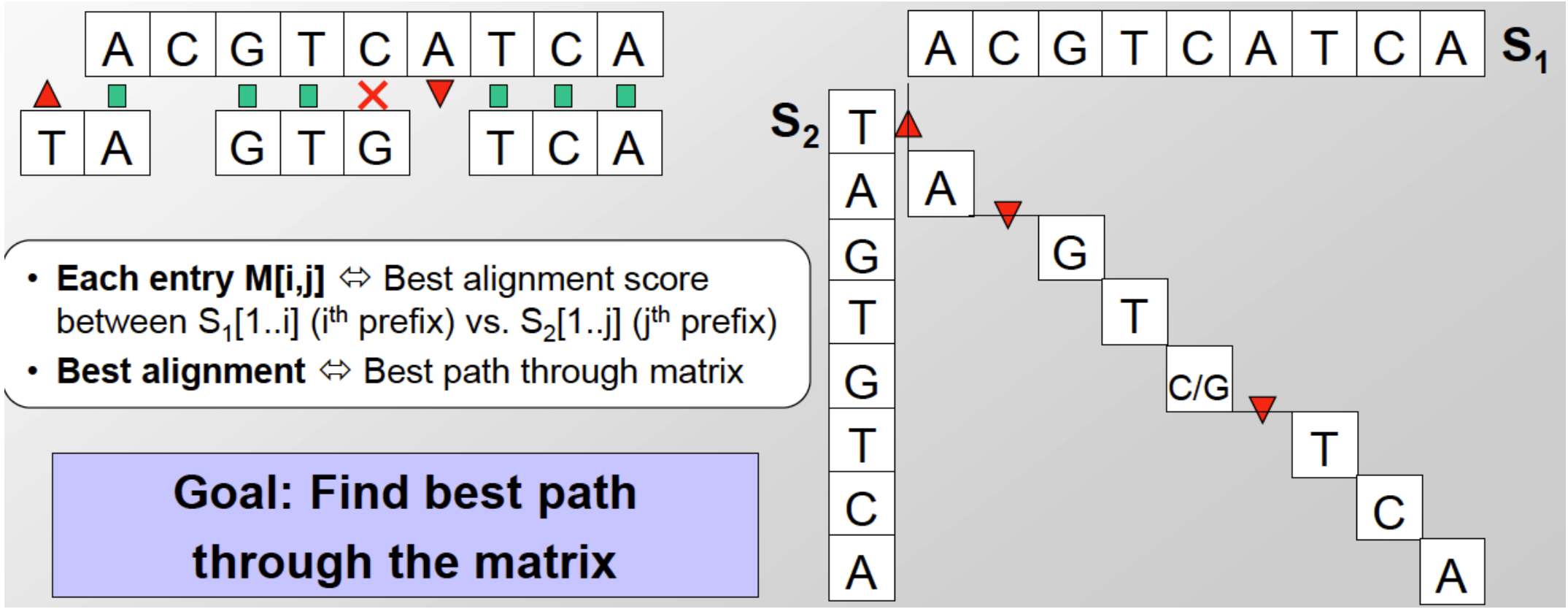






- Key idea: Calculate maximum alignment score of longer sequences based on previously-computed scores of shorter sequences
  - Assume all shorter alignments are already computed
  - Show that you can use them to compute longer alignment using them
  - By induction, can compute alignments of any length this way
- Implementation: Store all these scores in a matrix M(S1prefix, S2prefix)
  - Fill the matrix systematically from the top left to the bottom right

Path through the matrix: Alignment of S1and S2



Pair sequence alignment illustration

Initialize the scoring matrix

		T	G	T	T	A	C	G	G
	0	0	0	0	0	0	0	0	0
G	0								
G	0								
T	0								
T	0								
G	0								
A	0								
C	0								
T	0								
A	0								

Substitution matrix: $S(a_i, b_j) = \begin{cases} +3, & a_i = b_j \\ -3, & a_i \neq b_j \end{cases}$

Gap penalty: $W_k = kW_1$, $W_1 = 2$

Fill the scoring matrix

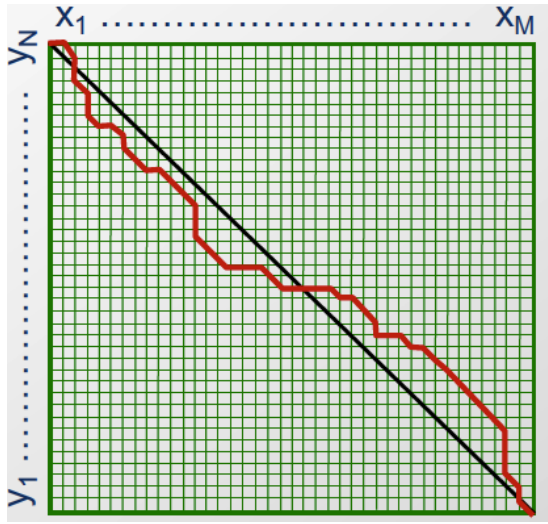
		T	G	T	T	A	C	G	G
	0	0	0	0	0	0	0	0	0
G	0	0	3	1	0	0	0	3	3
G	0	0	3	1	0	0	0	3	6
T	0	3	1	6	4	2	0	1	4
T	0	3	1	4	9	7	5	3	2
G	0	1	6	4	7	6	4	8	6
A	0	0	4	3	5	10	8	6	5
C	0	0	2	1	3	8	13	11	9
T	0	3	1	5	4	6	11	10	8
A	0	1	0	3	2	7	9	8	7

Substitution matrix: $S(a_i, b_j) = \begin{cases} +3, & a_i = b_j \\ -3, & a_i \neq b_j \end{cases}$

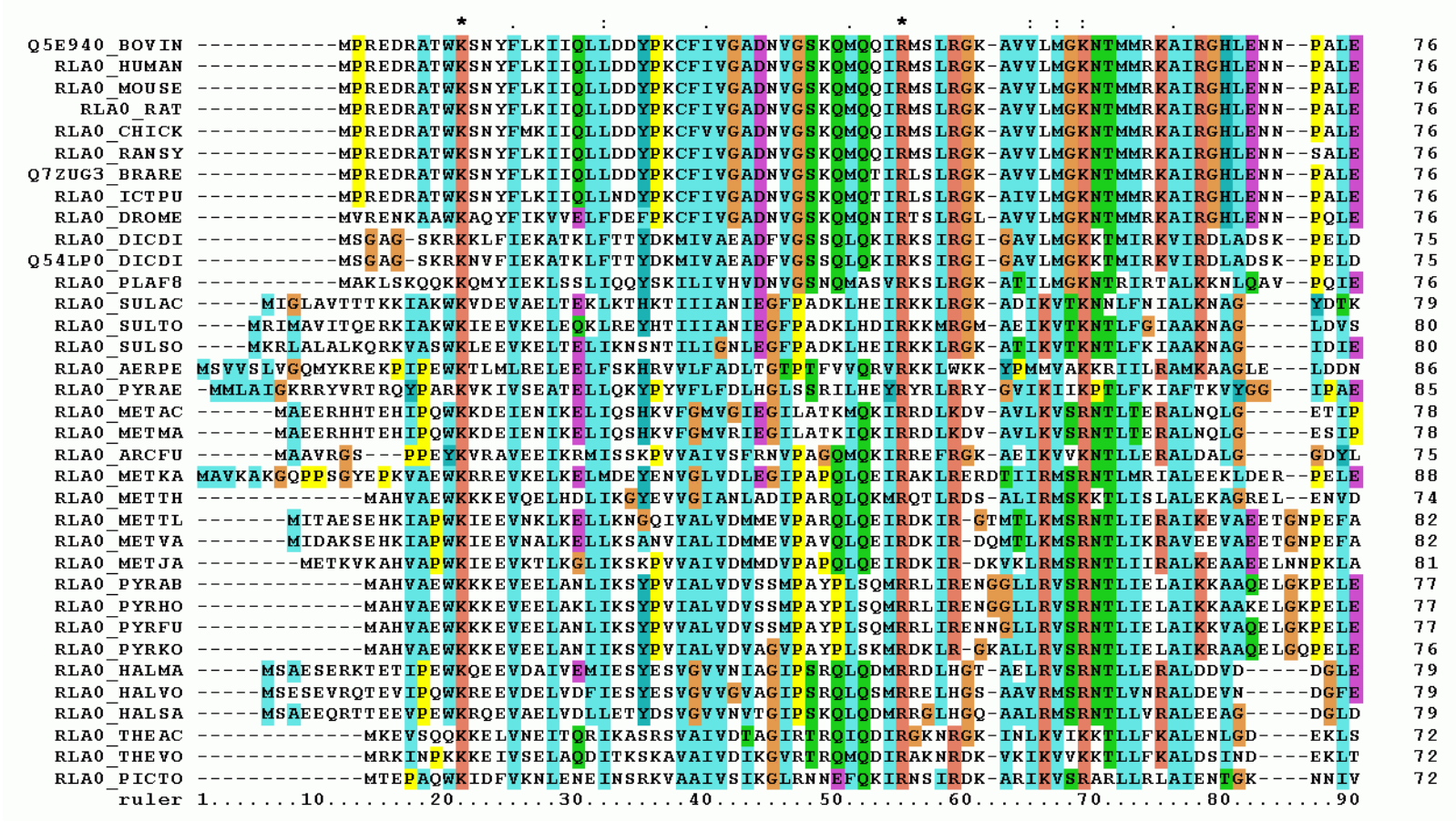
Gap penalty: $W_k = kW_1$, $W_1 = 2$

		T	G	T	T	A	C	G	G
	0	0	0	0	0	0	0	0	0
G	0	0	3	1	0	0	0	3	3
G	0	0	3	1	0	0	0	3	6
T	0	3	1	6	4	2	0	1	4
T	0	3	1	4	9	7	5	3	2
G	0	1	6	4	7	6	4	8	6
A	0	0	4	3	5	10	8	6	5
C	0	0	2	1	3	8	13	11	9
T	0	3	1	5	4	6	11	10	8
A	0	1	0	3	2	7	9	8	7

3	6	9	7	10	13
G	T	T	-	A	C
I	I	I		I	I
G	T	T	G	A	C



Pair -> multiple sequence alignment

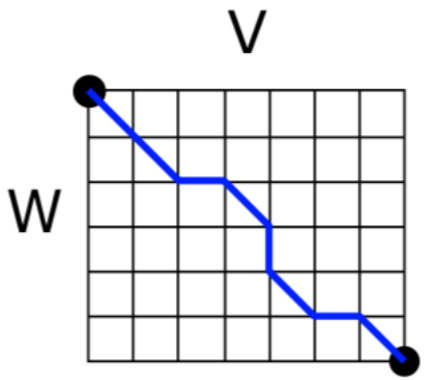


"Pairwise alignment whispers ... multiple alignment shouts out loud".

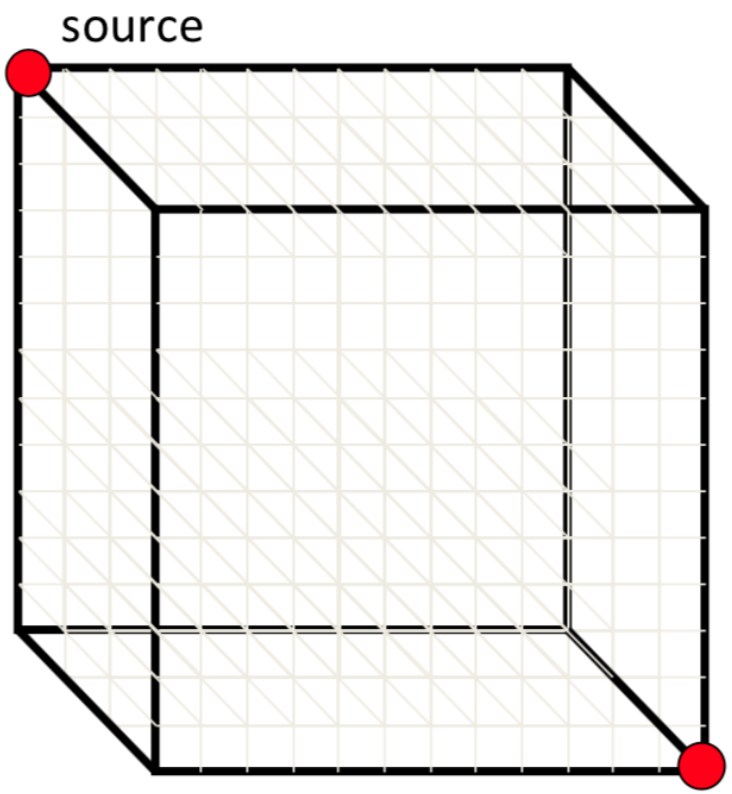
Hubbard, Lesk, Tramontano, Nature Structural Biology 1996.

Aligning Three Sequences

- Same strategy as pairwise edit distance
- Use 3-D cube, with each axis representing
- Alignment is a path from source to sink



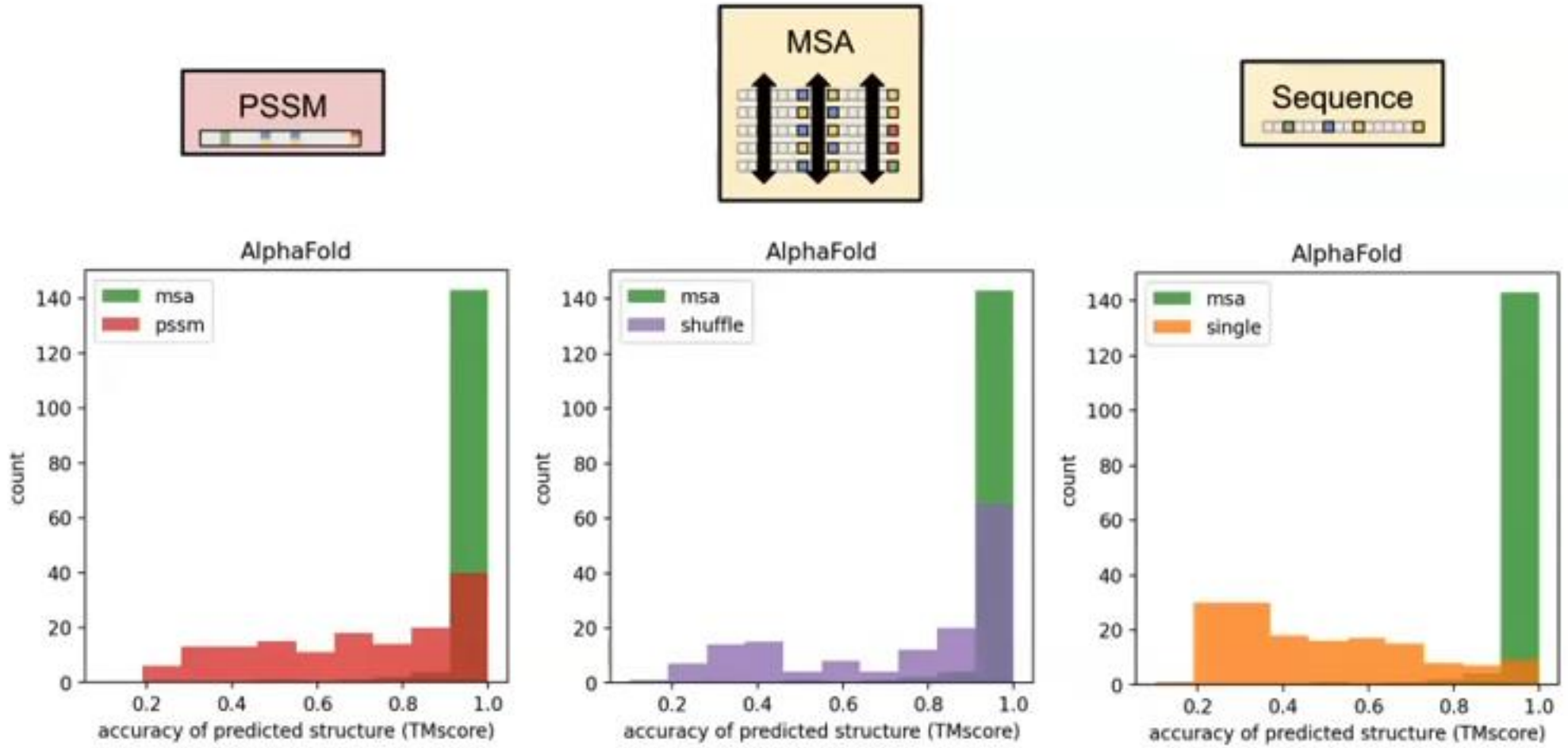
2-D Edit Graph



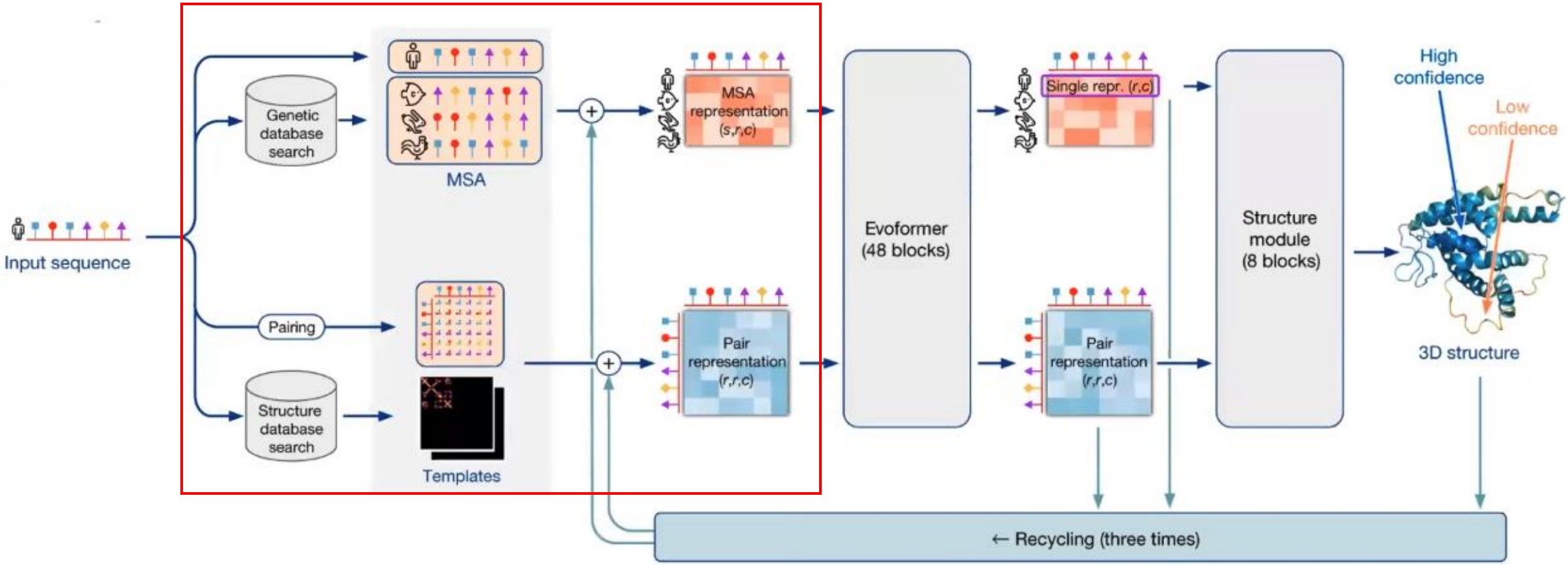
3-D Edit Graph

Take CS466(Fall) and CS581(Spring) if you’re interested in sequence alignment and phylogeny!

MSA co-evolution information is crucial

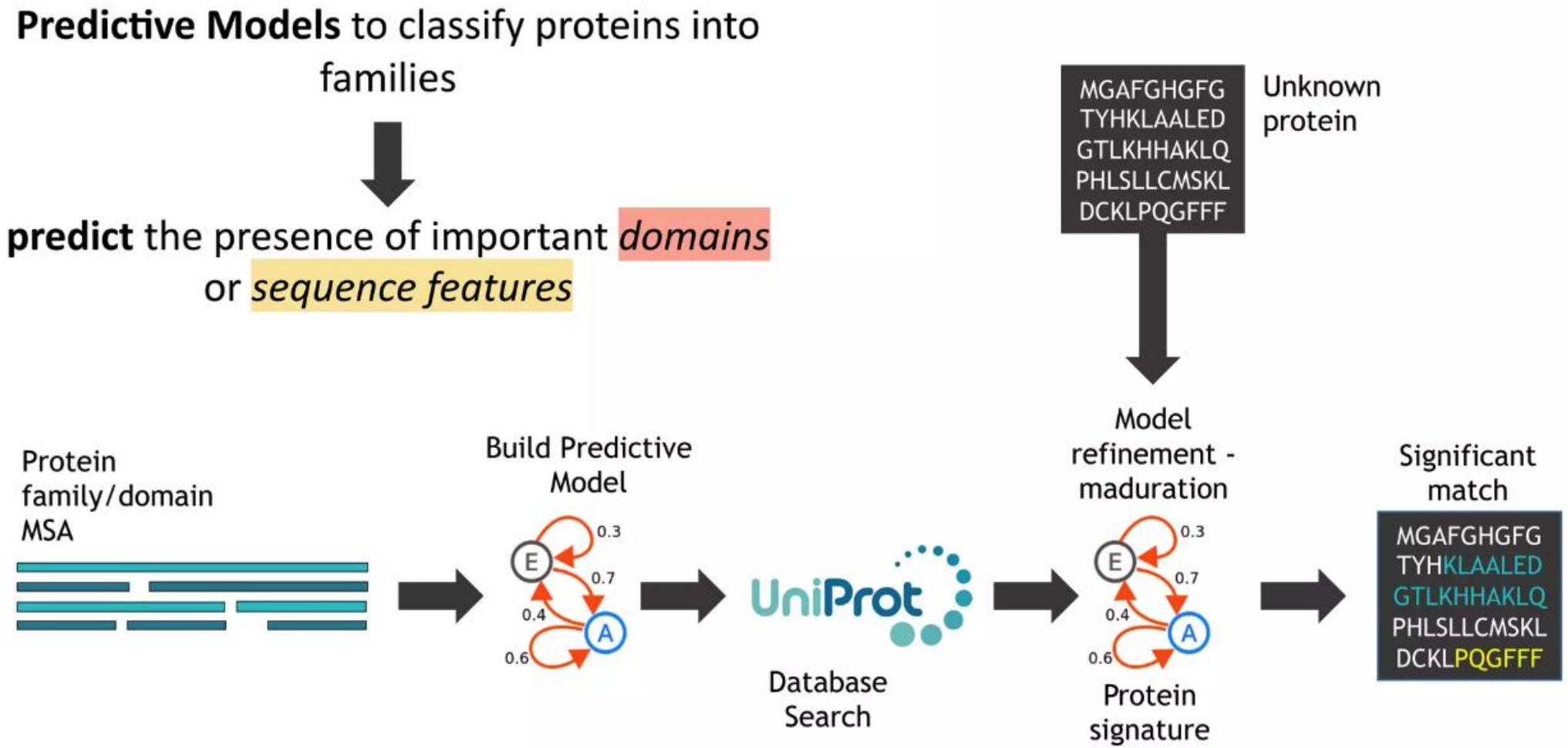


AlphaFold2 Architecture



Jumper et al., 2021

Protein sequence signatures

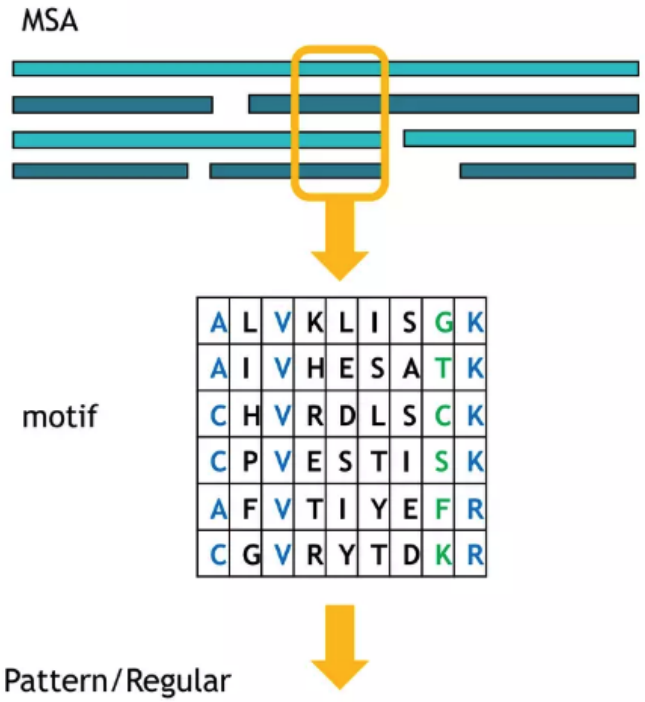


Modeling MSA: PSSM

1) Patterns

Regular expressions to model:

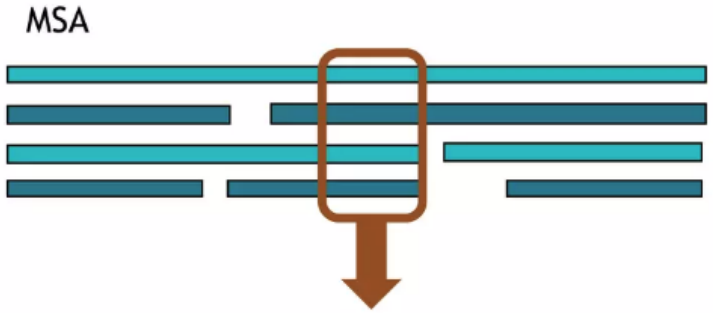
→ Sequence features



2) Profiles

MSA to PSSM:

→ Familes and Domains



PSSM

sequence positions

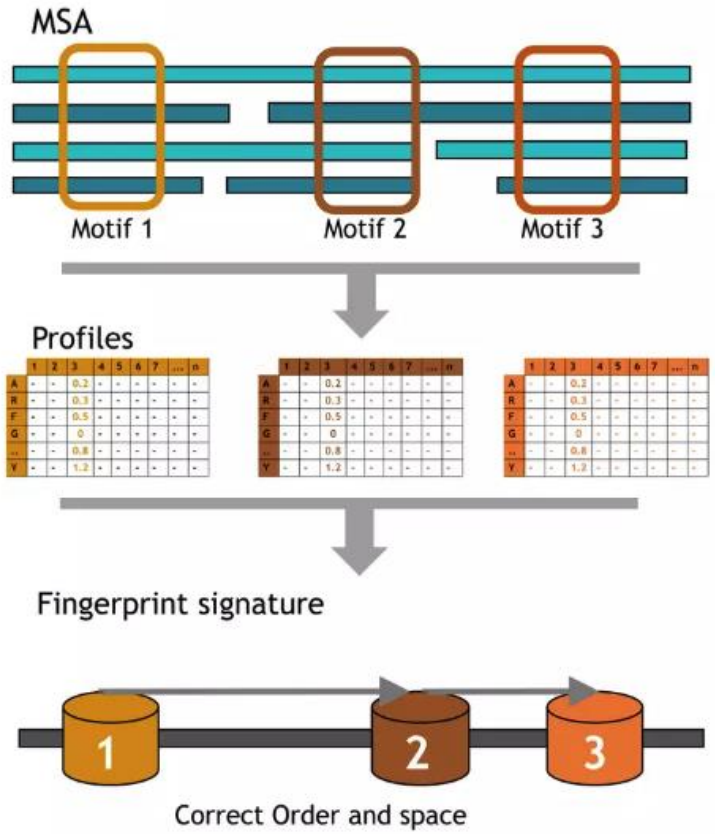
amino acids	1	2	3	4	5	6	7	..	n
A	-	-	0.2	-	-	-	-	-	-
R	-	-	0.3	-	-	-	-	-	-
F	-	-	0.5	-	-	-	-	-	-
G	-	-	0	-	-	-	-	-	-
..	-	-	0.8	-	-	-	-	-	-
Y	-	-	1.2	-	-	-	-	-	-

position-specific scoring matrix

Modeling MSA: PSSM

3) Fingerprints

MSA → Profiles → Fingerprint :
→ Families



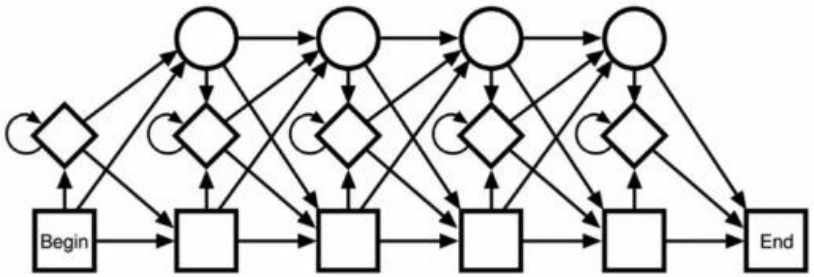
4) HMMs

MSA to PSSM to model:
→ Homologous sequences

"Probabilistics profiles"

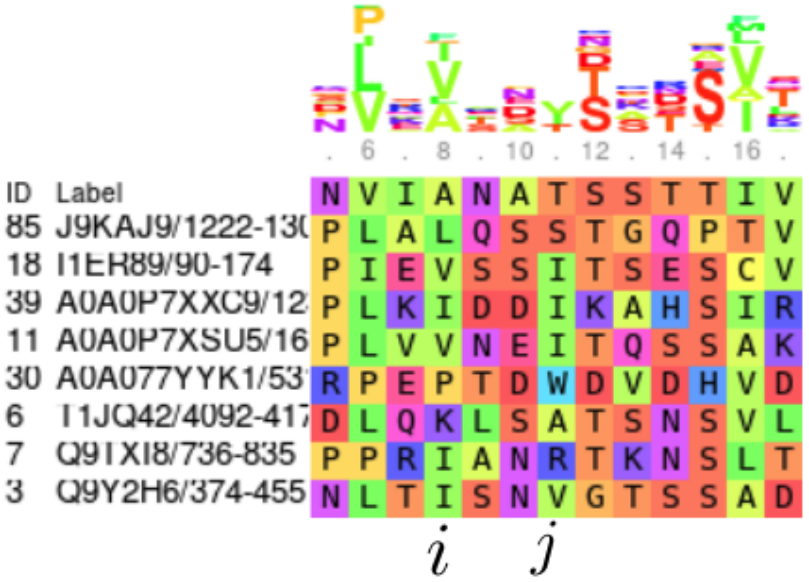
Convert multiple sequence alignments into position-specific scoring matrix (PSSMs).

	1	2	3	4	5	6	7	..	n
A	-	-	0.2	-	-	-	-	-	-
R	-	-	0.3	-	-	-	-	-	-
F	-	-	0.5	-	-	-	-	-	-
G	-	-	0	-	-	-	-	-	-
..	-	-	0.8	-	-	-	-	-	-
Y	-	-	1.2	-	-	-	-	-	-



How to measure correlations in MSA

PSSM assumes independence across position, and does not capture pairwise covariance



i, j Positions α, β Residue types

$f^{ij}_{\alpha\beta}$ Bivariate marginal (frequency)

f^{i}_{α} Univariate marginal (frequency)

(Example: $f^{4,7}_{\mathrm{V,I}} = 0.22$)

Correlations: $$C^{ij}_{\alpha\beta} = f^{ij}_{\alpha\beta} - f^{i}_{\alpha} f^{j}_{\beta}$$

Observed pairwise frequency

Expected pairwise frequency if positions vary independently

How to measure correlations in MSA

Mutual Information (MI)

$$MI^{ij} = \sum_{\alpha\beta} f^{ij}_{\alpha\beta} \log \frac{f^{ij}_{\alpha\beta}}{f^{i}_{\alpha} f^{j}_{\beta}}$$

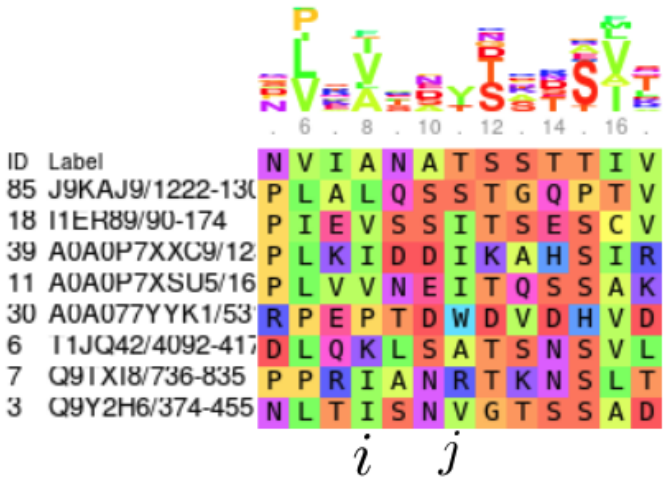
Can be interpreted as the "neg. log likelihood" of generating the distribution $f^{ij}_{\alpha\beta}$ when sampling from the distribution $f^{i}_{\alpha} f^{j}_{\beta}$

Statistical Coupling (SCA)

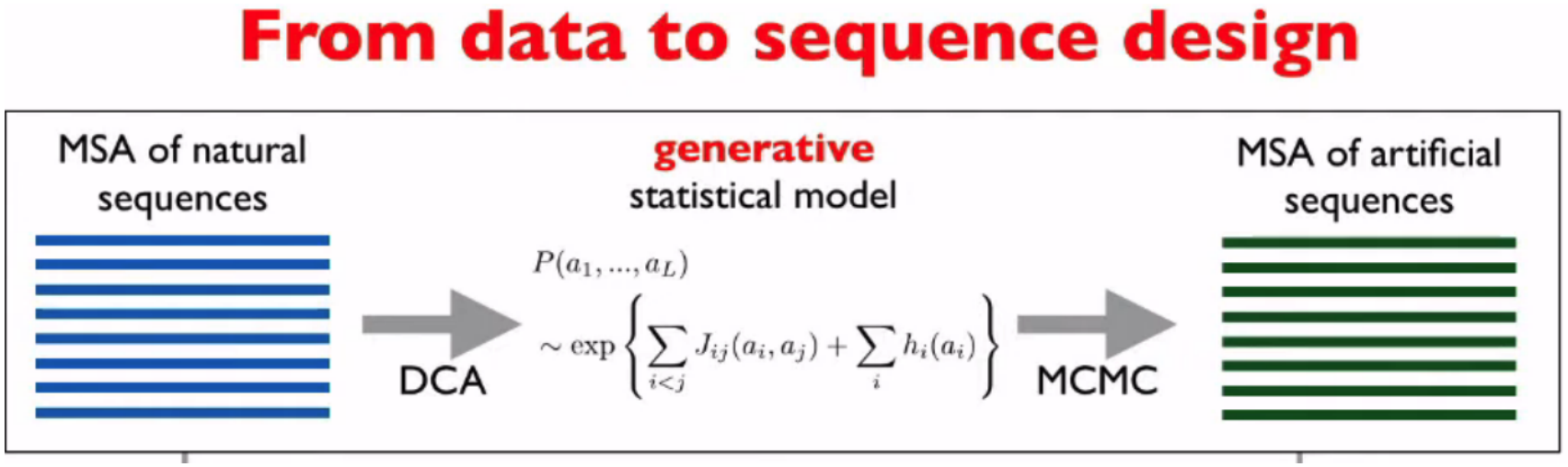
$$\Delta\Delta G^{ij} = kT \sqrt{\sum_{\alpha} \left(\log \frac{P^{i|\delta j}_{\alpha}}{\bar{P}^{i|\delta j}_{\alpha}} - \log \frac{P^{i}_{\alpha}}{\bar{P}^{i}_{\alpha}} \right)^2}$$

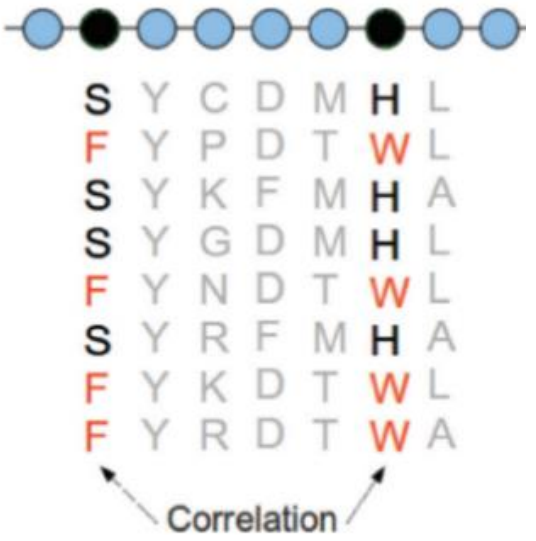
Probability of α at i excluding sequences with mutation at j

Bar means average over all positions

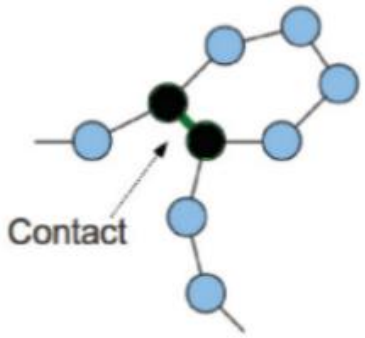


Modeling MSA: Potts model





Can we build a graphical model that captures sequence covariance?



Modeling MSA: Potts model

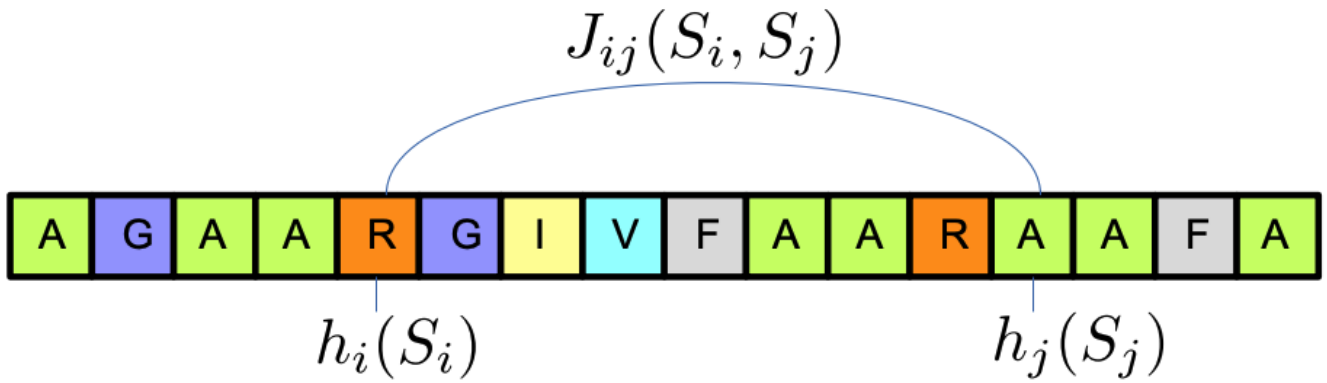
$$P(s) = \frac{1}{Z} \exp - \left(\sum_{i=1}^{N} h_{s_i}^{i} + \sum_{1 \le i < j \le N} J_{s_i s_j}^{ij} \right)$$

"Fields" (L x q) "Couplings" ((L x q)^2)

L = sequence length (eg 200)

q = # of residue types (eg 20)

Note similarity of fields to PSSMs



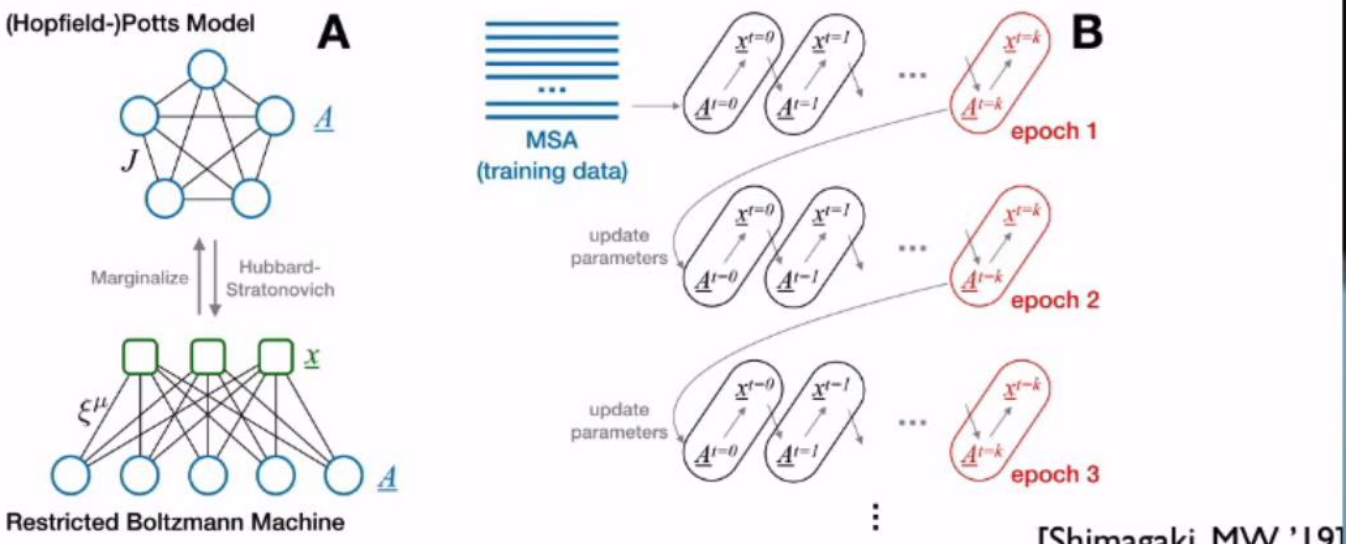
Potts parameters interpretated as energy contributions from each position/pair

Fitting graphical model parameters:
- Belief Propagation, Susceptibility Propagation
- Mean Field inference
- Pseudolikelihood Methods + Conjugate Gradient Descent
- Cluster Expansion
- Monte Carlo + Quasi-Newton Optimization

Potts Model via Restricted Boltzmann Machine (Energy based)

$$P(a_1, ..., a_L) = \frac{1}{Z} \exp\left\{ \sum_{\mu=1}^{p} \sum_{i,j=1}^{L} \xi_i^\mu(a_i)\xi_j^\mu(a_j) + \sum_{i=1}^{L} h_i(a_i) \right\}$$

$$= \frac{1}{\tilde{Z}} \int \prod_\mu dx^\mu \exp\left\{ -\sum_{\mu=1}^{p} \frac{(x^\mu)^2}{2} + \sum_{\mu=1}^{p} \sum_{i=1}^{L} x^\mu \xi_i^\mu(a_i) + \sum_{i=1}^{L} h_i(a_i) \right\}$$

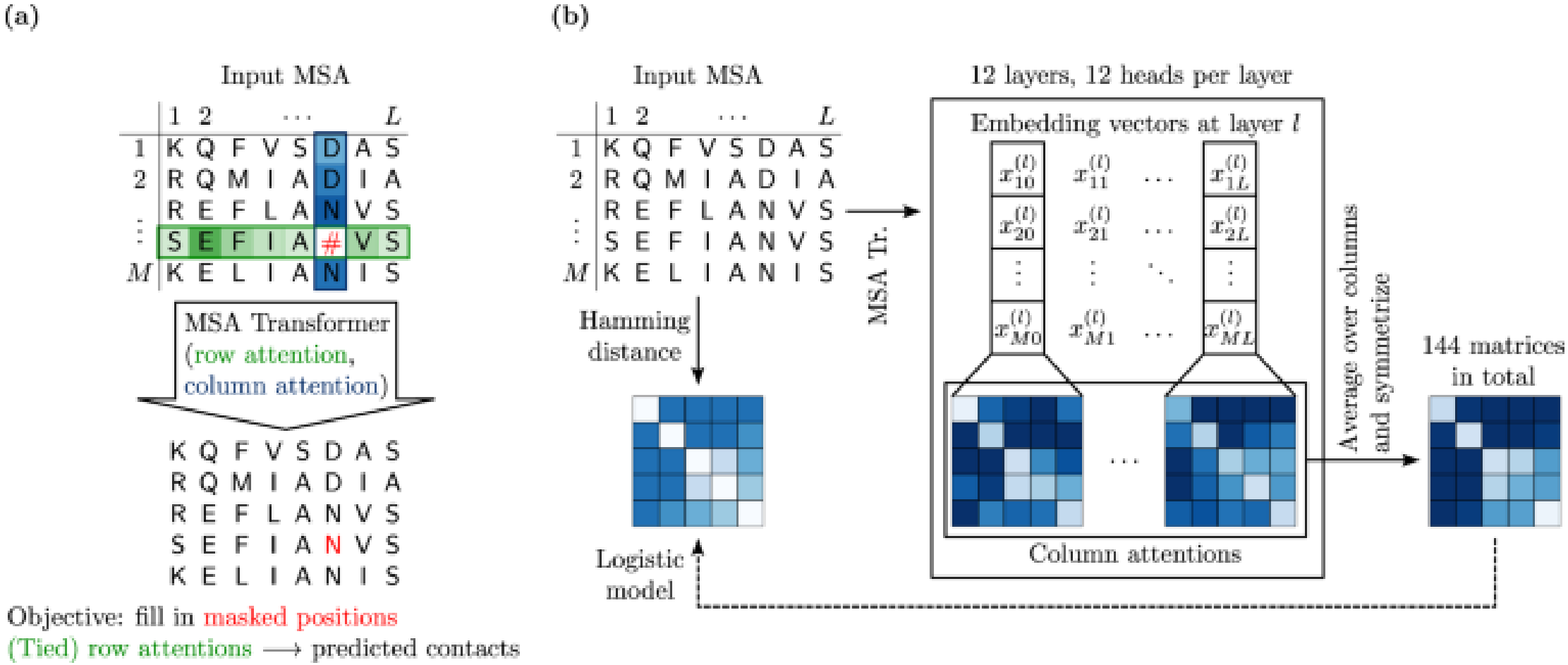


[Shimagaki, MW '19]

cf. [Tubiana, Cocco, Monasson '19]

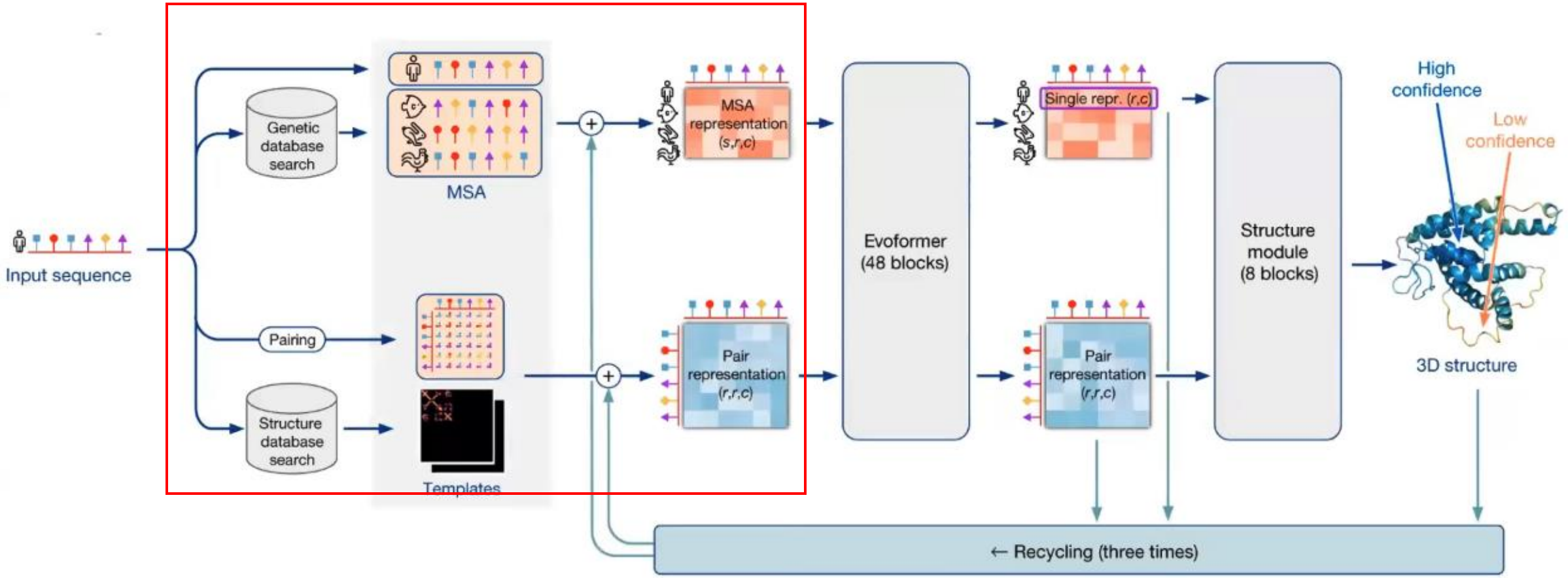
Deep learning with MSA

- MSA Transformer



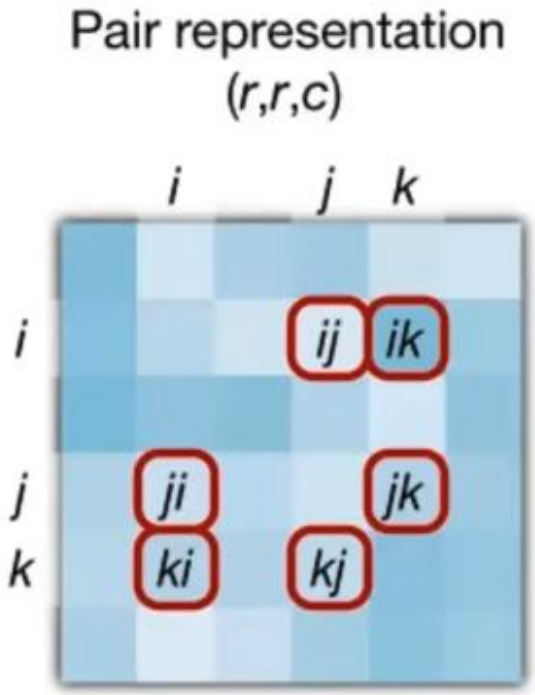
Deep learning with MSA

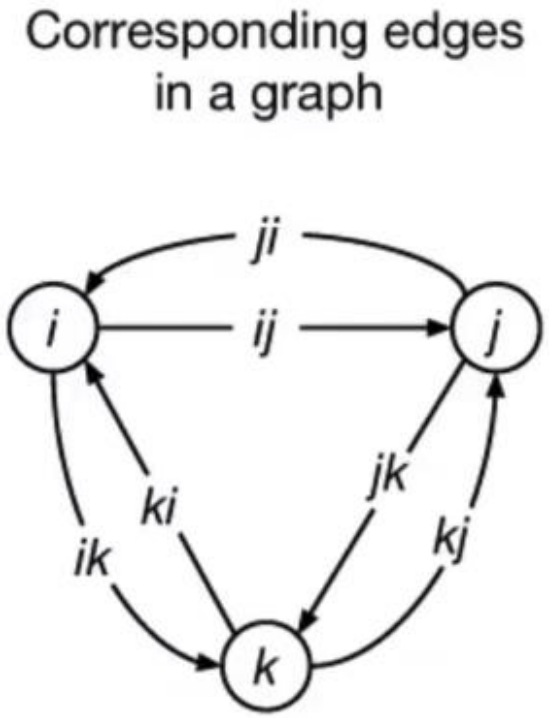
- Alphafold Evoformer: extracting co-evolution information for structure prediction

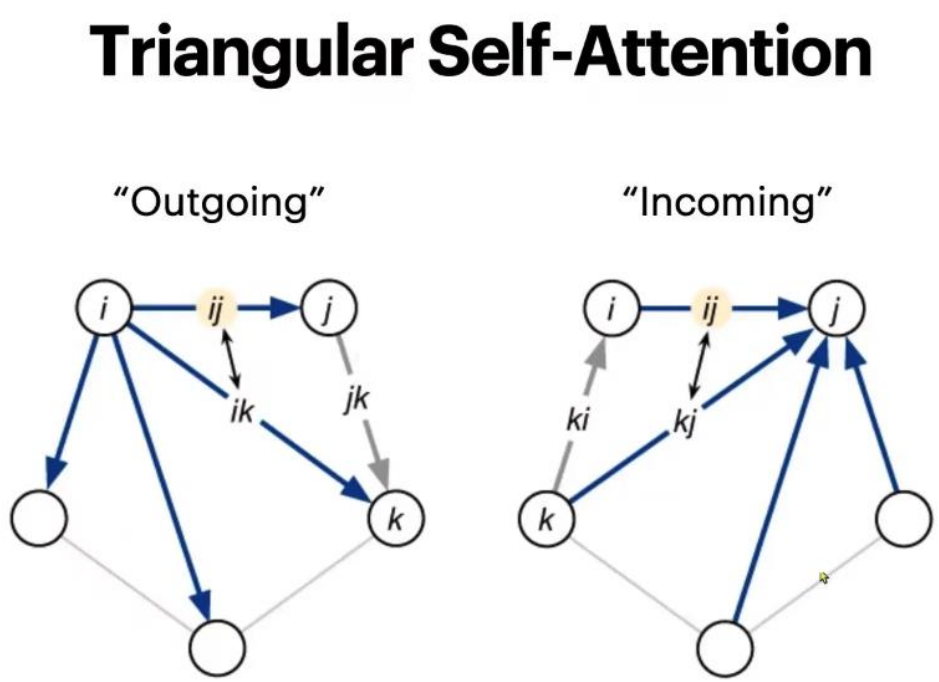


Jumper et al., 2021

Pair representation in AF2







Protein Language Models

- Instead of MSA, how about directly learning evolution from entire protein sequence database with language model?

Autoregressive (predict next token):

Natural language:	`To be, or not to be, that is the _____`
Protein sequence:	`GSGHMAKKYSIGLAIGTNSVGL_`

Masked (predict masked out tokens):

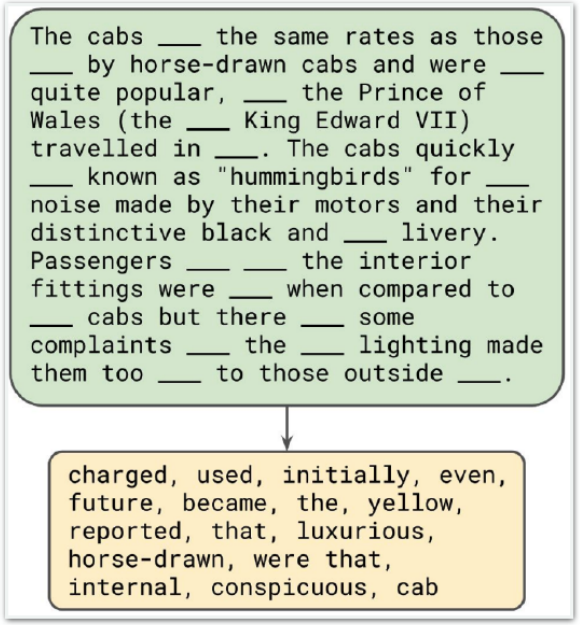
Natural language:	`To _, or not to _, that is the _`
Protein sequence:	`GSGHM_KKYSIGVG_AVITD_PSK`

Language models



BERT (Encoder-only)

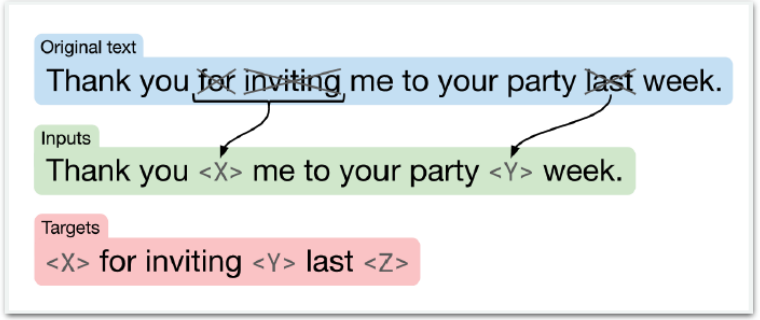
Devlin et al., 2018



Masked token prediction

T5 (Encoder-decoder)

Raffel et al., 2019

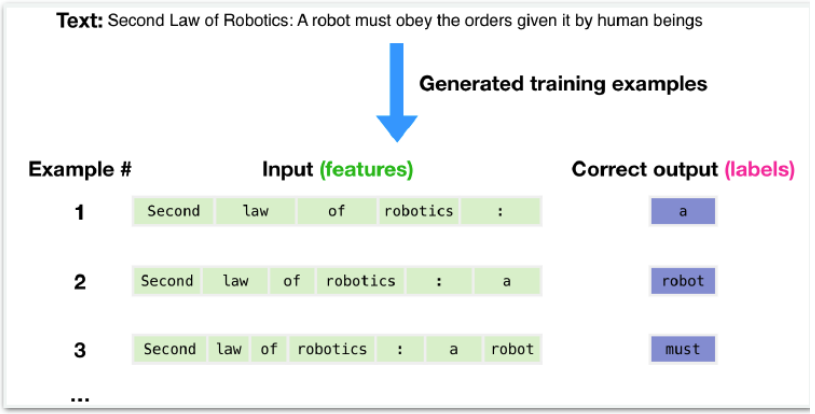


Denoising span-mask prediction

OpenAI

GPT - 4

Decoder-only

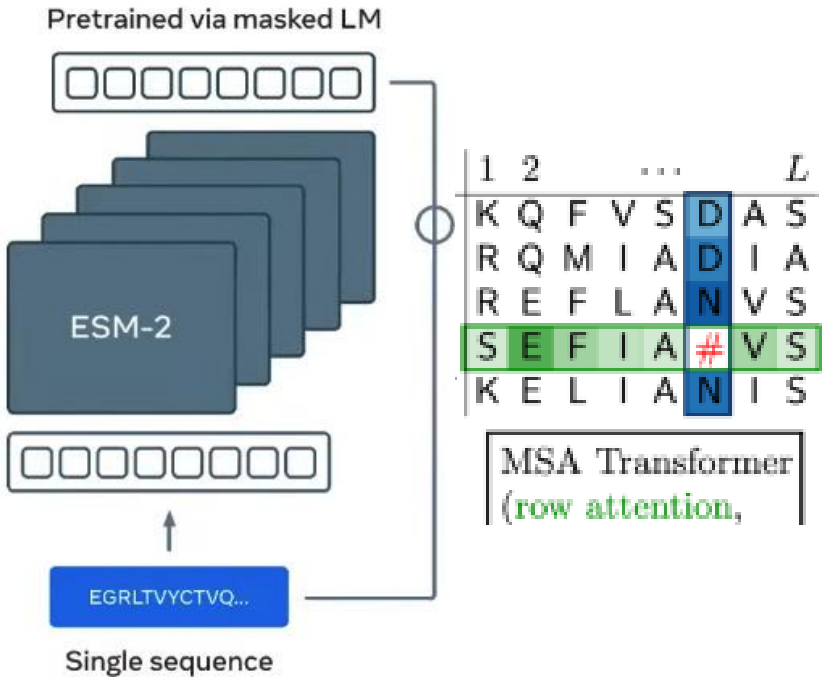


Next token prediction

Protein Language Models (pLM)

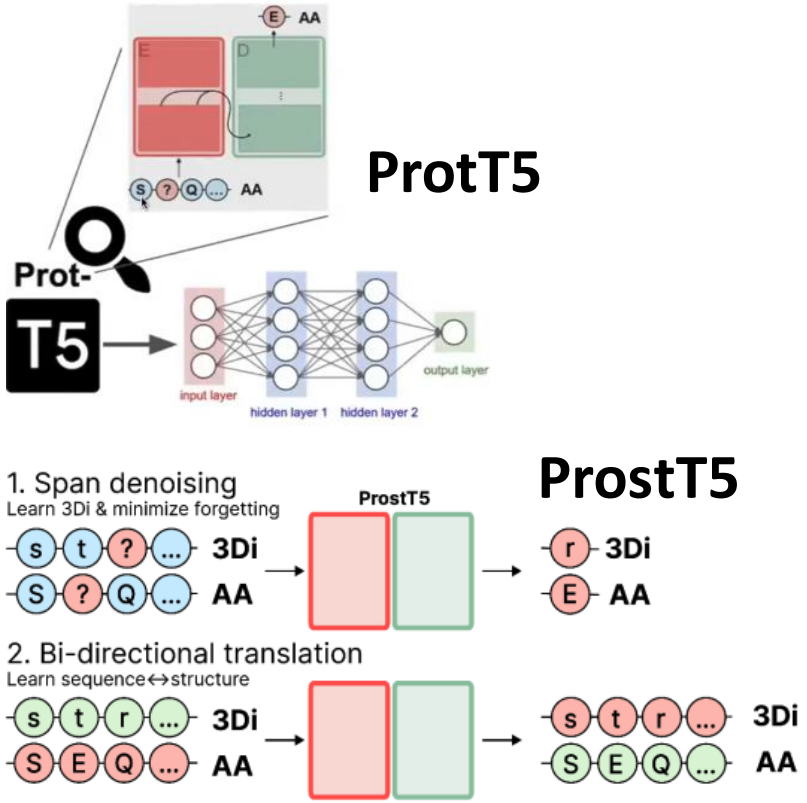


BERT (Encoder-only)



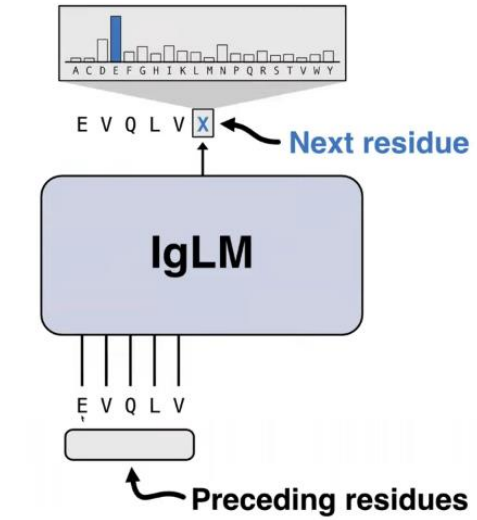
Masked token prediction

T5 (Encoder-decoder)



Denoising span-mask prediction

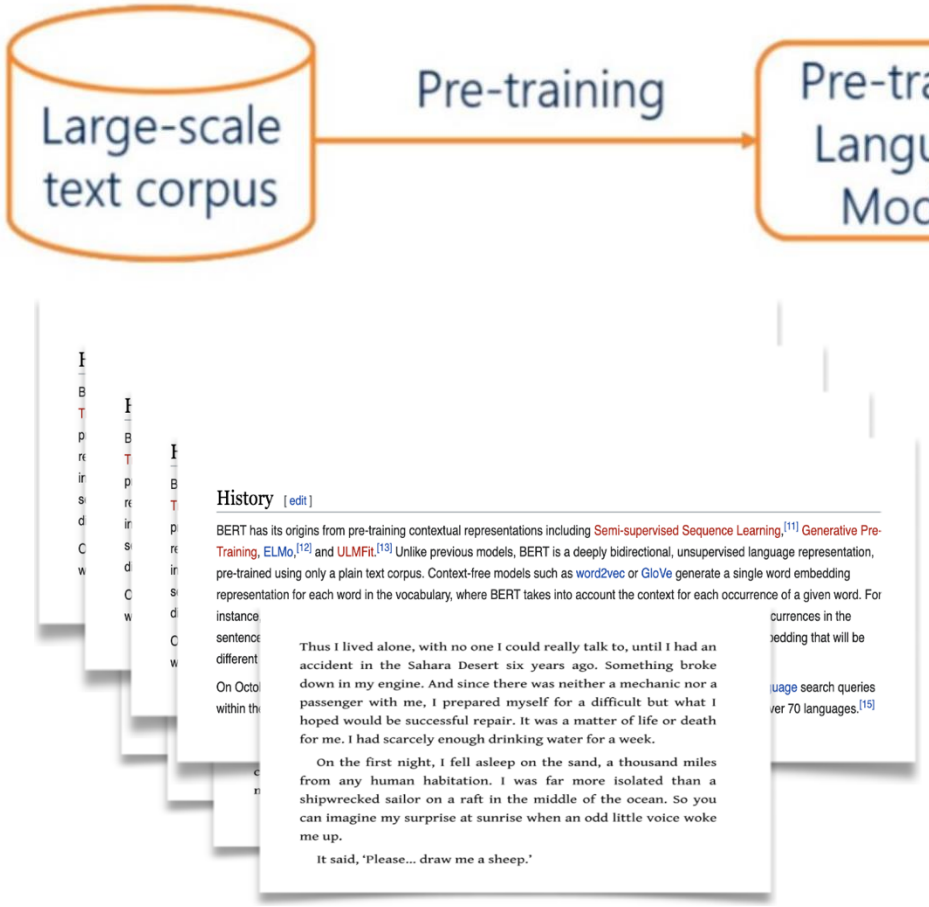
Decoder-only



Progen

Next token prediction

Modern learning paradigm: transfer learning

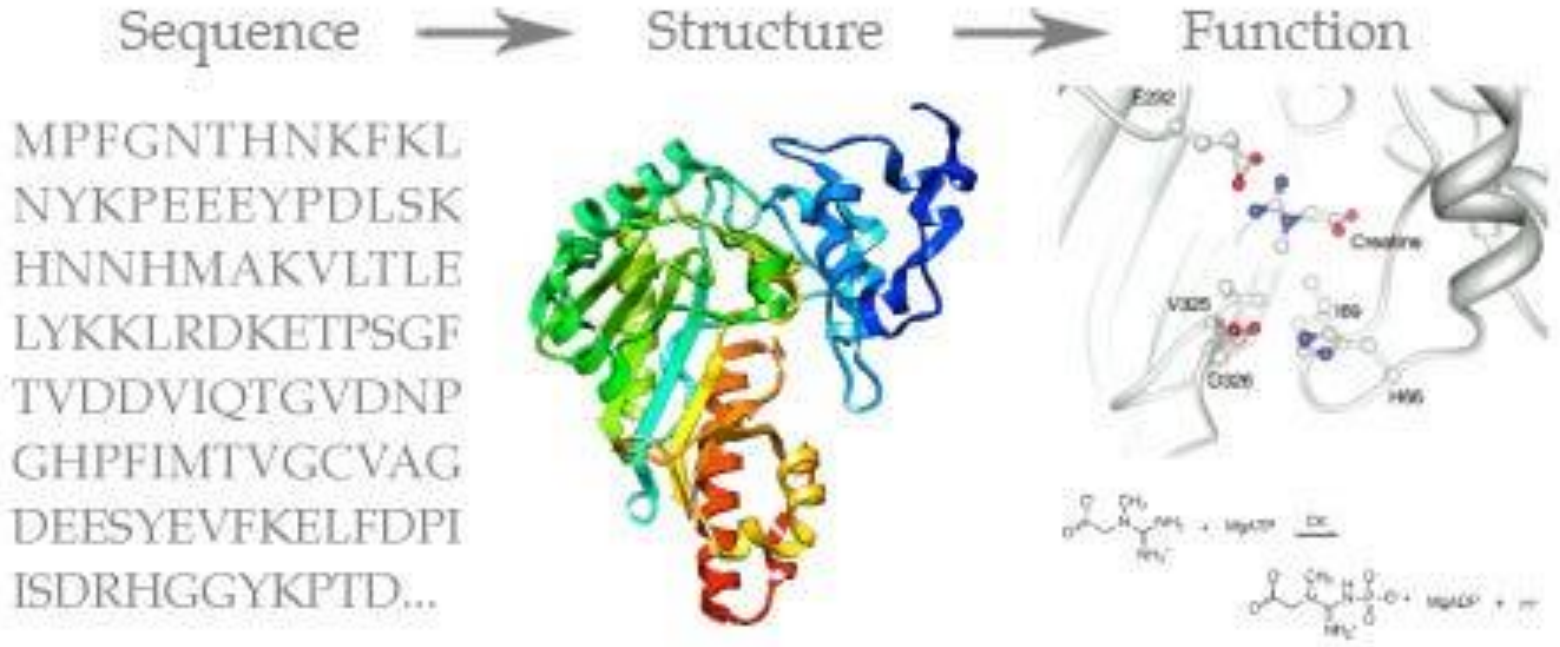


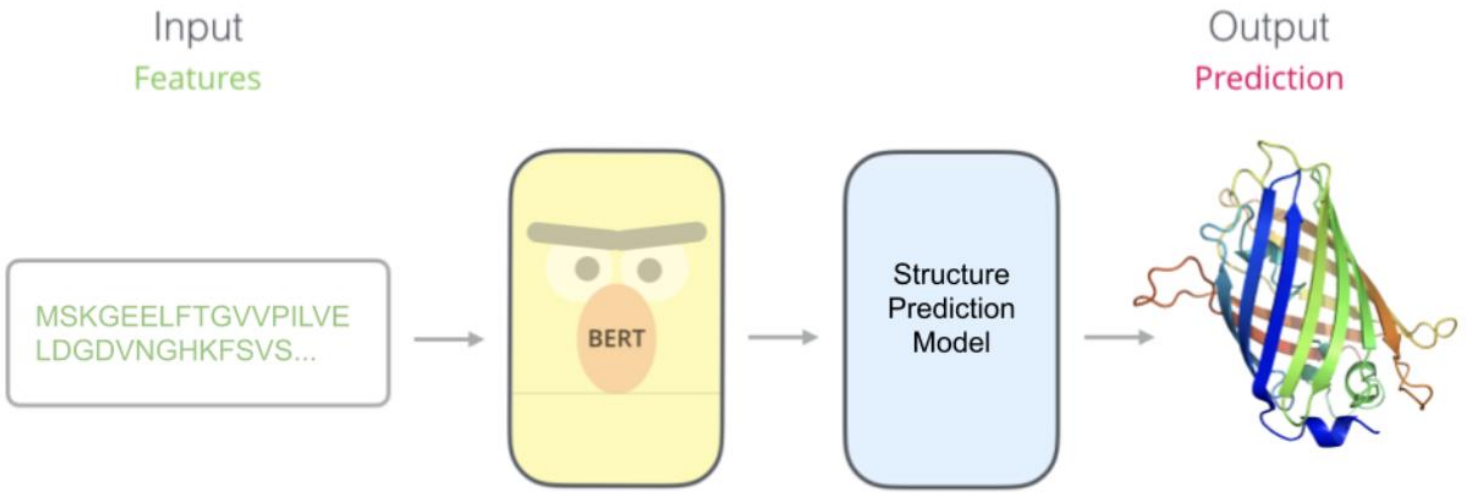
Sentiment prediction

NER

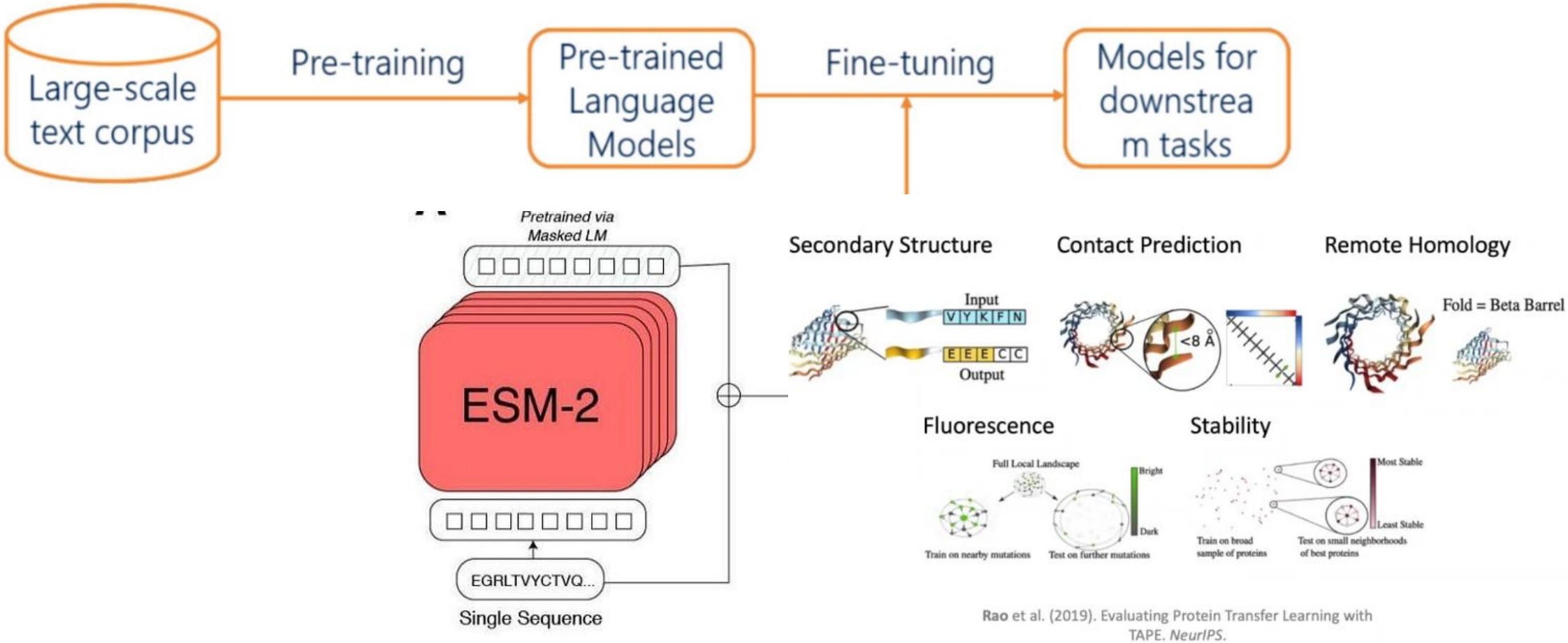
……

Central dogma





Modern learning paradigm: transfer learning

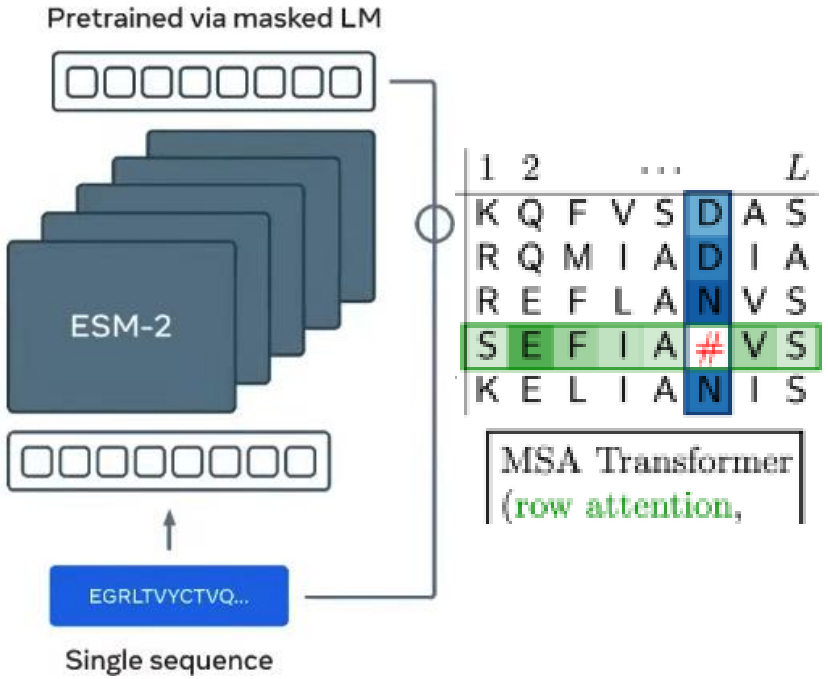


Rao et al. (2019). Evaluating Protein Transfer Learning with TAPE. *NeurIPS*.

Protein Language Models (pLM)



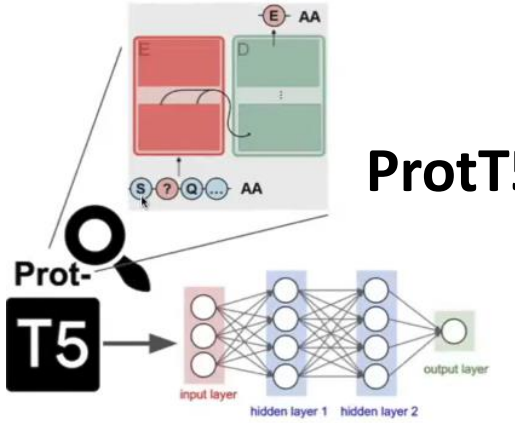
BERT (Encoder-only)

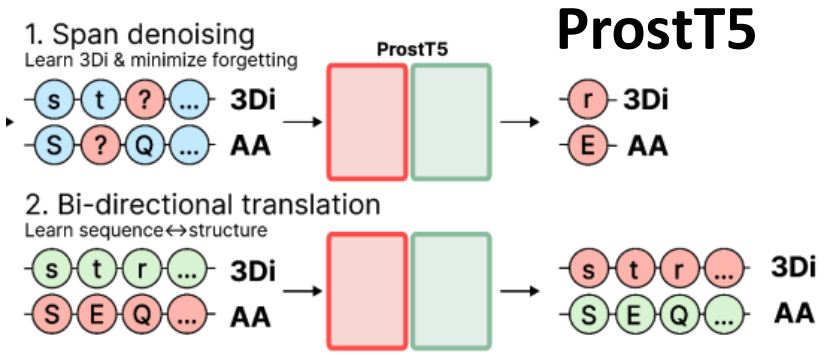


Masked token prediction

T5

T5 (Encoder-decoder)



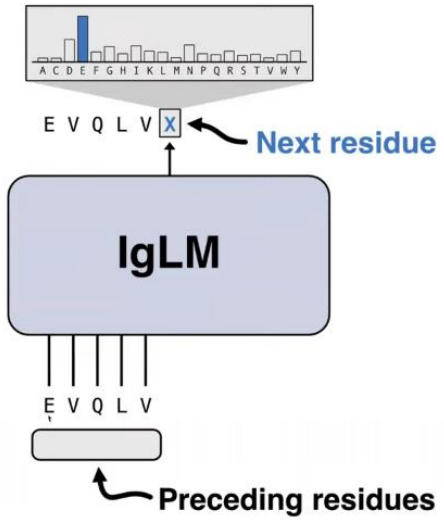


Denoising span-mask prediction

OpenAI

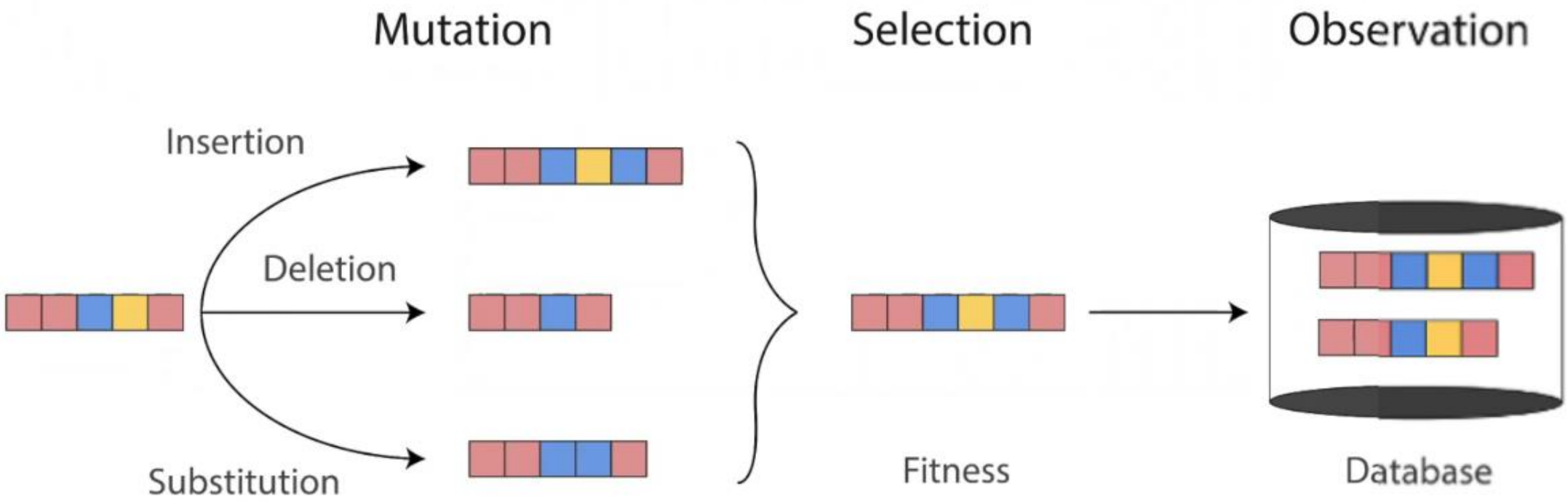
GPT - 4

Decoder-only

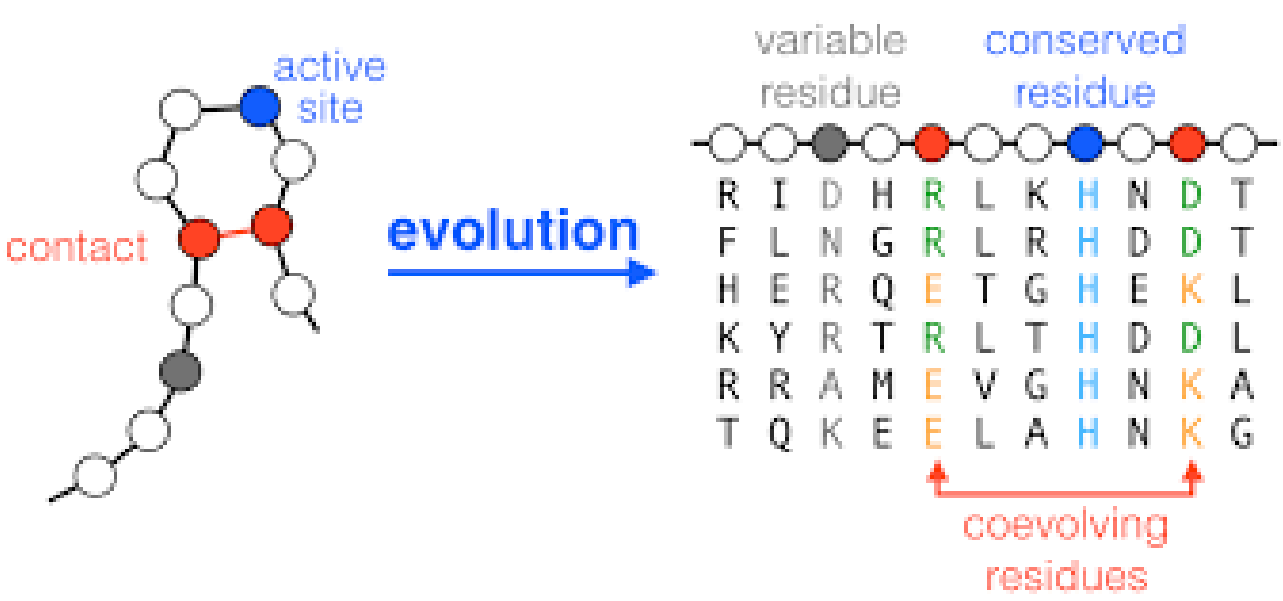


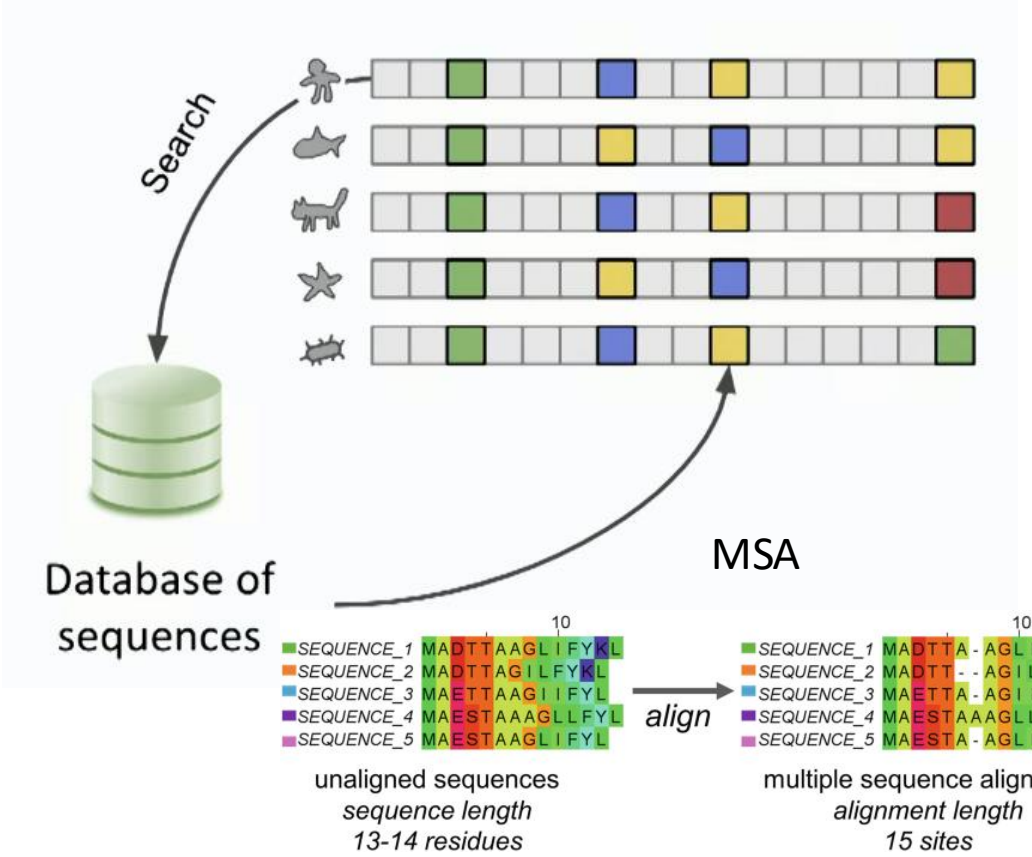
Next token prediction

Evolution of protein sequences



Modeling protein sequence: co-evolution is important





By measuring **co-evolution**, we can infer contacts in proteins!

Independent Sites (PSSM) Model

Sequence Logo



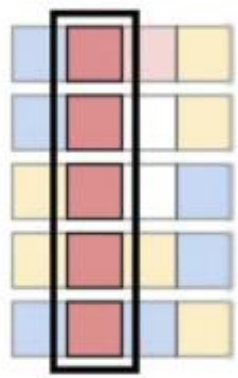
Likelihood

$$\mathcal{L}(x;\theta) = \prod_{i=1}^{L} p(x_i \mid \theta_i)$$

Graph Representation (MRF)

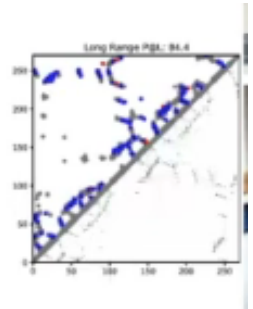


Multiple Sequence Alignment



Pairwise Sites (Potts) Model

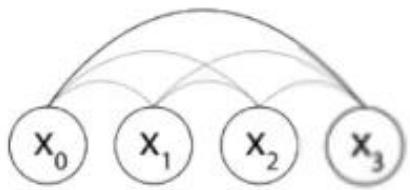
Contact Map



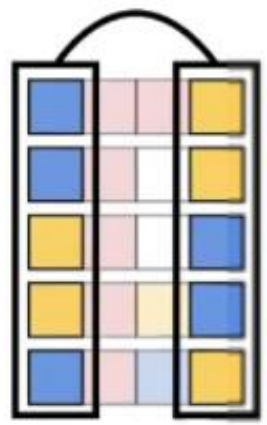
pseudo-Likelihood

$$\mathcal{L}_{\mathrm{PLL}}(x;\theta) = \prod_{i=1}^{L} p(x_i \mid x_{j\neq i}, \theta_i, \theta_{ij})$$

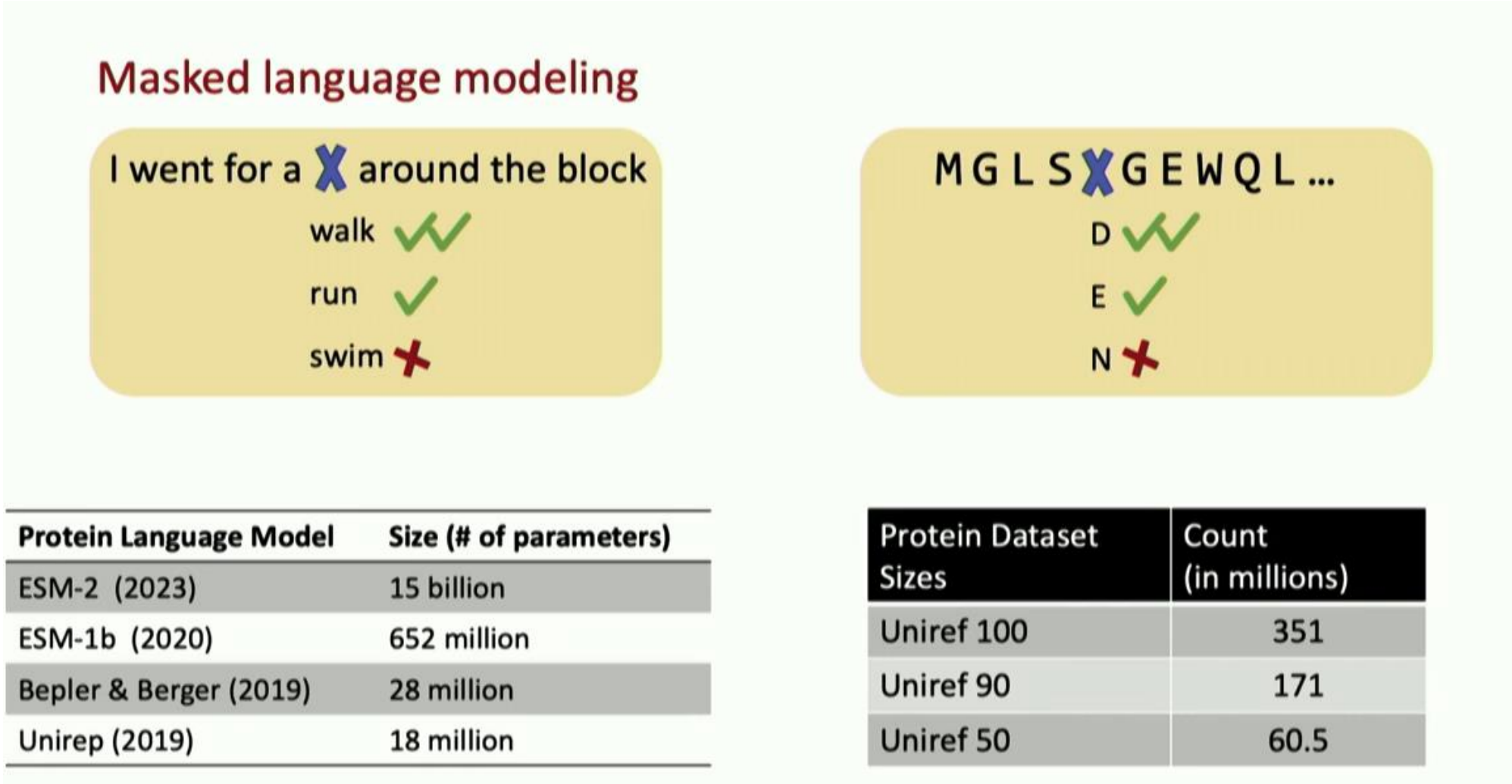
Graph Representation (MRF)



Multiple Sequence Alignment



Applying BERT MLM pretraining to protein



Protein Language Model	Size (# of parameters)
ESM-2 (2023)	15 billion
ESM-1b (2020)	652 million
Bepler & Berger (2019)	28 million
Unirep (2019)	18 million

Protein Dataset Sizes	Count (in millions)
Uniref 100	351
Uniref 90	171
Uniref 50	60.5

Earlier effort on protein language model prior to 2019



Yang et al. (2018) use a doc2vec style model for protein engineering



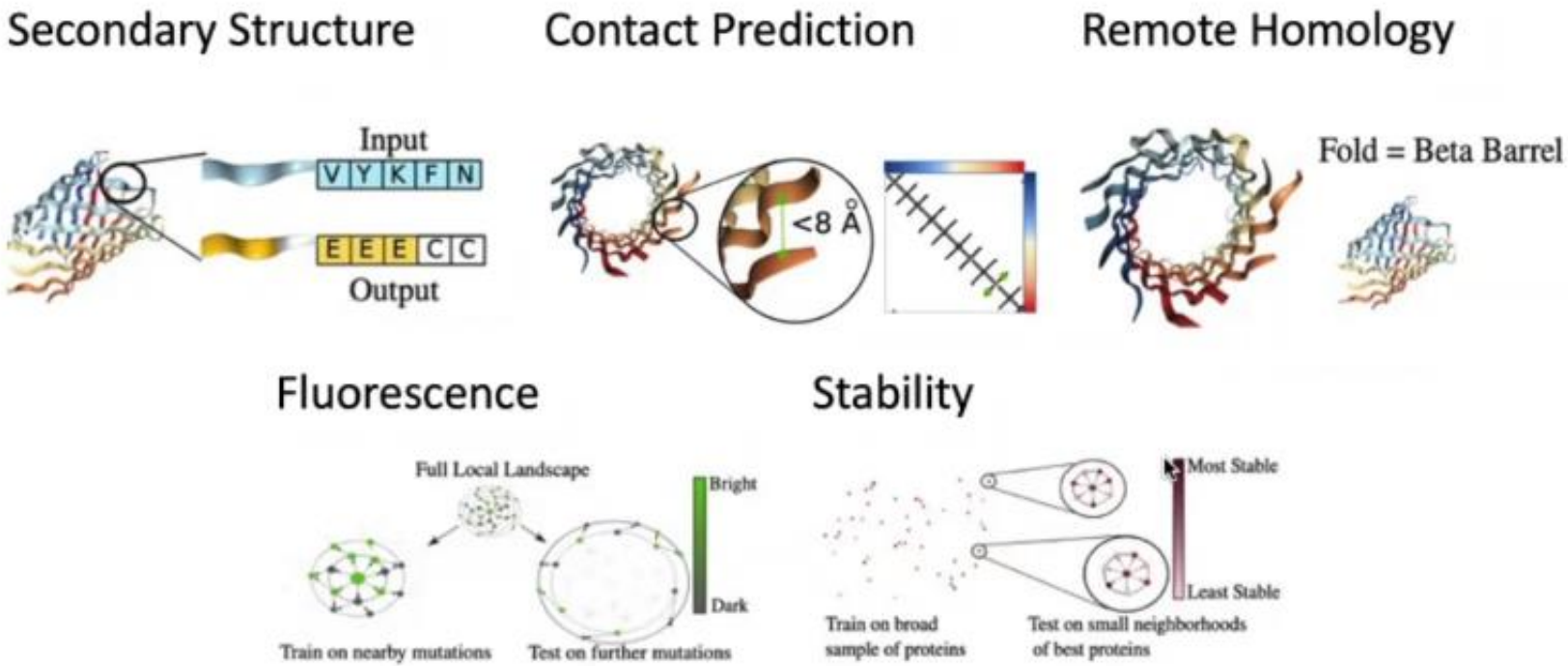
Bepler & Berger (2017) train a 5 layer LSTM on proteins, evaluated on both structural and protein engineering tasks



Alley et al. (2018) use an unsupervised unidirectional mLSTM for protein engineering

Tasks Assessing Protein Embeddings (TAPE)

- ML Approach: Use Benchmark Tasks
- Different areas of protein biology



Rao et al. (2019). Evaluating Protein Transfer Learning with TAPE. *NeurIPS*.

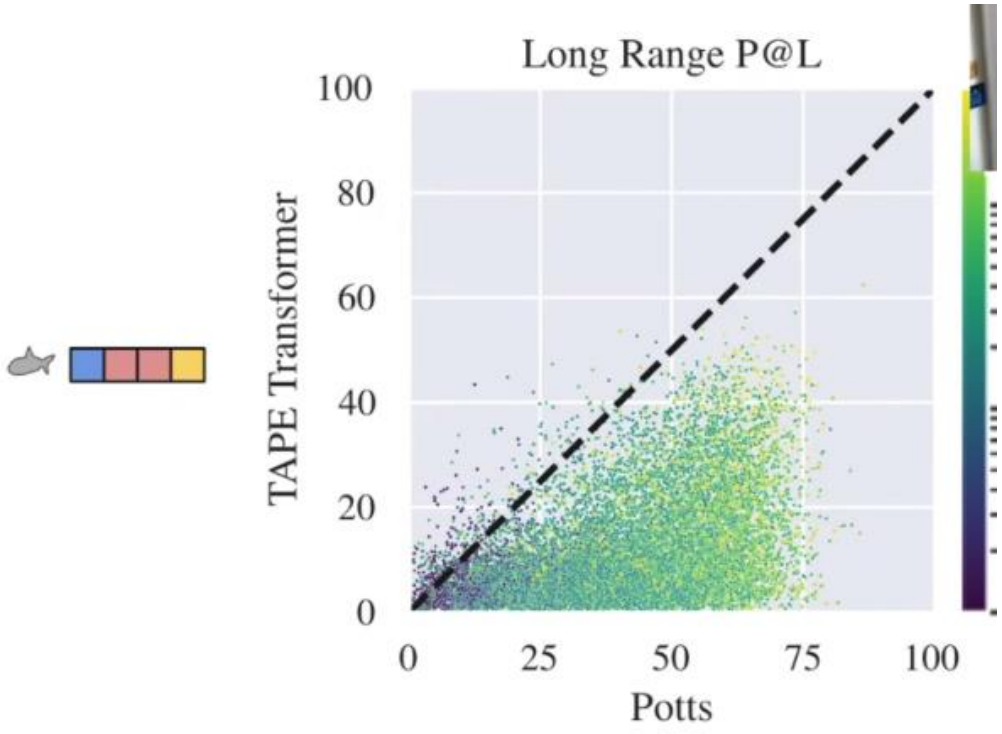
TAPE Benchmark Results

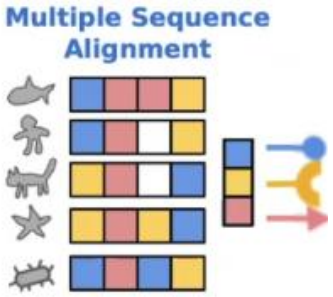
Table 2: Results on downstream supervised tasks

Method		Structure		Evolutionary	Engineering	
		SS	Contact	Homology	Fluorescence	Stability
No Pretrain	Transformer	0.70	0.32	0.09	0.22	-0.06
	LSTM	0.71	0.19	0.12	0.21	0.28
	ResNet	0.70	0.20	0.10	-0.28	0.61
Pretrain	Transformer	0.73	0.36	0.21	**0.68**	**0.73**
	LSTM	0.75	0.39	**0.26**	0.67	0.69
	ResNet	0.75	0.29	0.17	0.21	**0.73**
Supervised [11]	LSTM	0.73	0.40	0.17	0.33	0.64
UniRep [12]	mLSTM	0.73	0.34	0.23	0.67	**0.73**
Baseline	One-hot	0.69	0.29	0.09	0.14	0.19
	Alignment	**0.80**	**0.64**	0.09	N/A	N/A

Rao et al. (2019). Evaluating Protein Transfer Learning with TAPE. *NeurIPS*.

TAPE vs Potts model





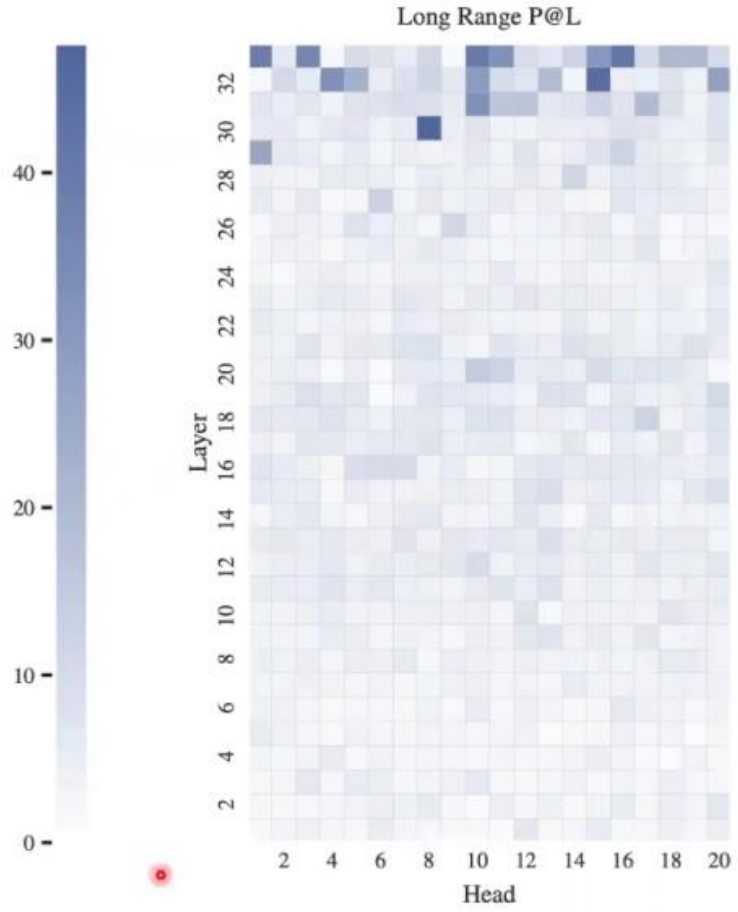
(2021). Transformer protein language models are unsupervised structure leaners. *ICLR*.

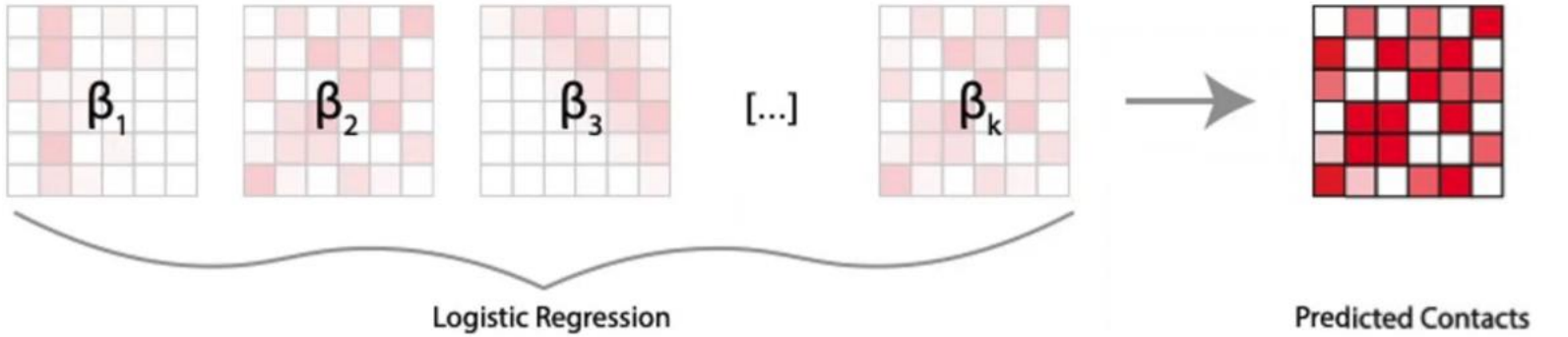
TAPE is able to predict structure in some degree but still underperform Potts model

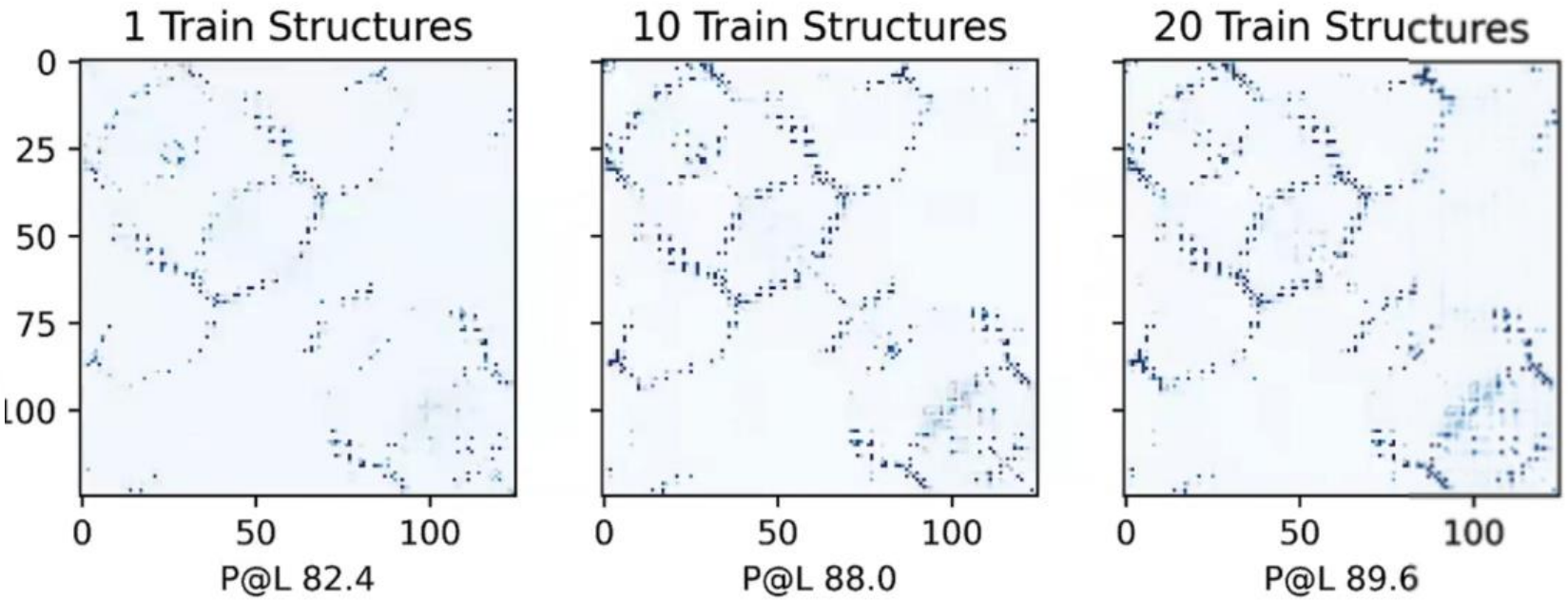
What about larger models?

- TAPE Transformer
  - 40M parameters
  - Trained on protein domains
- ESM-1b
  - Rives et al. 2020
  - 650M parameters
  - Trained on full proteins

ESM1 attention heads learn to predict contacts

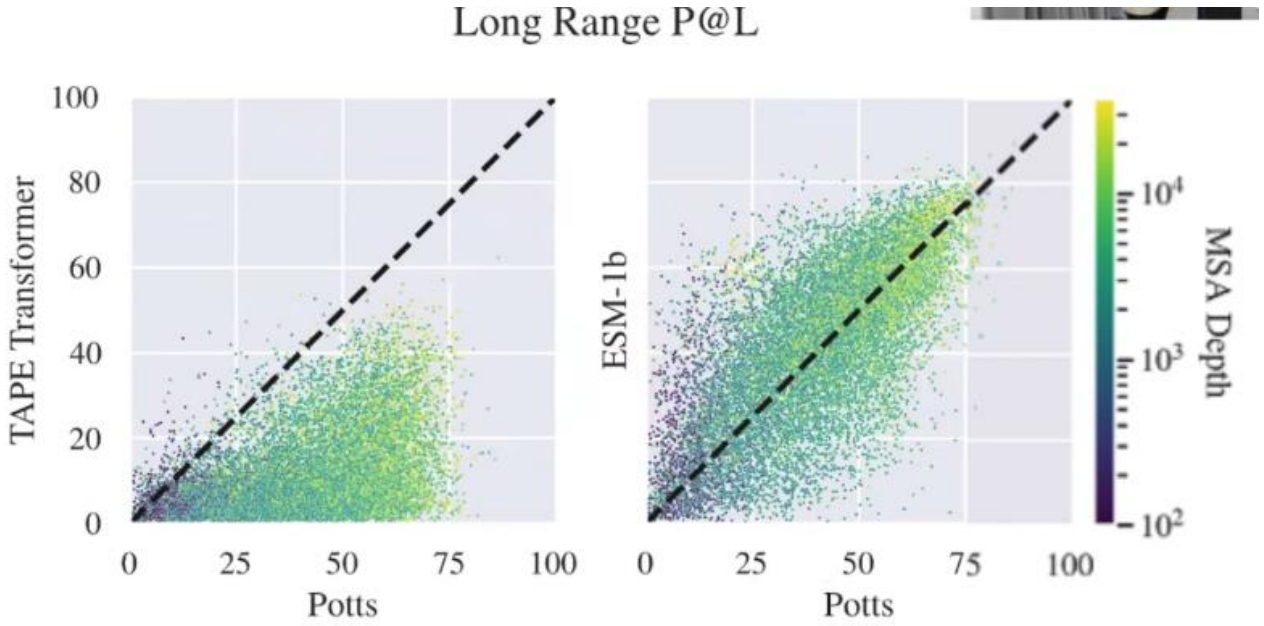


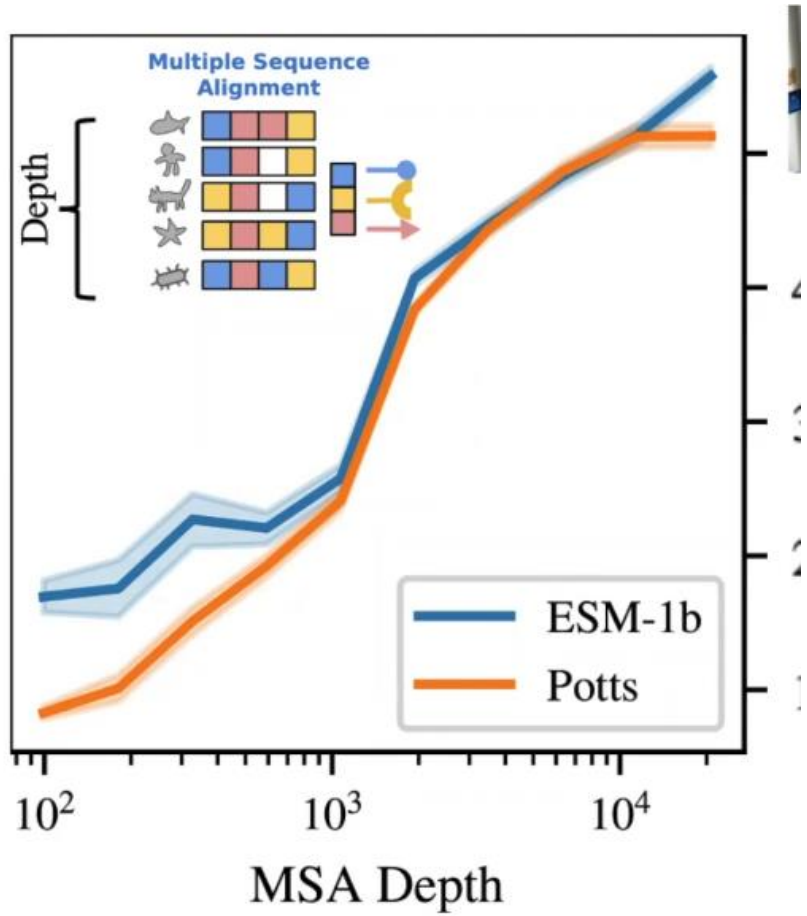




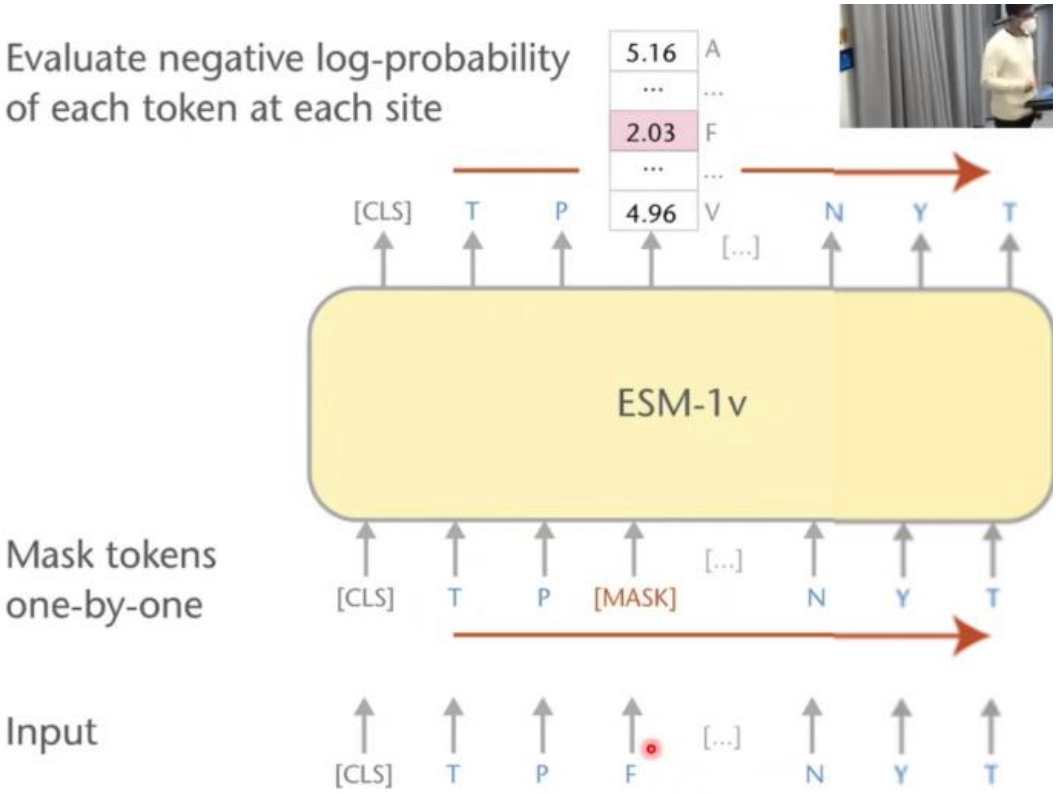
ESM1 outperform Potts model

Model	Citation	P@L
Potts	Balarkishan 2011	39.3
TAPE Transformer	Rao 2019	11.2
ESM-1b	Rives 2020	**41.1**





Pseudo likelihood in MLM

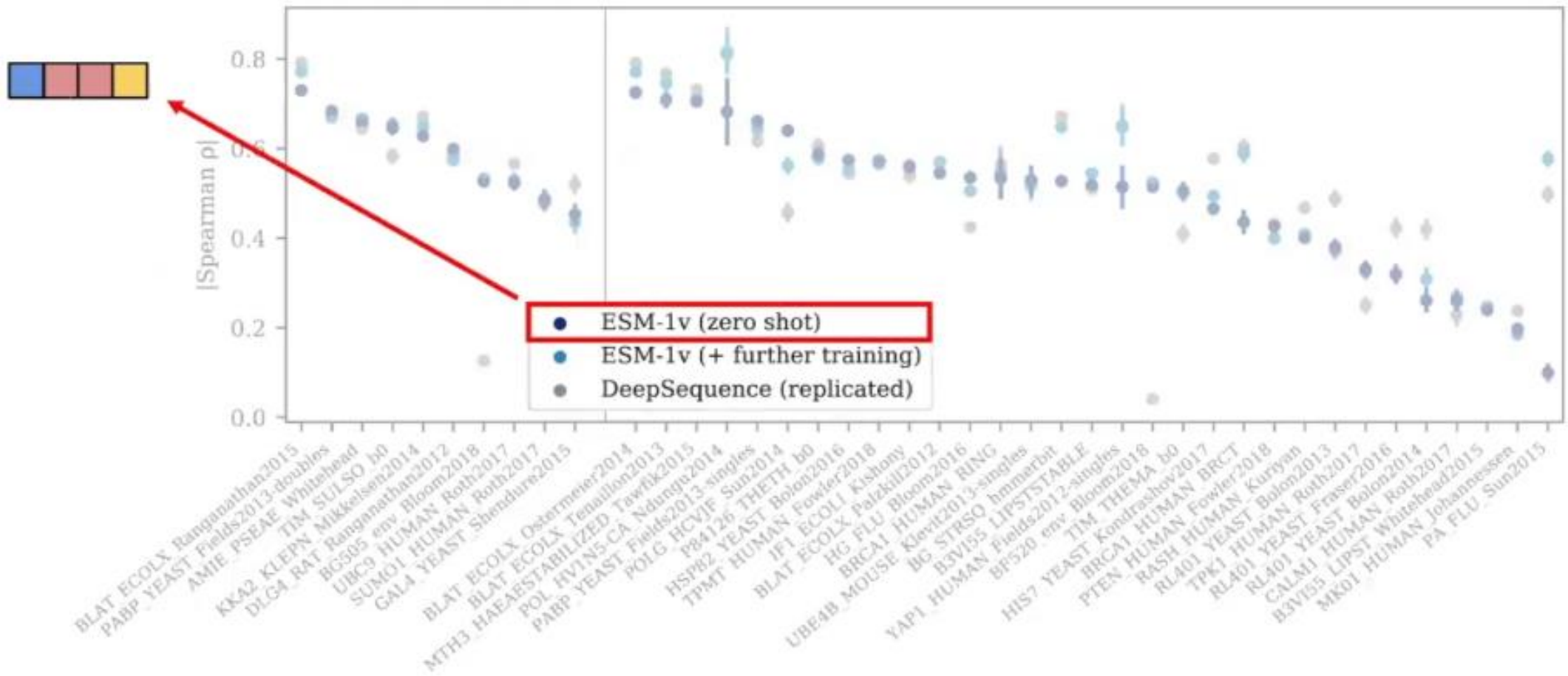


$$\mathcal{L}_{\text{Indep}}(x;\theta) = \prod_{i=1}^{L} p(x_i \mid \theta)$$

$$\mathcal{L}_{\text{PLL}}(x;\theta) = \prod_{i=1}^{L} p(x_i \mid x_{j\neq i}, \theta)$$

$$\mathcal{L}_{\text{MLM}}(x;\theta) = \mathop{\mathbb{E}}_{\text{mask}} \prod_{i\in\text{mask}} p(x_i \mid x_{j\notin\text{mask}}; \theta)$$

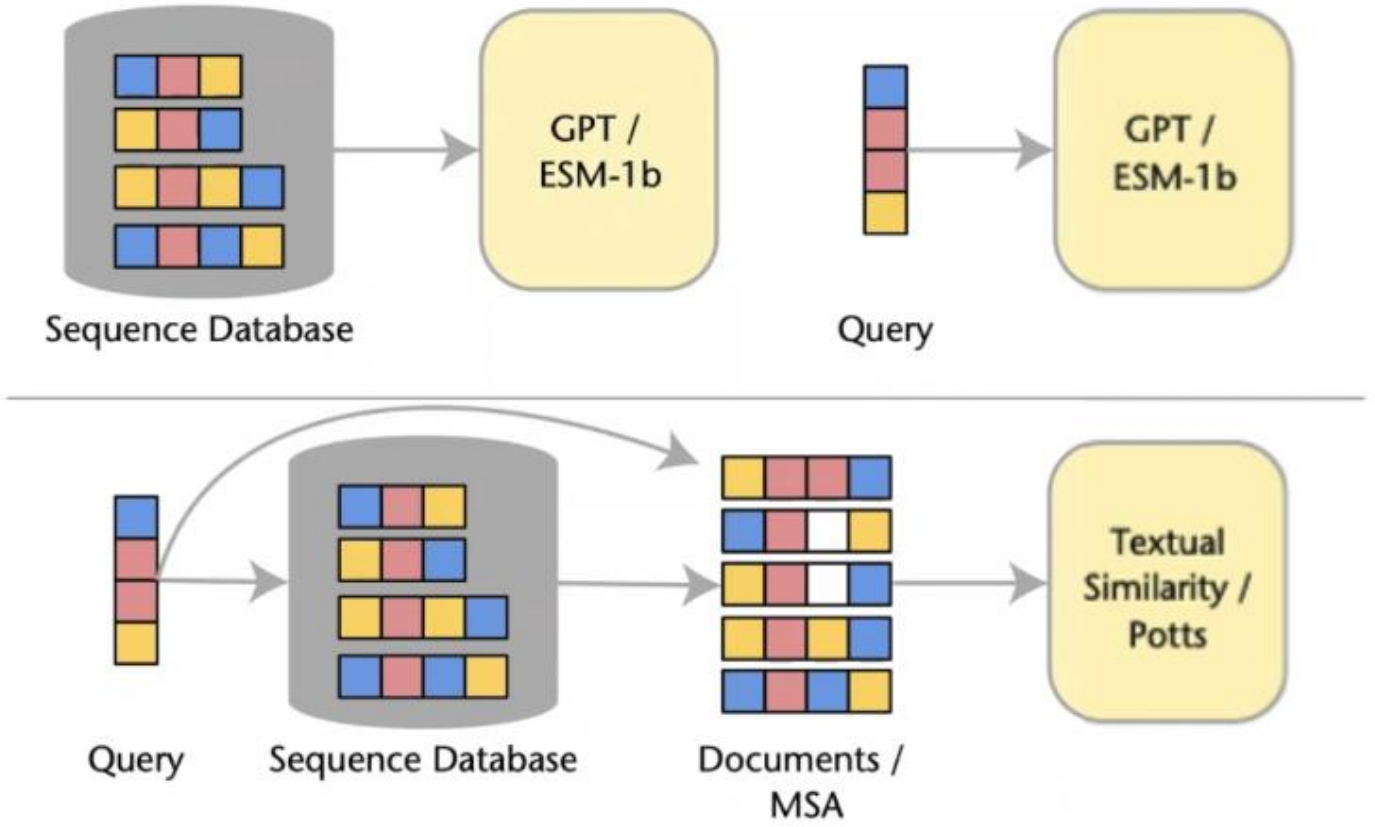
ESM-1v: Variant effect prediction



Meier et al. (2021). Language models enable zero-shot prediction of the effects of mutations on protein function. *NeurIPS*.

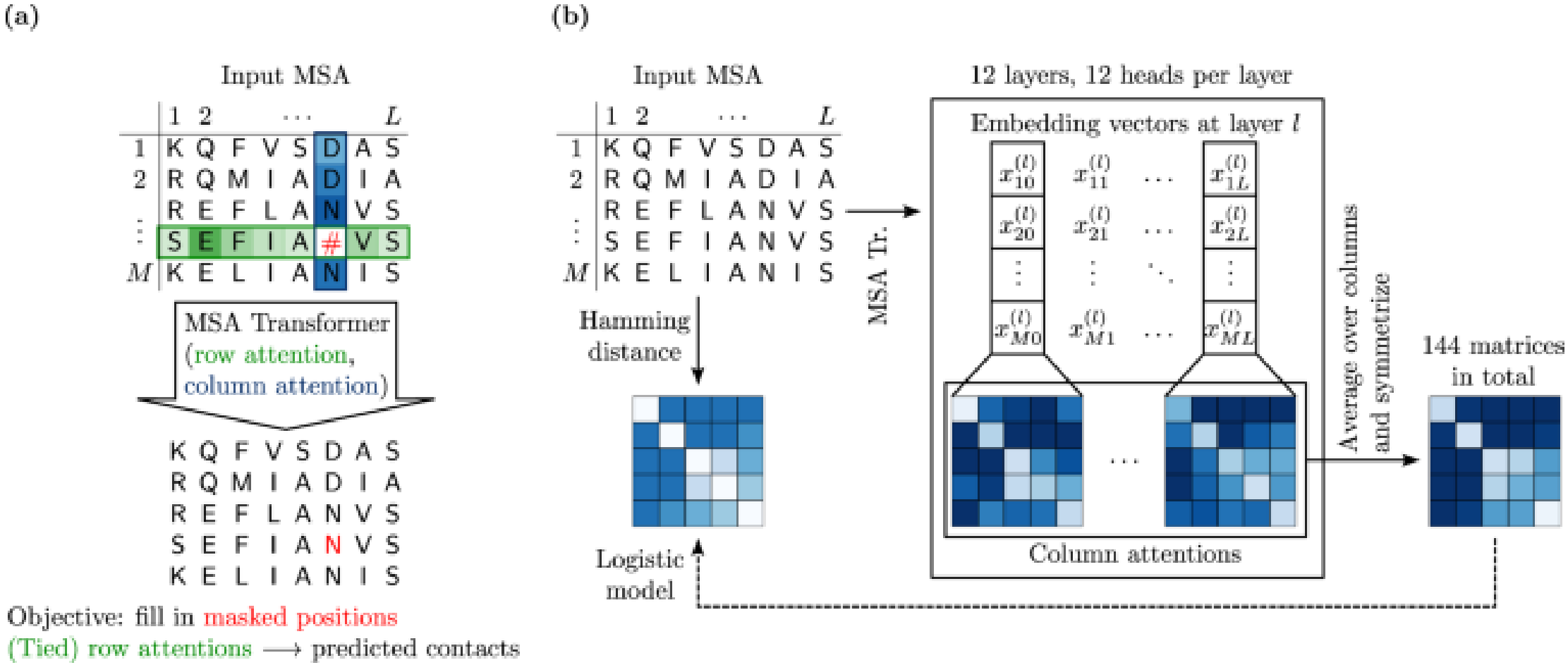
Modeling MSA instead of single sequence

Analogy:
Zero-shot
question
answering



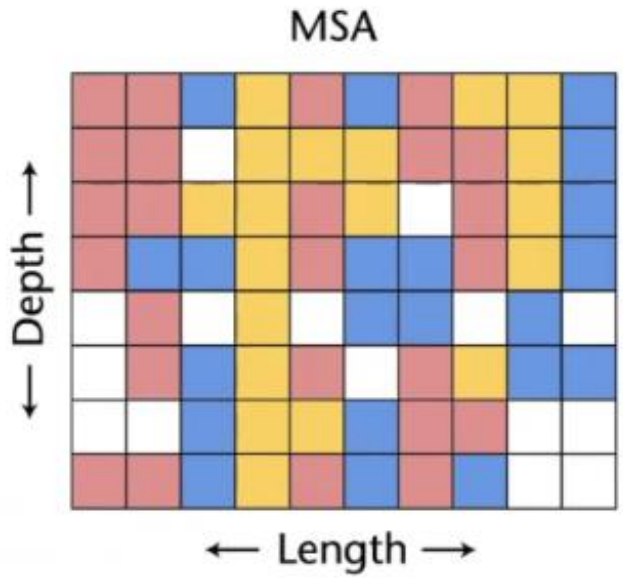
Deep learning with MSA

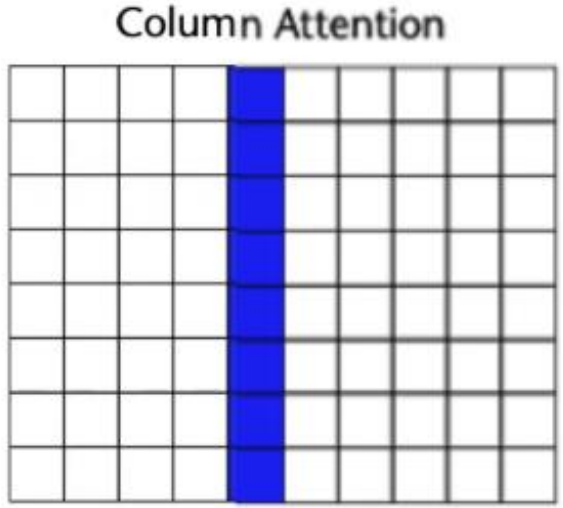
- MSA Transformer

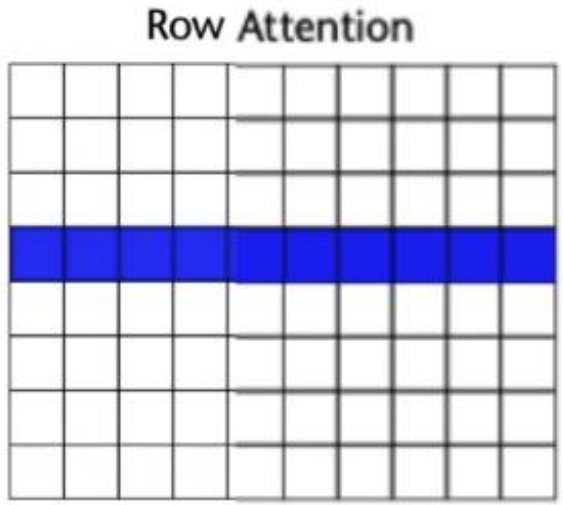


MSA Transformer: Adapt Data

- Axial Attention reduces memory (Ho et al. 2019)
- Further reduction by subsampling sequences
- Train with masked language modeling



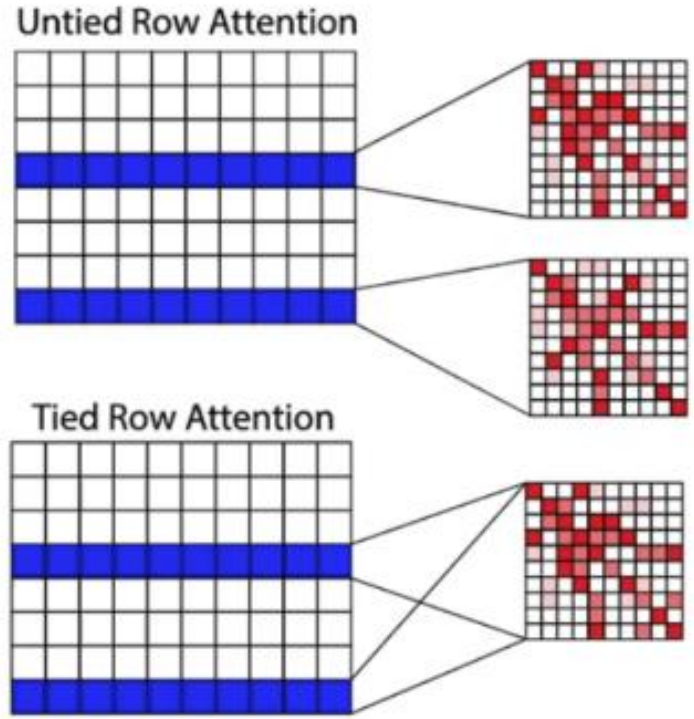




Rao et al. (2021). MSA Transformer. *ICML*.

MSA Transformer: Adapt Model

- Sequences share structure
- Tying attention maps helps learn shared structure
- Normalization required to prevent weight explosion

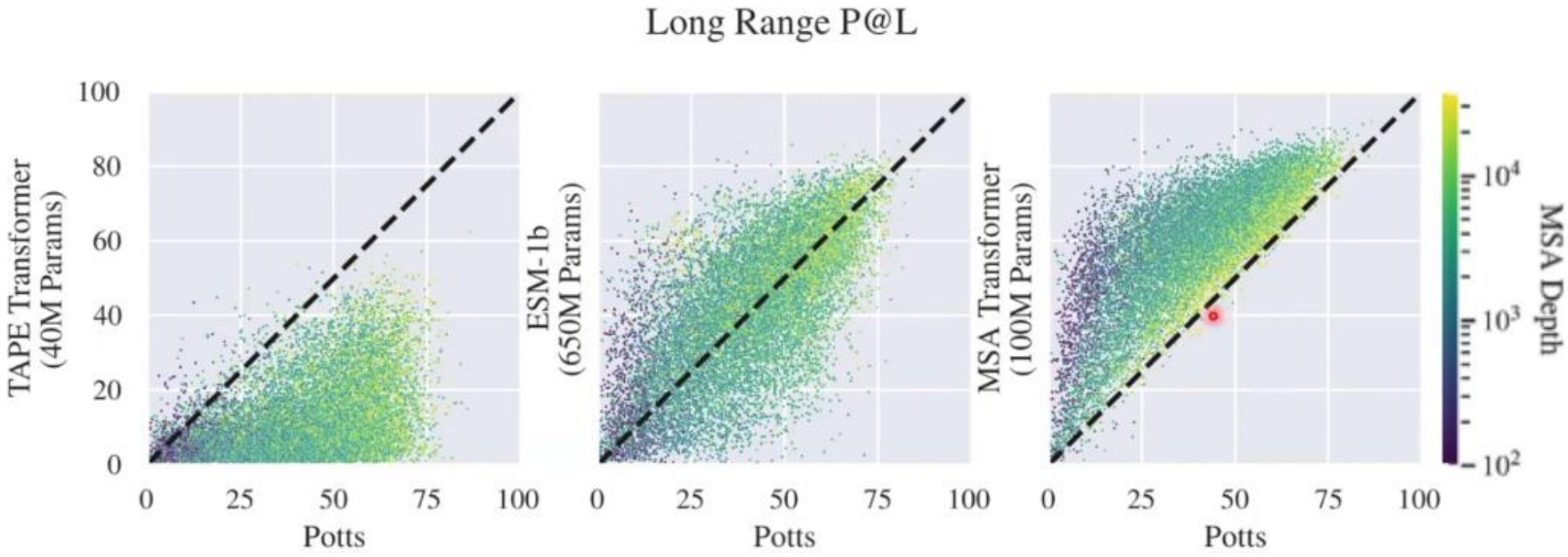


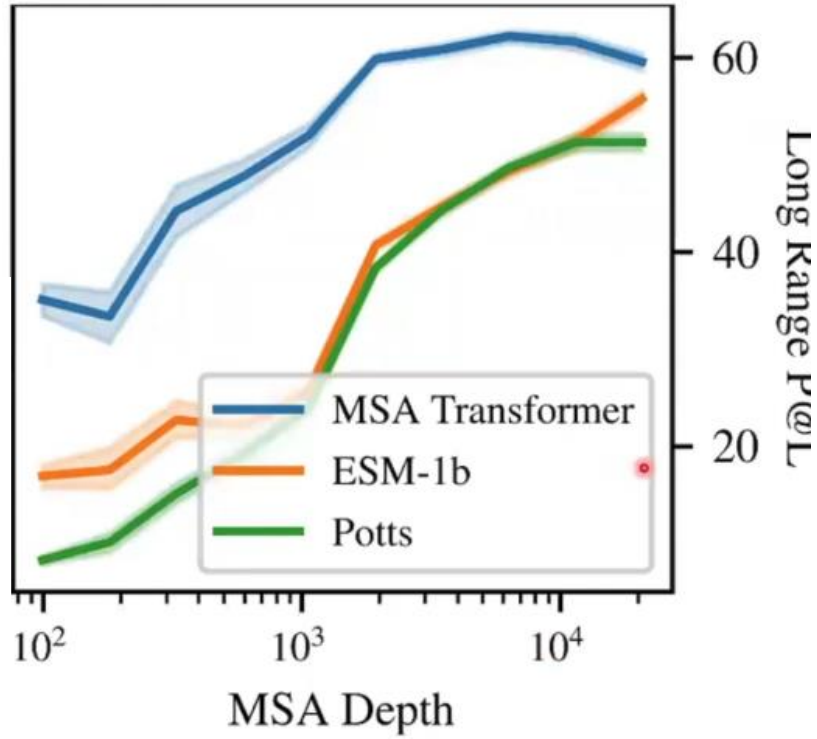
$$\sum_{m=1}^{M} \frac{Q_m K_m^T}{\lambda(M,d)}$$

Rao et al. (2021). MSA Transformer. *ICML*.

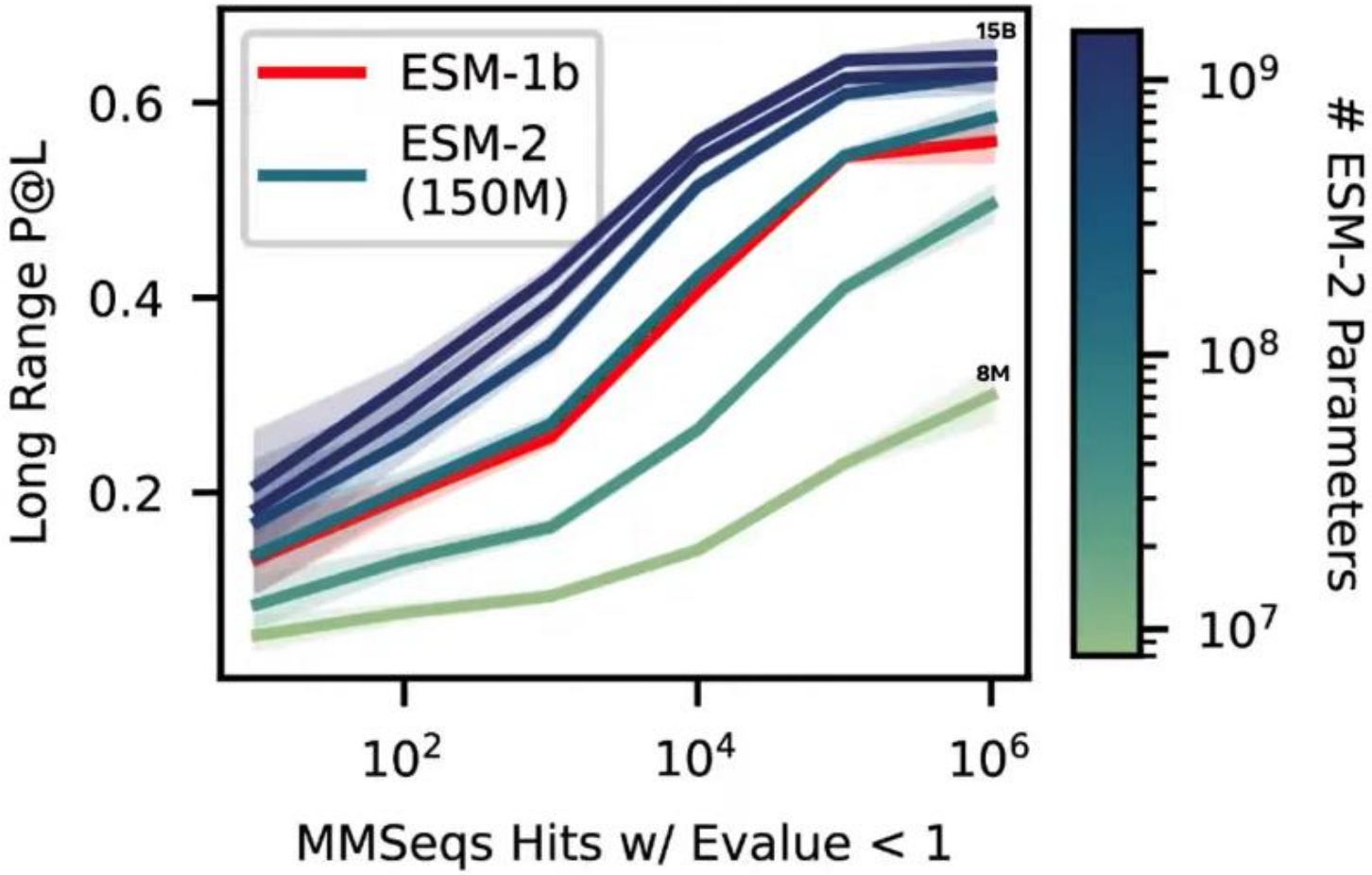
MSA Transformer outperform ESM-1b

	Model	# Params	# Updates	P@L
Baseline	Potts	N/A	N/A	39.3
	ESM-1b	650M	505k	41.1
MSA Transformer	Untied	100M	100k	42.1
	Mean	100M	100k	50.1
	Sqrt	100M	100k	56.3

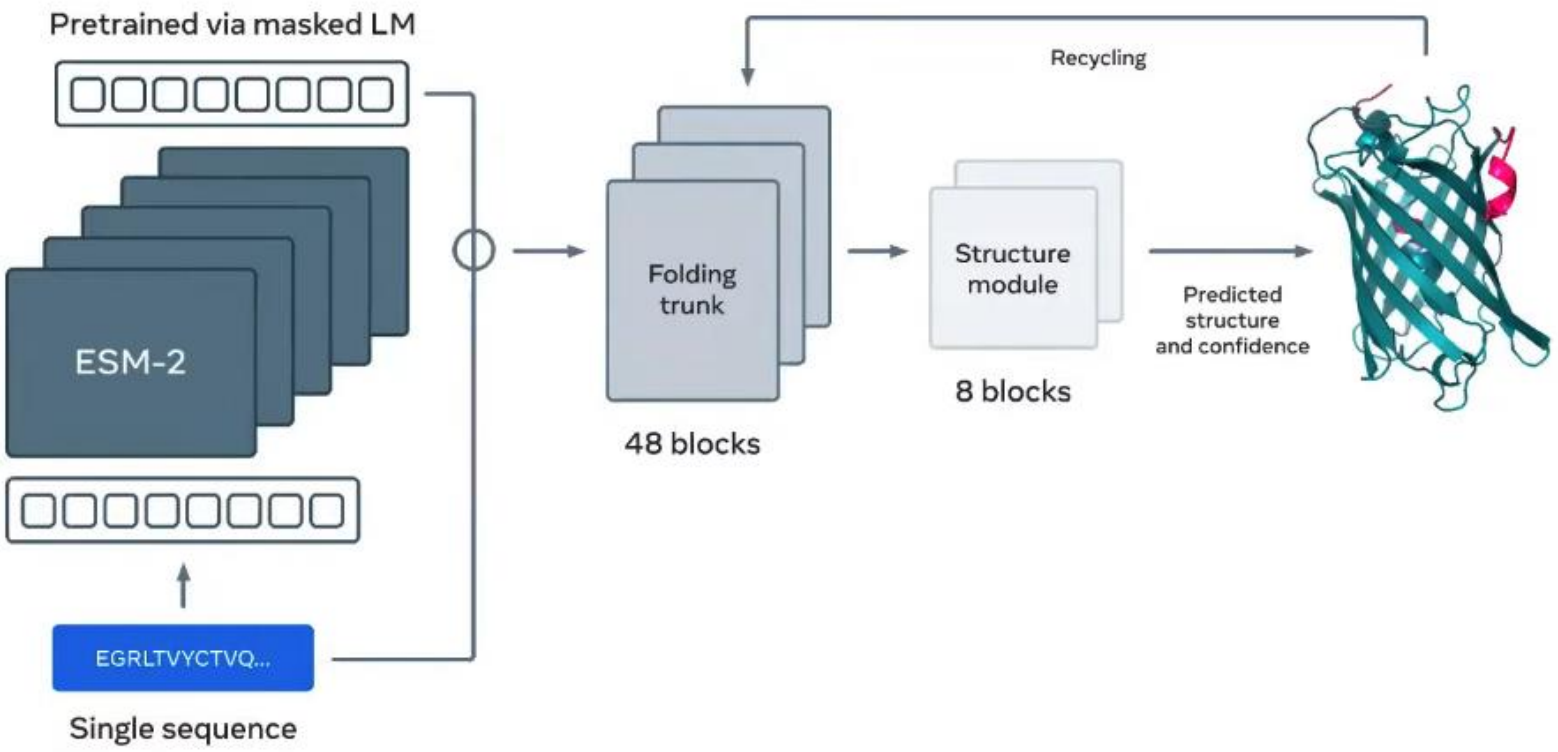


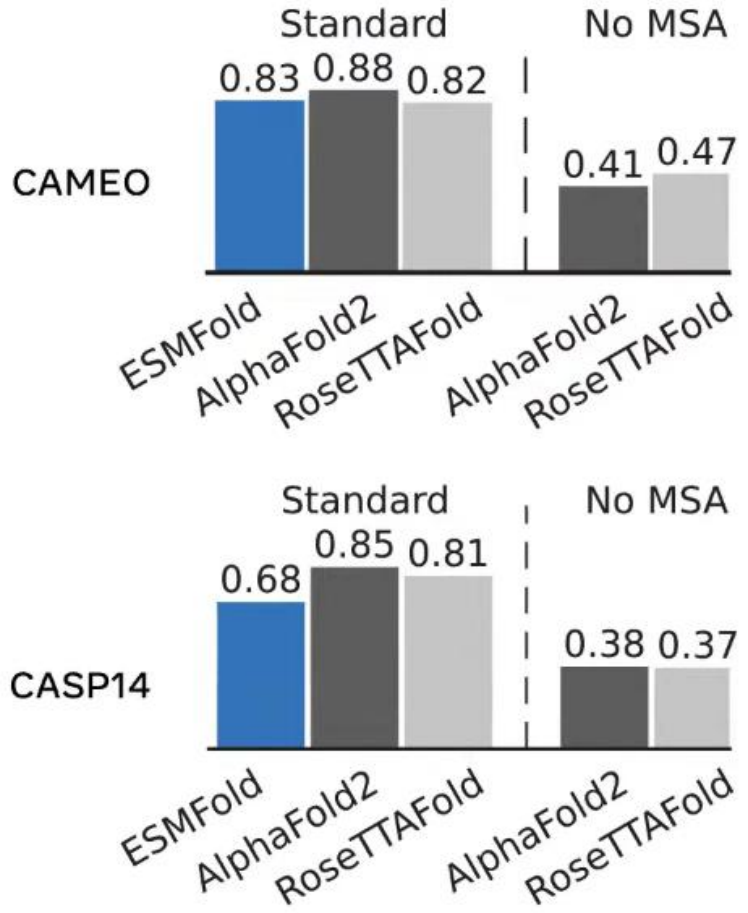


Further scale up: ESM2



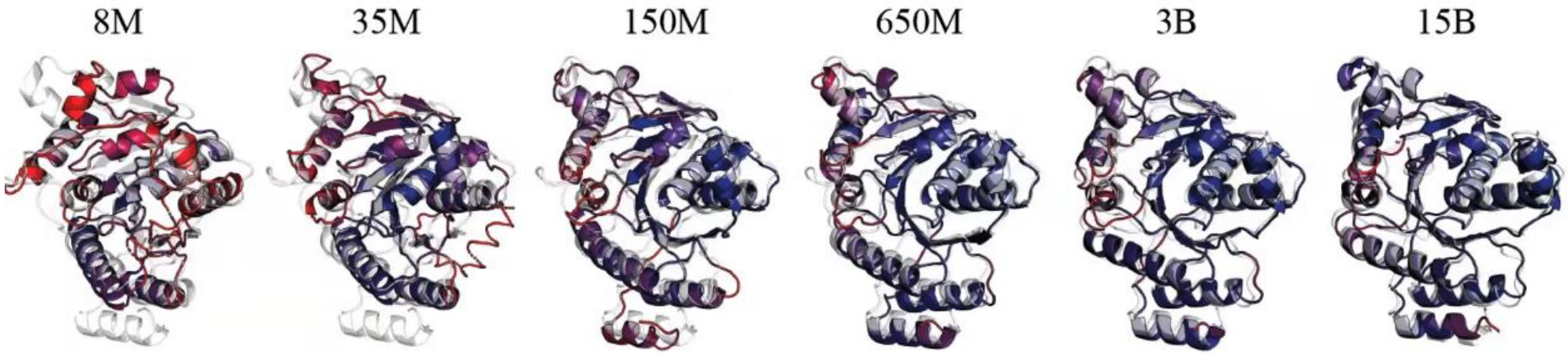
ESMFold: predict structure from single sequence



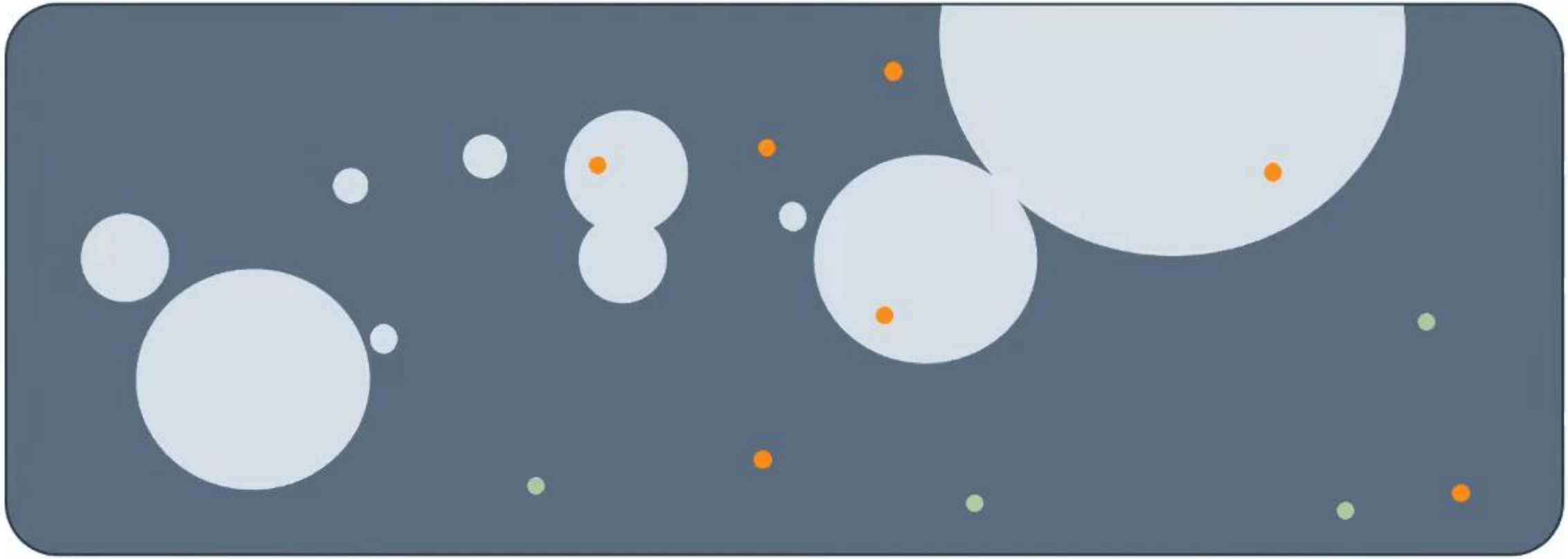


Lin, et al. 2023. "Evolutionary-scale prediction of atomic-level protein structure with a language model." Science [Preprint 2022]

Emergence of atomic level structure with scale



Language models generalize beyond natural proteins



Natural proteins
De novo target set
Generated proteins

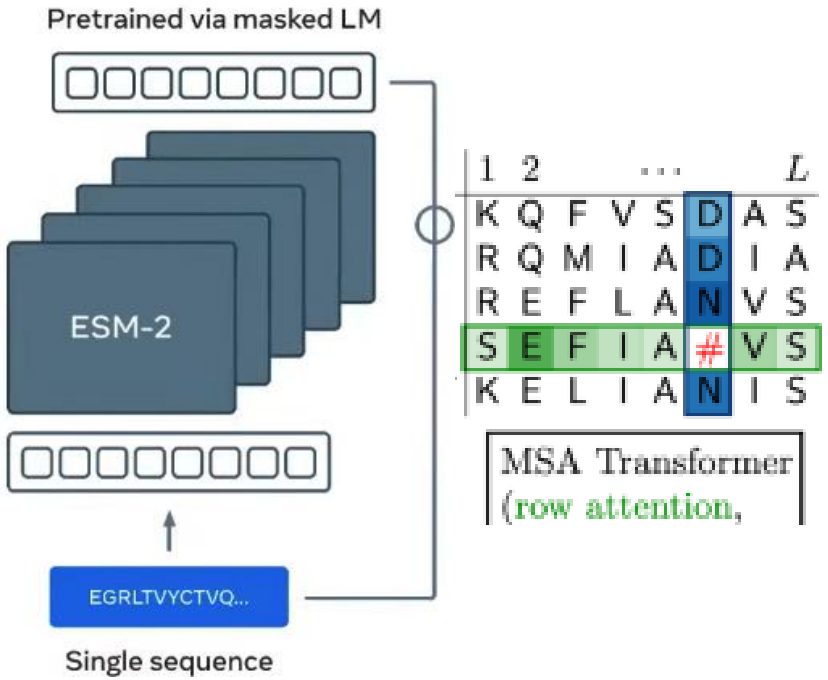
Sequence Space
Length 200 protein = $\mathbf{20^{200}}$

Verkuil, et al. 2022. "Language models generalize beyond natural proteins." [Preprint]

Protein Language Models (pLM)



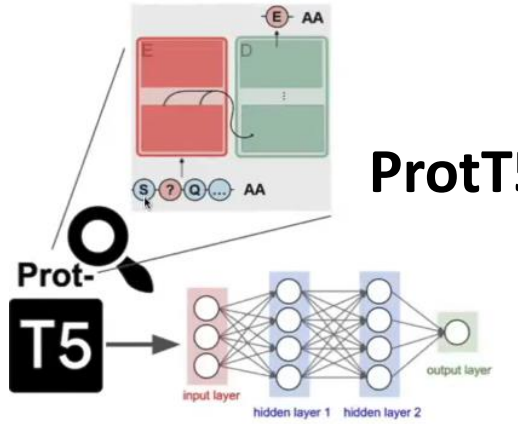
BERT (Encoder-only)

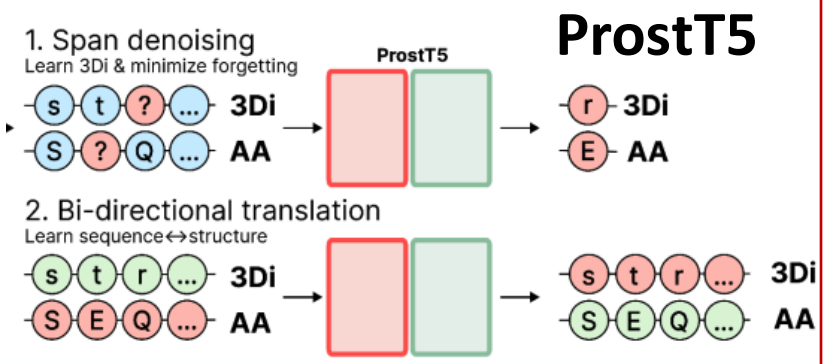


Masked token prediction

T5

T5 (Encoder-decoder)



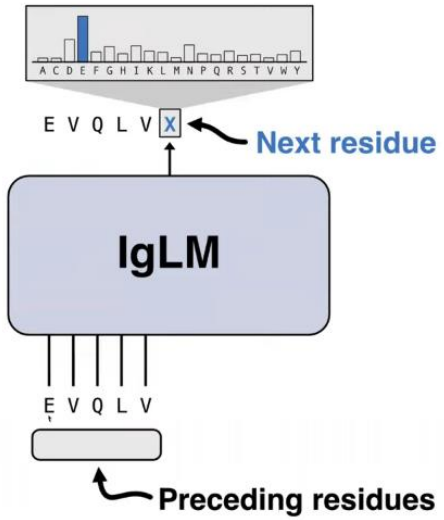


Denoising span-mask prediction

OpenAI

GPT - 4

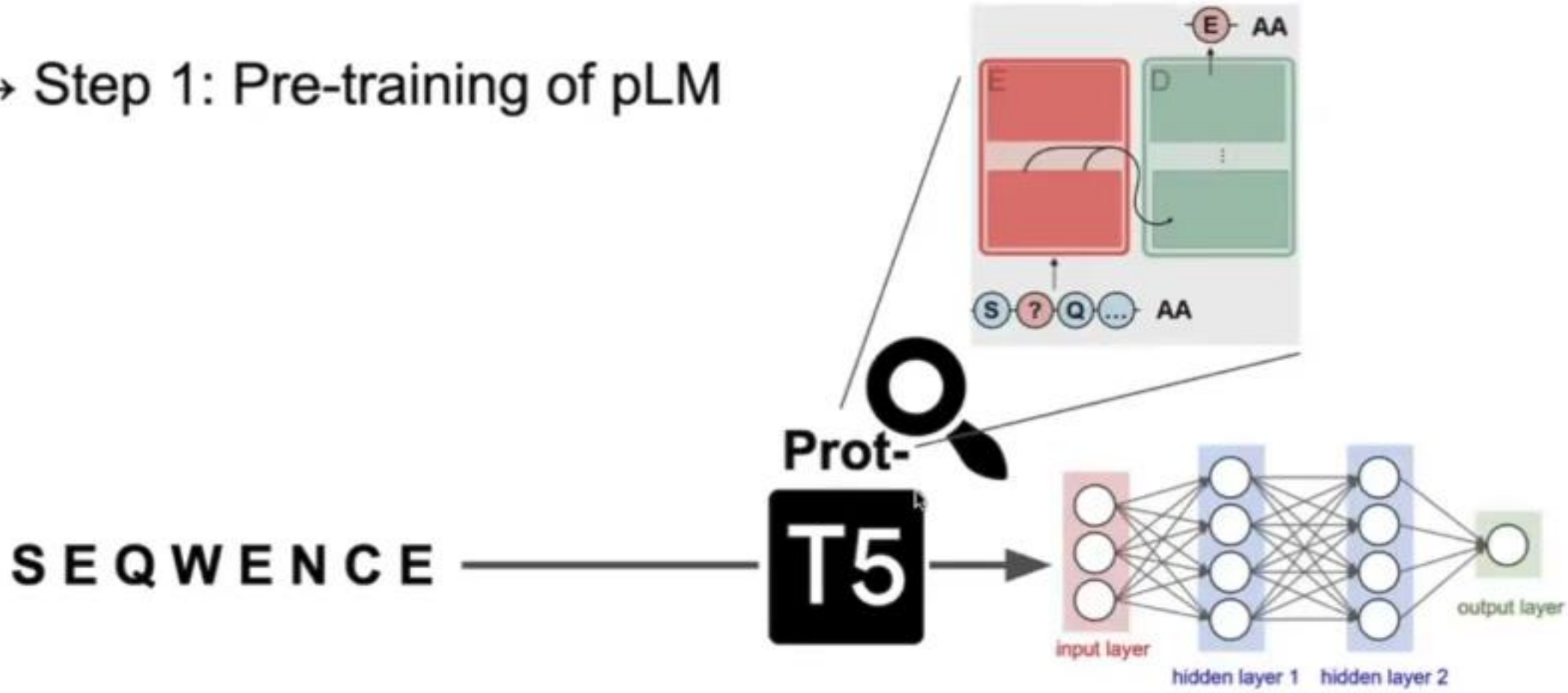
Decoder-only



Next token prediction

ProtT5/ProtParc

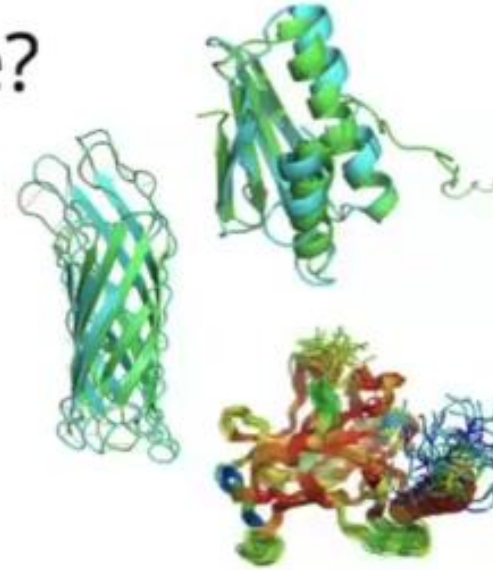
→ Step 1: Pre-training of pLM

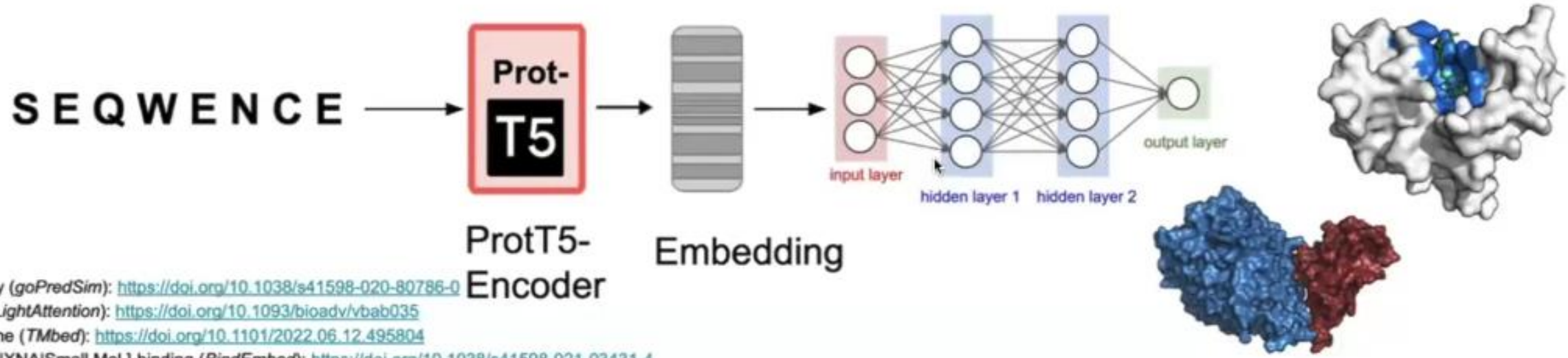


Relative position encoding

How to make protein sequences machine-readable?

→ Step 2: Feature extraction





ProtT5-Encoder

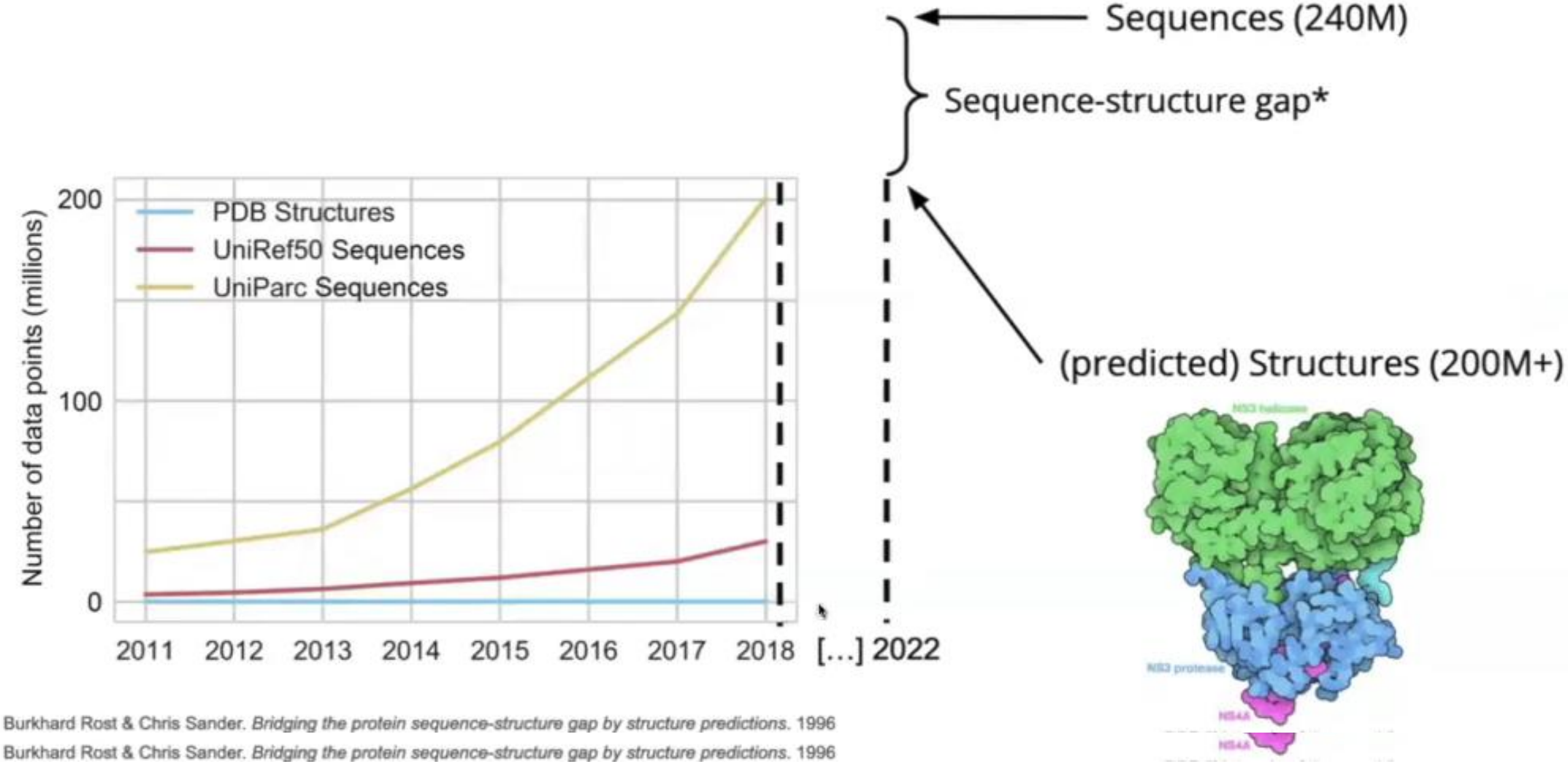
Embedding

Gene Ontology (*goPredSim*): https://doi.org/10.1038/s41598-020-80786-0
Subcell. Loc (*LightAttention*): https://doi.org/10.1093/bioadv/vbab035
Transmembrane (*TMbed*): https://doi.org/10.1101/2022.06.12.495804
Protein-[Metall|XNA|Small Mol.]-binding (*BindEmbed*): https://doi.org/10.1038/s41598-021-03431-4
Secondary Structure (ProtTrans): https://ieeexplore.ieee.org/abstract/document/9477085
3D Structure (*EMBER*): https://doi.org/10.1016/j.str.2022.05.001
Disorder (*SETH*): https://doi.org/10.1101/2022.06.23.497276
Conservation & SAV-effect: https://doi.org/10.1007/s00439-021-02411-y

… now also with PEFT/LoRA*

*Schmirler, Heinzinger, Rost. **Fine-tuning protein language models boosts predictions across diverse tasks.** bioRxiv. 2023

Post AF2 era: AFDB made more predicted structures available



Burkhard Rost & Chris Sander. *Bridging the protein sequence-structure gap by structure predictions*. 1996

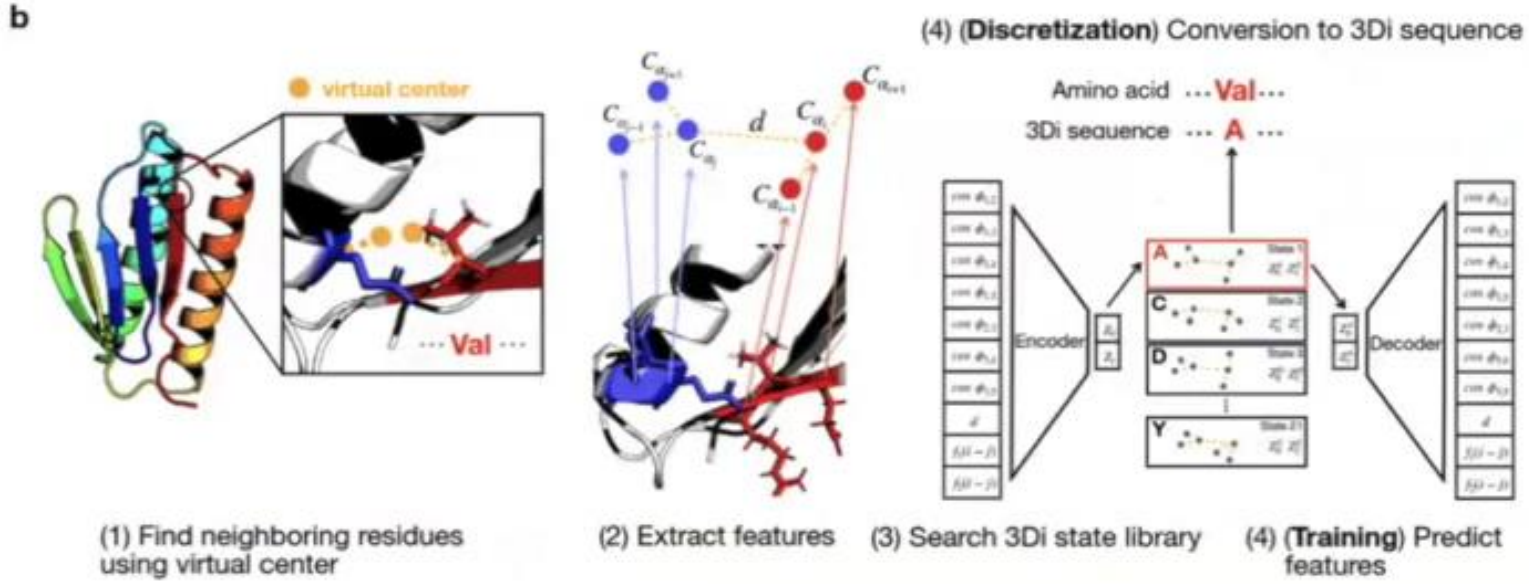
Burkhard Rost & Chris Sander. *Bridging the protein sequence-structure gap by structure predictions*. 1996

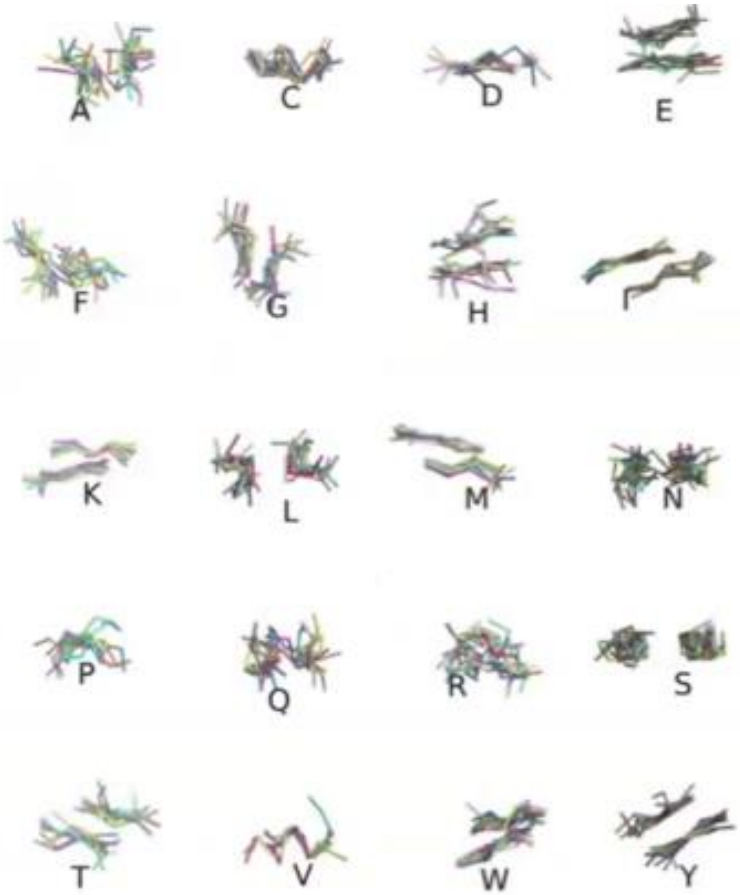
Tokenize structures?

FoldSeek

Fast structure-comparison tool.

Learnt mapping (VQ-VAE) converts 3D to 1D (then, MMSeqs2):

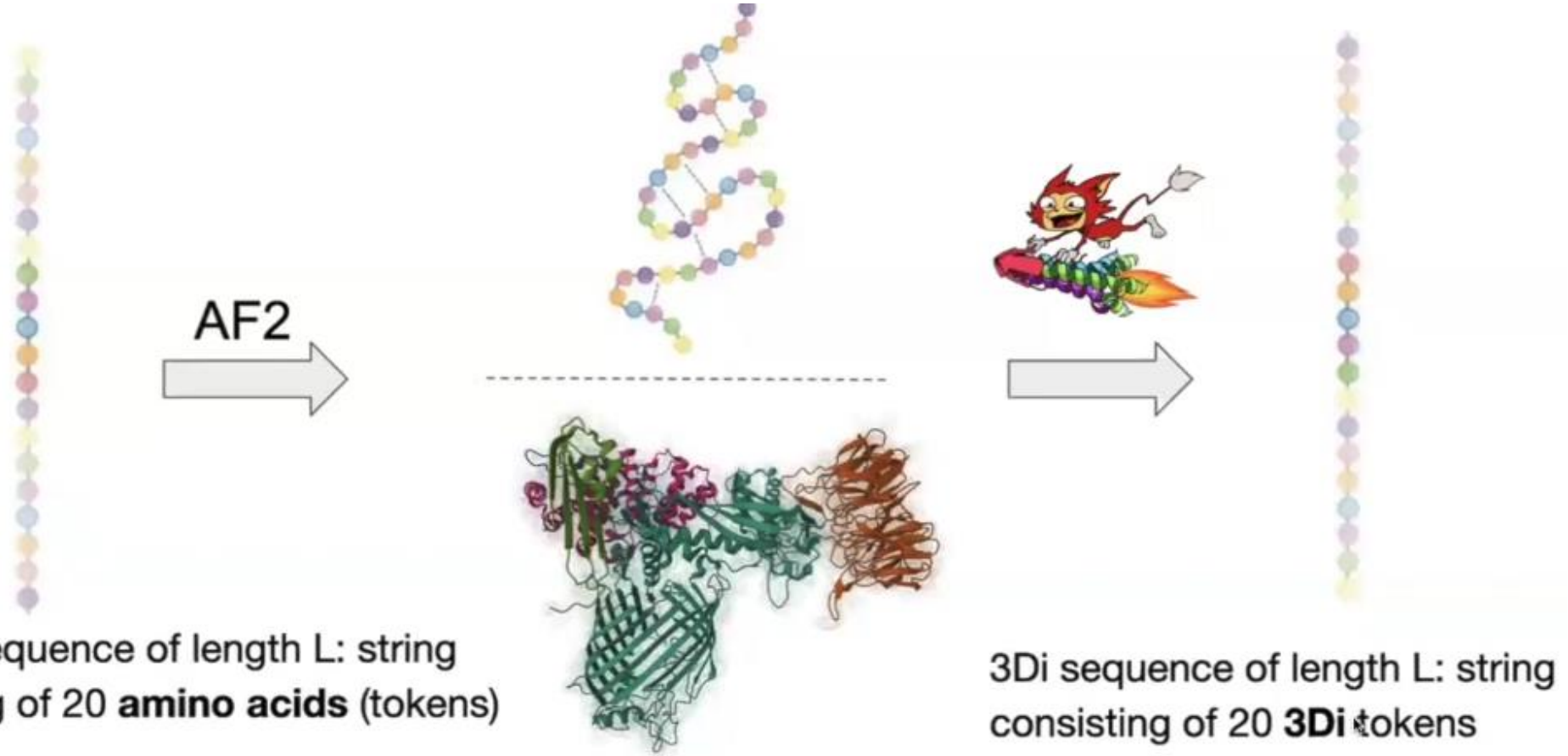




Not AAs, but 3Di tokens

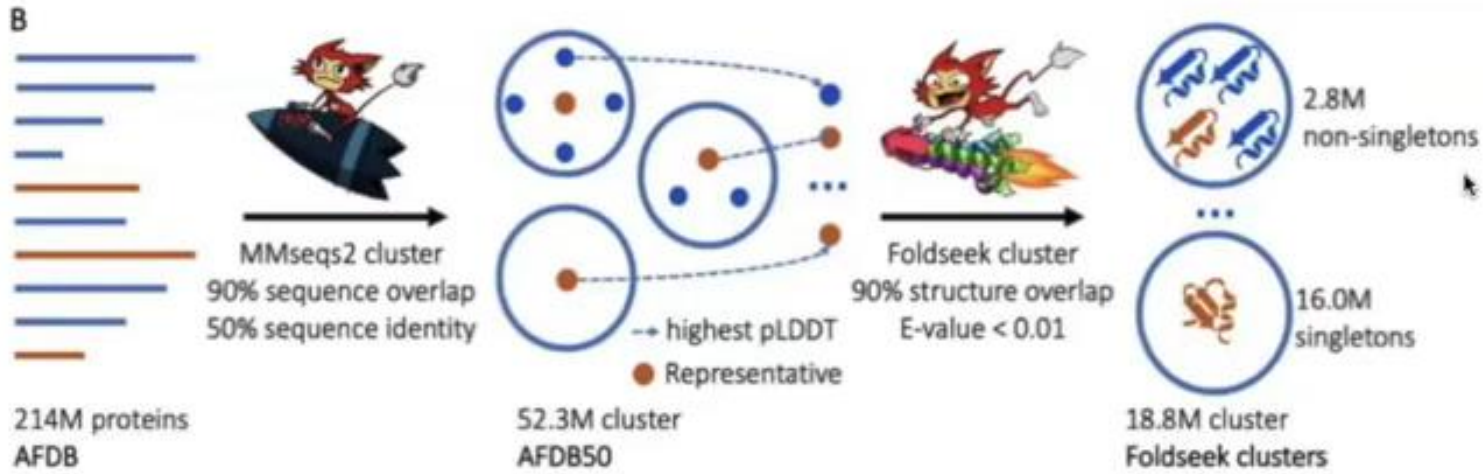
Van Kempen, Kim, Tumescheit, Mirdita, Lee, Gilchrist, Söding, Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology* (2024)

2 strings: sequence(aa) and structure (3di token)



Van Kempen, Kim, Tumescheit, Mirdita, Lee, Gilchrist, Söding, Steinegger. Fast and accurate protein structure search with Foldseek. *Nature Biotechnology* (2024)
John Jumper, and Demis Hassabis *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature*. (2021)

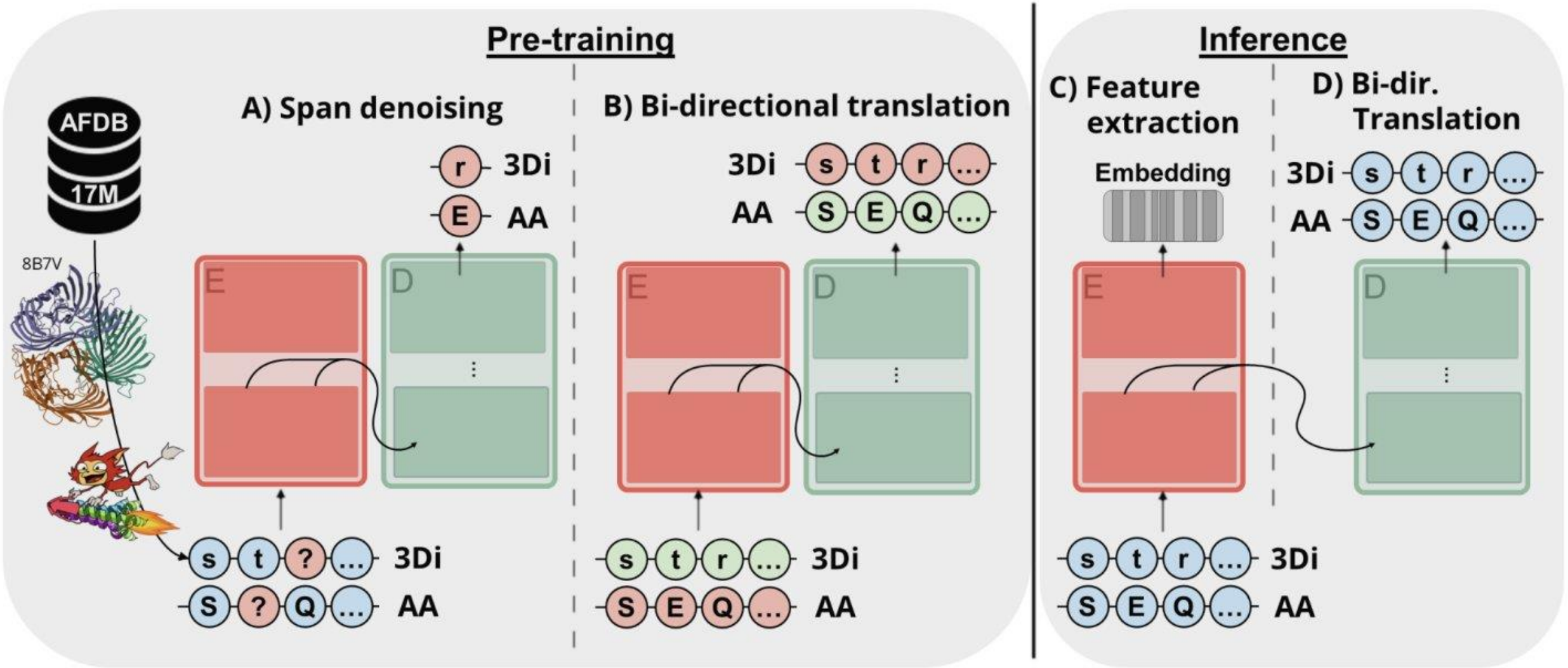
Step #1: Dataset



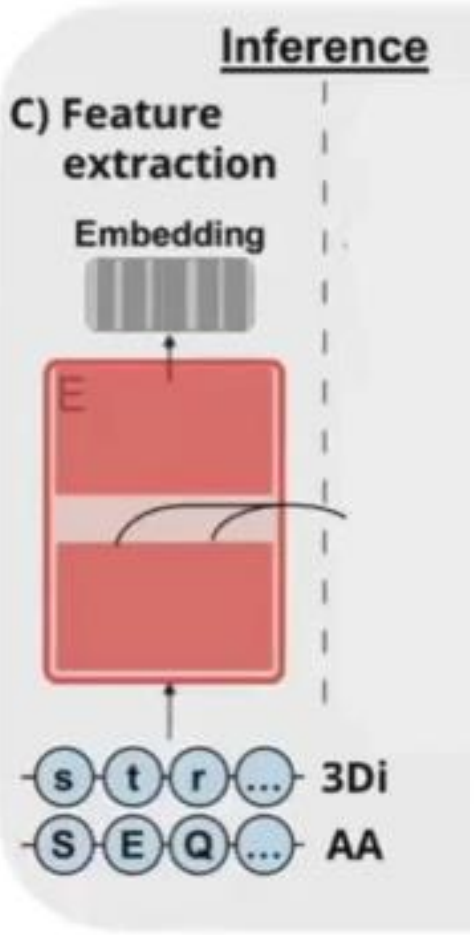
1. Start: AFDB (**214M**)
2. Sequence similarity clustering (**52M**)
3. Structure similarity clustering (**19M**)
   —*
4. Expand clusters by 20 most diverse members
5. Filter by pLDDT (>70), length (>30) and 3Di-token diversity (<95% identical tokens)
   → **17M** sequence-structure pairs or 34M individual samples: `SEQWENCE,strctare`

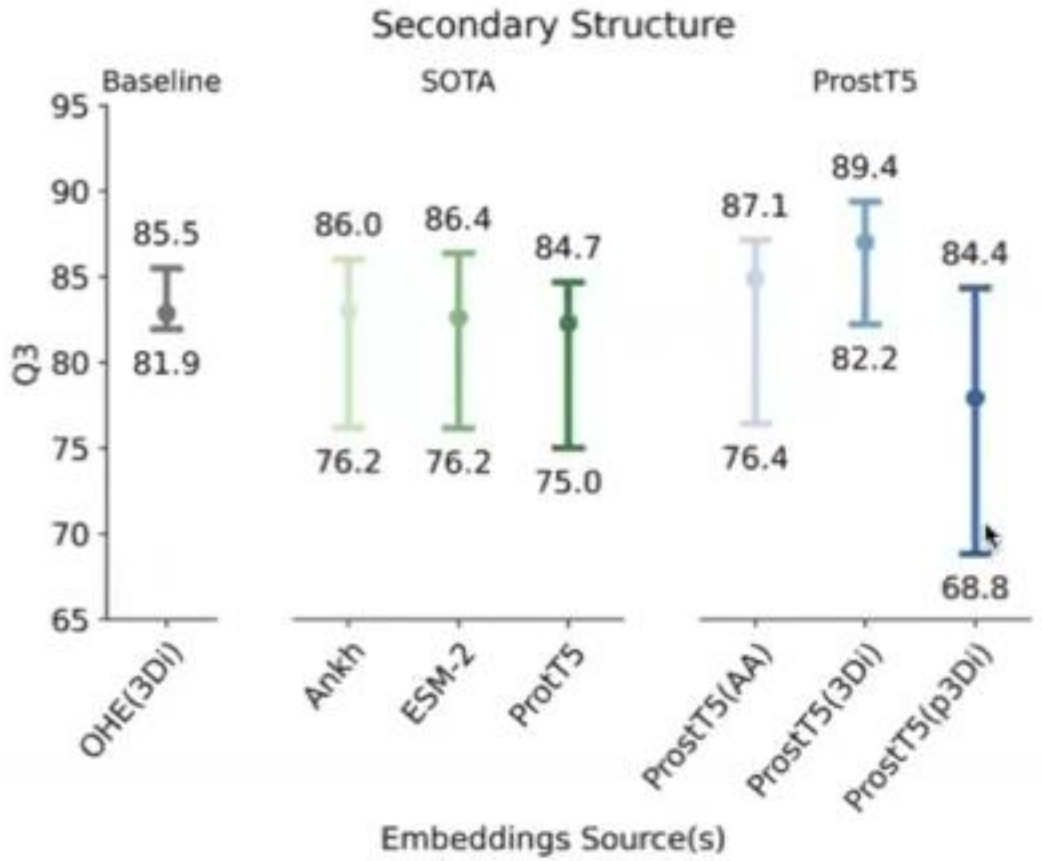
Barrio-Hernandez, Yeo, Jänes, Mirdita, Gilchrist, Wein, Varadi, Velankar, Beltrao & Steinegger. *Clustering predicted structures at the scale of the known protein universe*. Nature (2023)

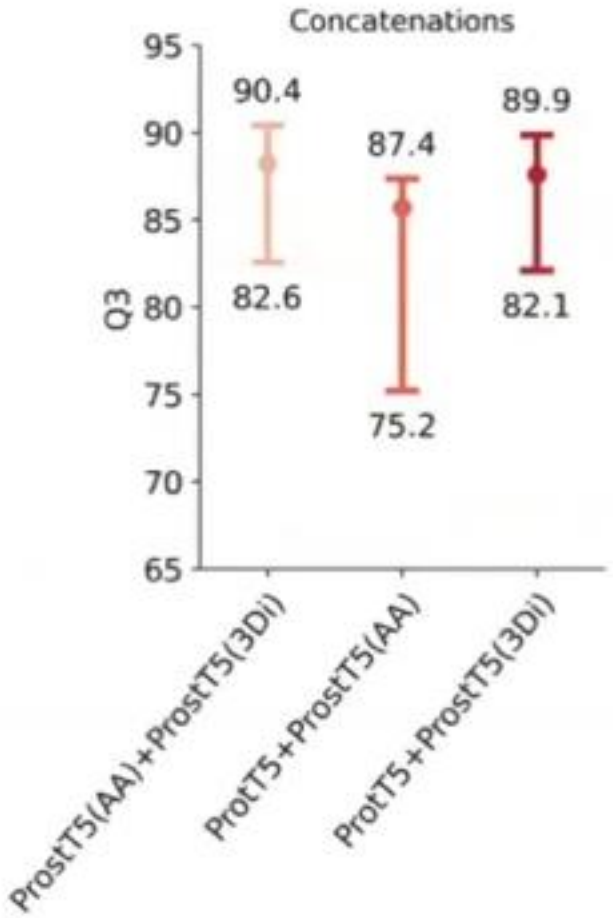
ProstT5: sequence <-> structure token



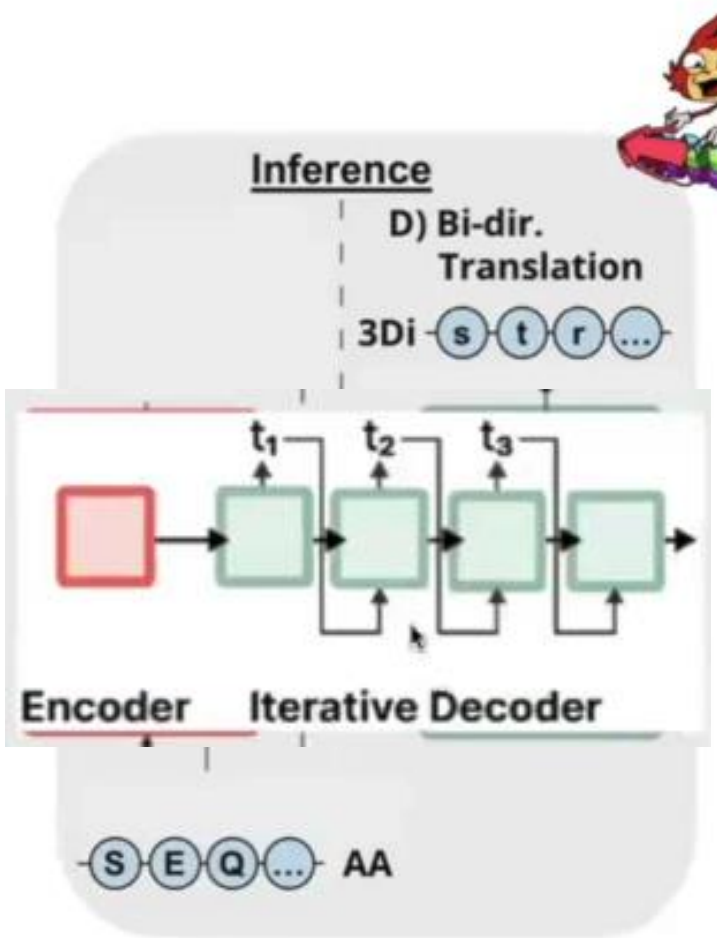
Transfer learning performance

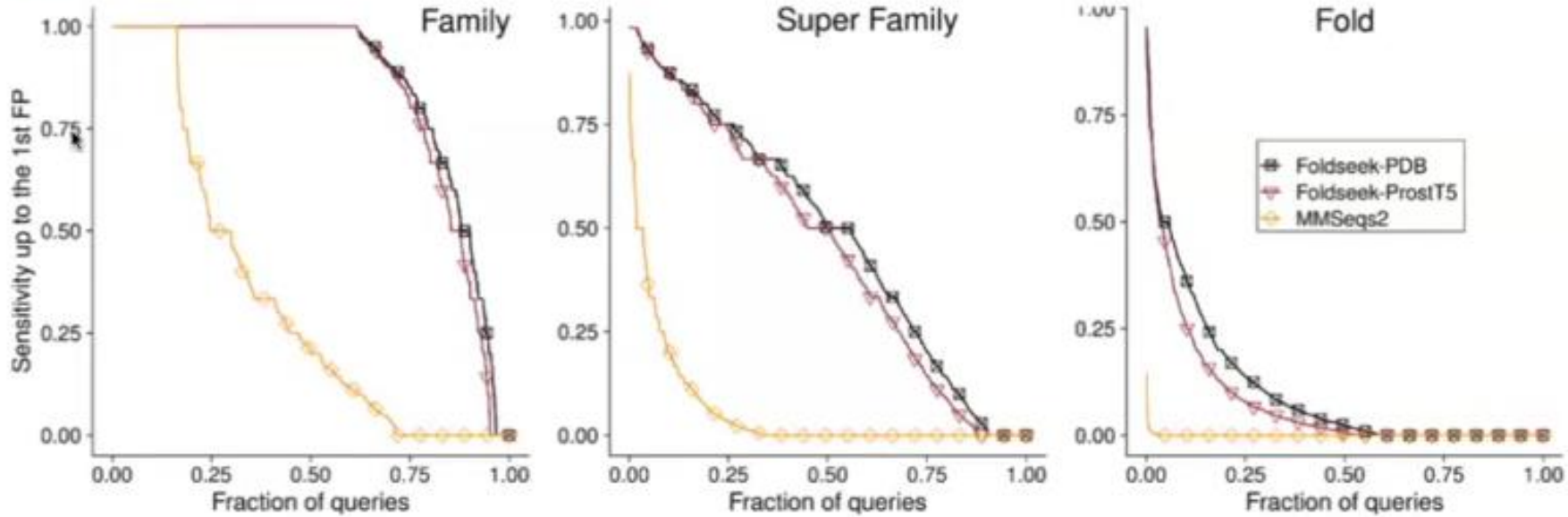






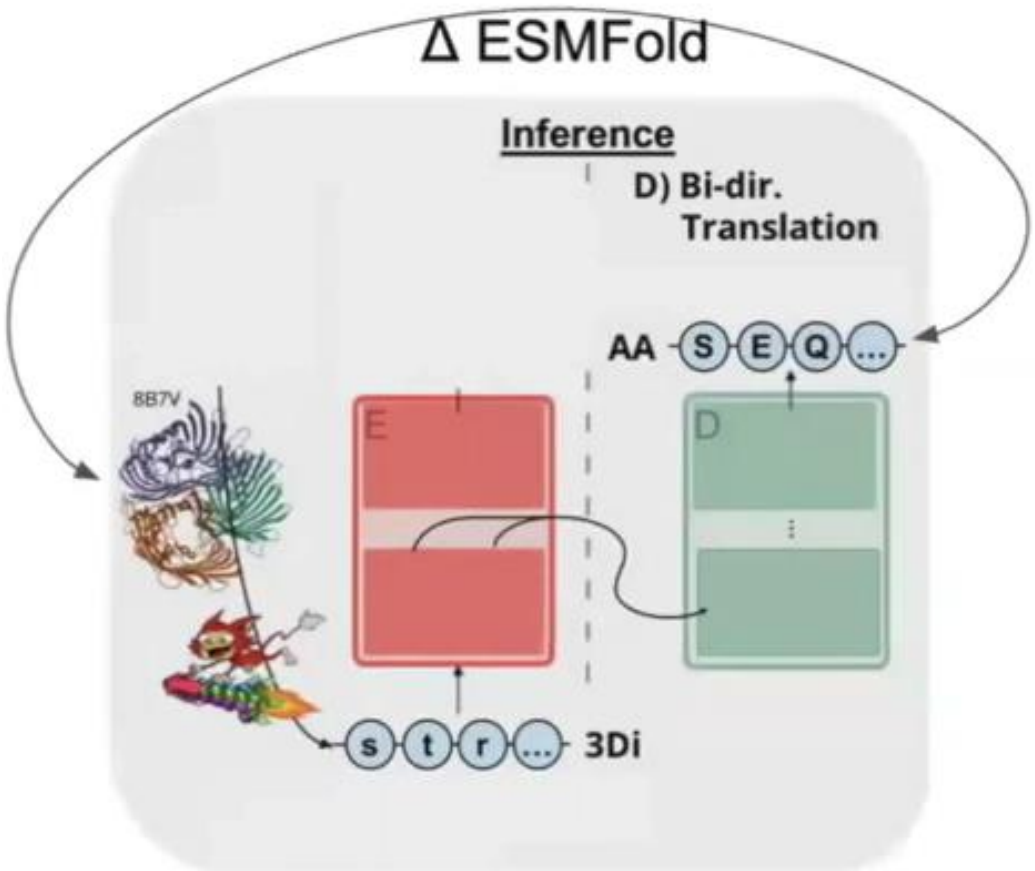
Use-case #2: Translate AA → 3Di





→ Remote homology detection at structure comparison accuracy without actual 3D prediction!

Use-case #3: Translate 3Di → AA (Inverse folding)



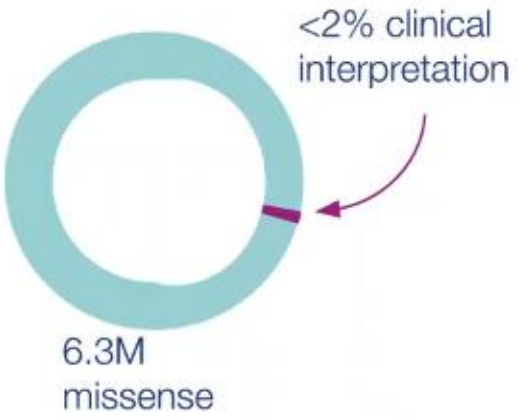
	Native/ESMFold	*ProstT5*	*ProteinMPNN*
IDDT ↑	0.78 ±0.01	0.72 ±0.01	**0.77 ±0.01**
RMSD ↓	2.55 ±0.01	2.90 ±0.01	**2.61 ±0.01**
TM-score ↑	0.62 ±0.02	0.58 ±0.02	**0.61 ±0.02**
PIDENT	100 ±0	21.9 ±0.9	**29.6 ±1**
Entropy ↓	0.13 ±0.01	0.20 ±0.01	0.39 ±0.03

Dauparas, J., & Baker, D. et al. Robust deep learning–based protein sequence design using ProteinMPNN. *Science* (2022)

Downstream applications of PLM

Human variant annotation

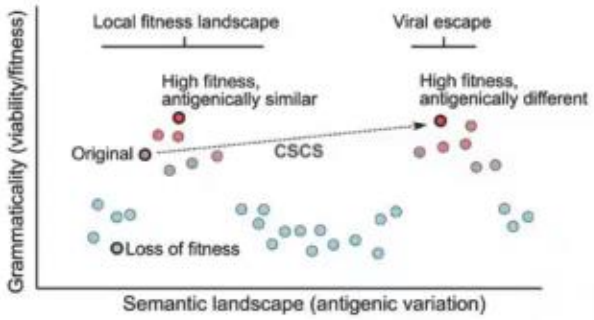
- The large majority of human variants[1] have no known interpretation



- Example: **EVE**[2], protein-specific alignment-based generative models for mutation effects prediction

Viral escape prediction

- Viral escape mutations are the ones that both **maintain fitness** while **disrupting Ab binding**



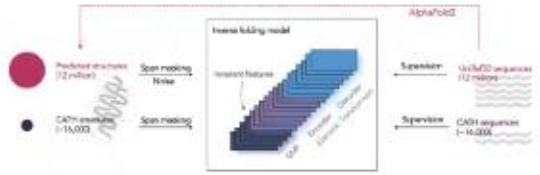
- Example: **Hie et al.**[3], use a single LLM to decompose escape in terms **semantic & grammaticality changes**

Protein design

- Generating **novel** yet **fit** sequences, conditioning on:
  - **Labels**: Madani et al., Progen[4]

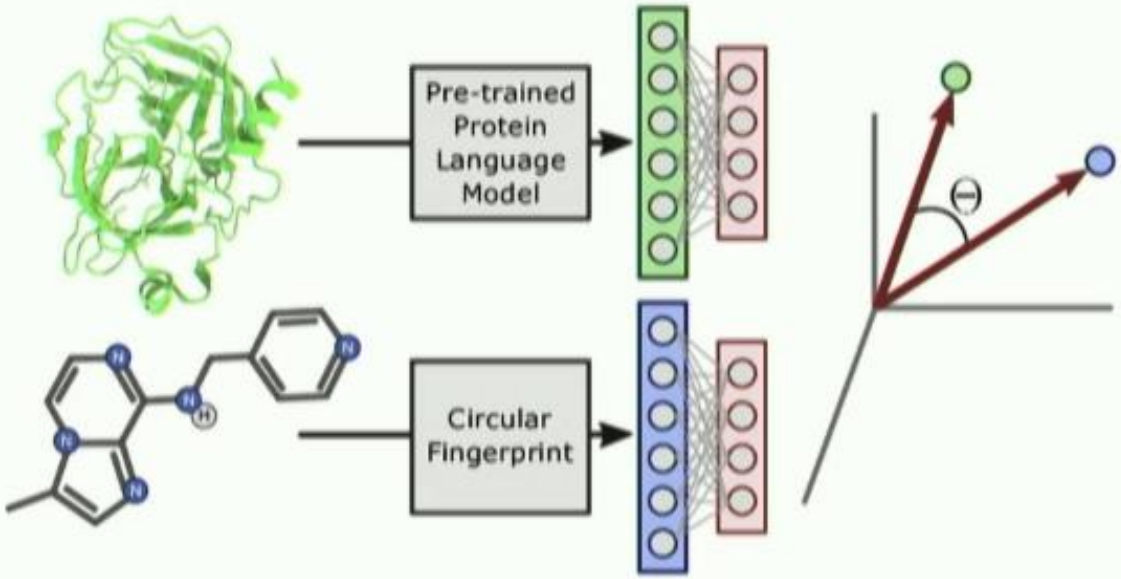


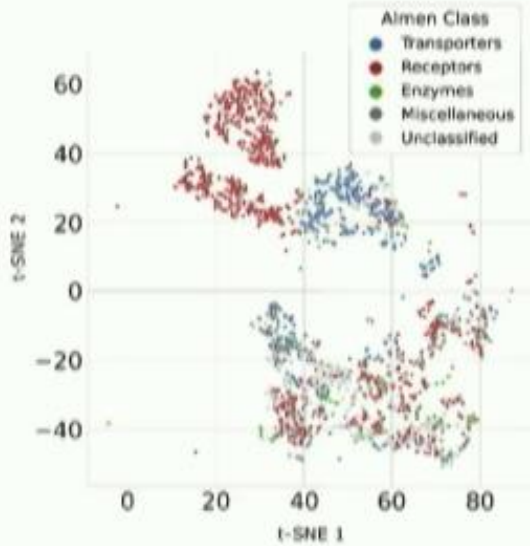
  - **Structure** (Inverse folding): Ingraham et al[5], Hsu et al[6].

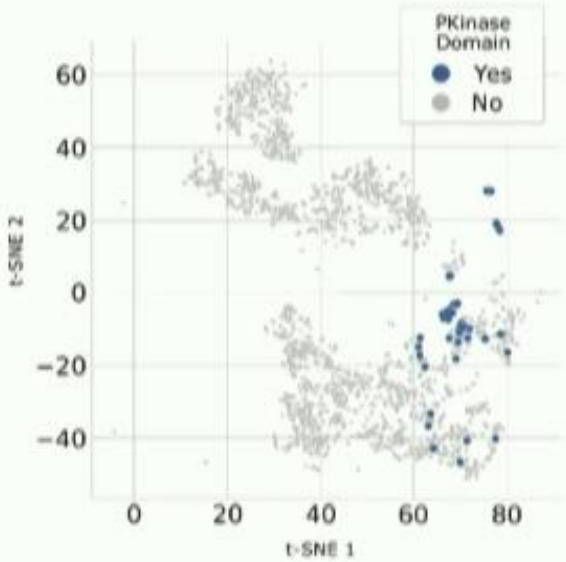


1. Landrum & Kattman. ClinVar at five years: Delivering on the promise. Hum Mutat 39, 1623-1630.
2. Frazer, Notin, Dias et al. Disease variant prediction with deep generative models of evolutionary data. Nature, 2021.
3. Hie et al. Learning the language of viral evolution and escape. Science, 2021.
4. Madani et al. ProGen: Language Modeling for Protein Generation. 2020.
5. Ingraham et al. Generative Models for Graph-Based Protein Design. NeurIPS, 2019.
6. Hsu et al. Learning inverse folding from millions of predicted structures. 2022.

Transfer learning to predict drug interactions







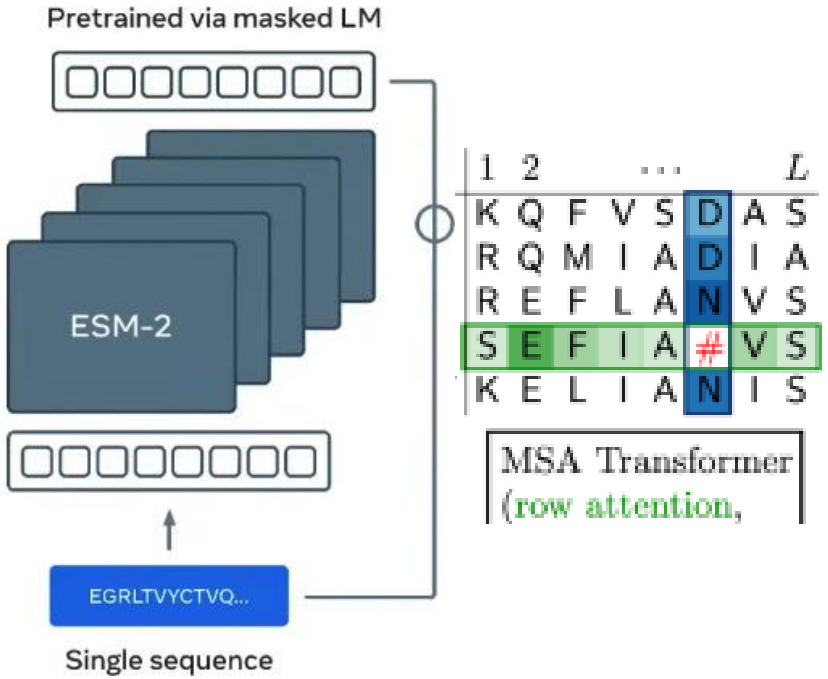
Explainable: Drugs and proteins mapped to the same space

Fast: Can screen 100 millions interactions in a day

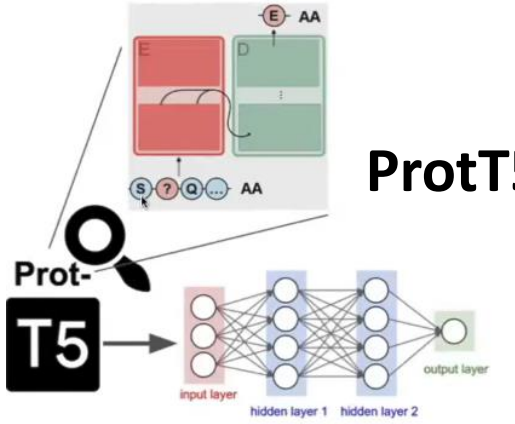
Protein Language Models (pLM)

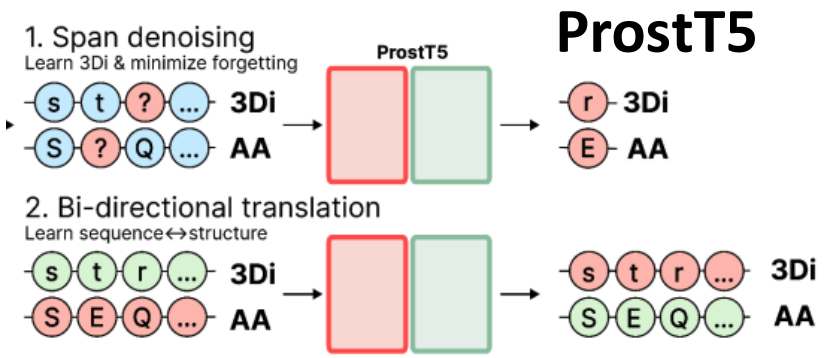


BERT (Encoder-only)



T5 (Encoder-decoder)





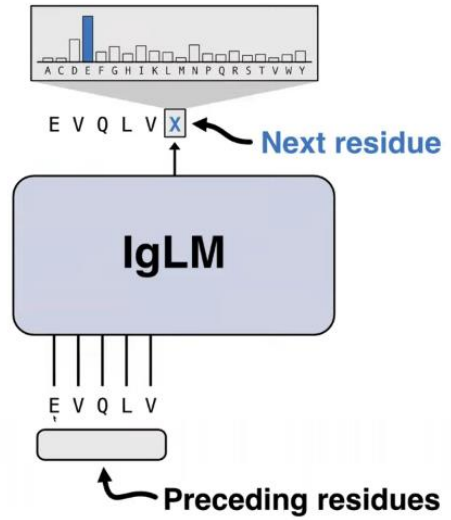
Masked token prediction

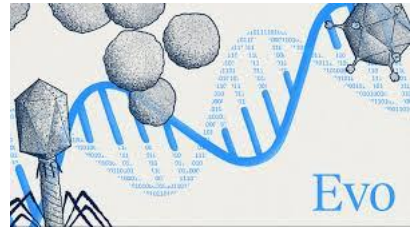
Denoising span-mask prediction

OpenAI

GPT - 4

Decoder-only

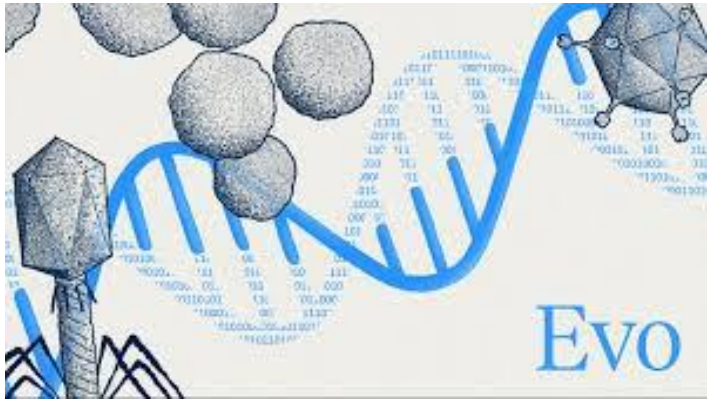


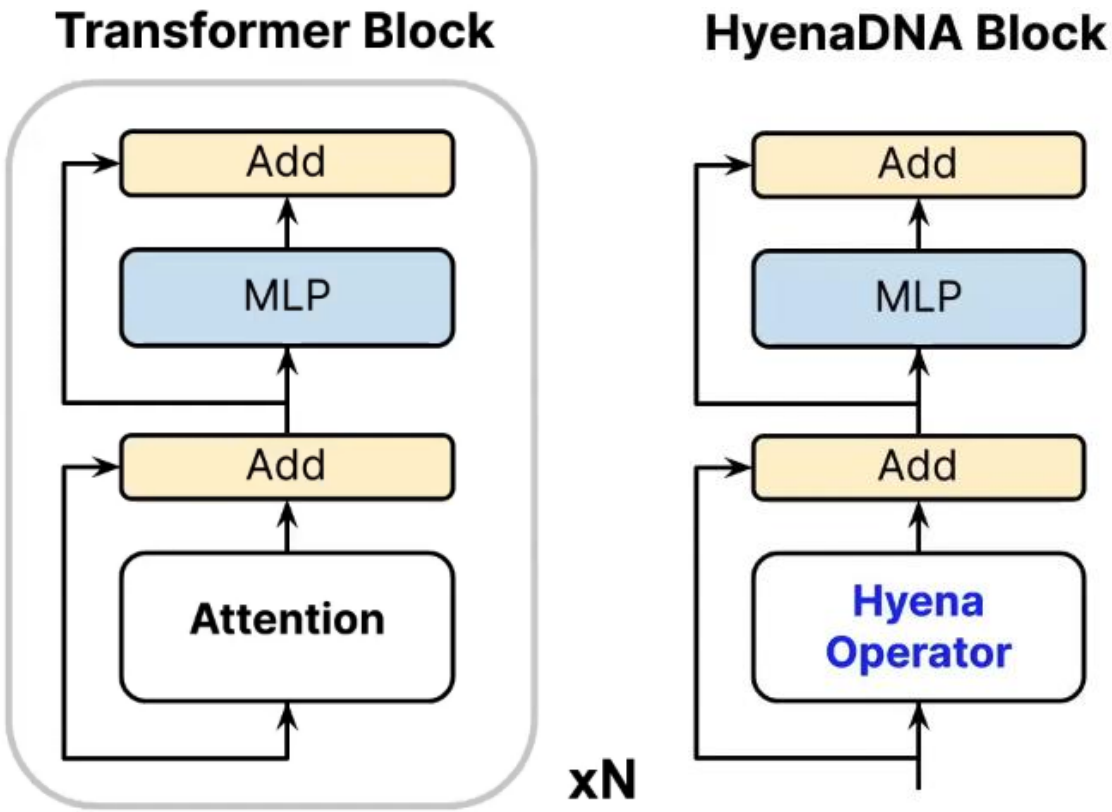


Next token prediction

Hyena: hybrid of SSM + Attention

Trained on





1. **Long convolution ->**

   (global context)

2. **Elemwise gate ->**

   (data-dependent)

HyenaDNA/EVO

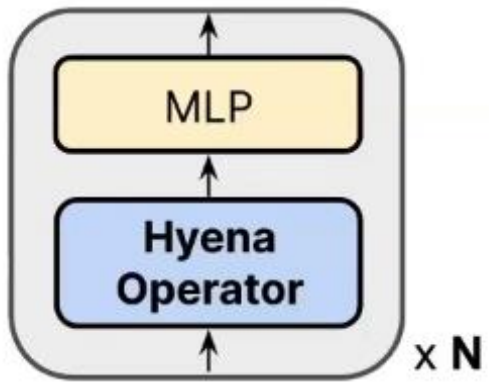
Sample from Human Genome



A C G T A C G T

Next token prediction

A C G T A C G T



A C G T A C G ?

(Single Nucleotide Tokens)

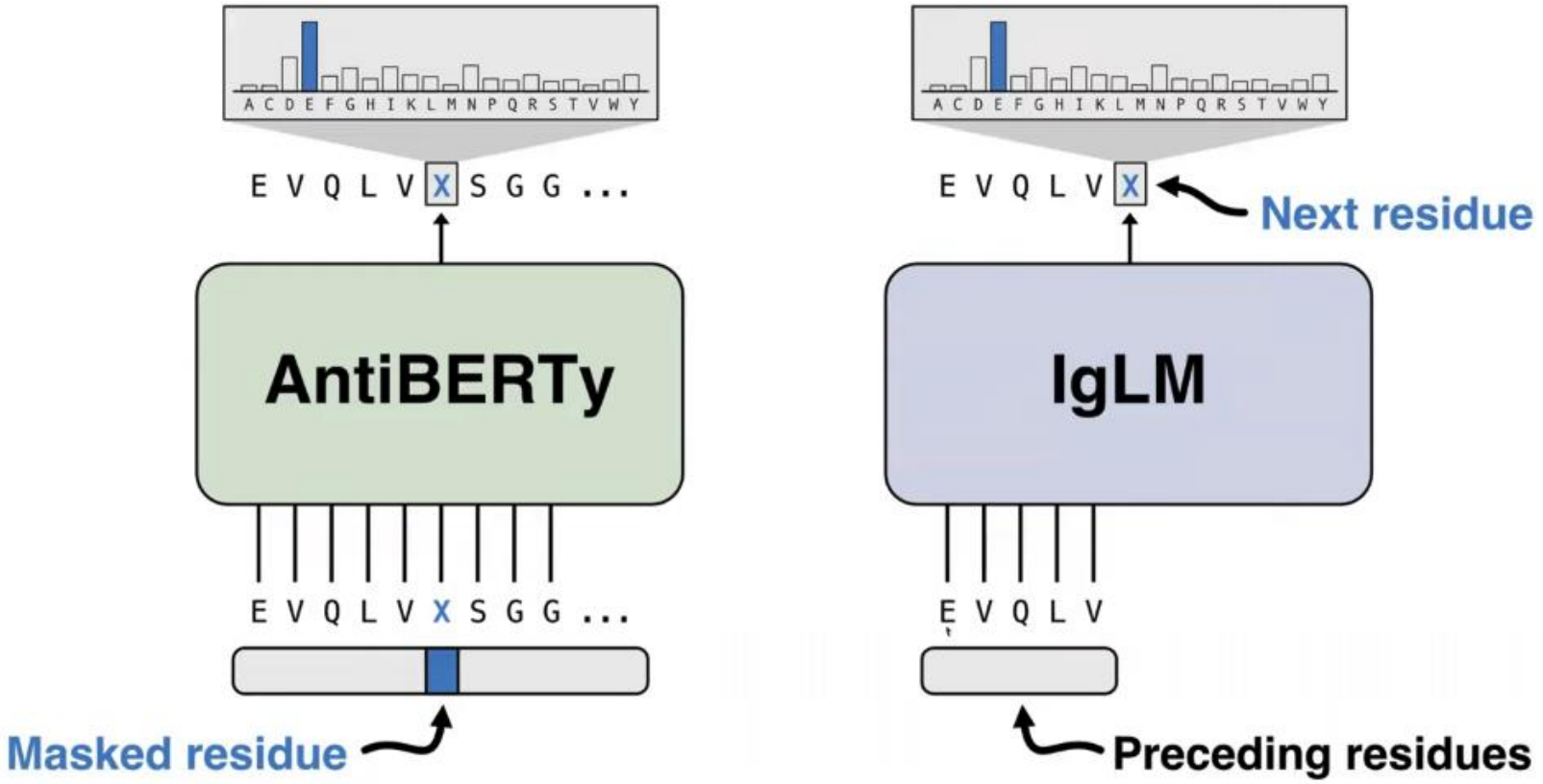
Downstream tasks

Regulatory element classification (short)

Chromatin profile, species classification (long)

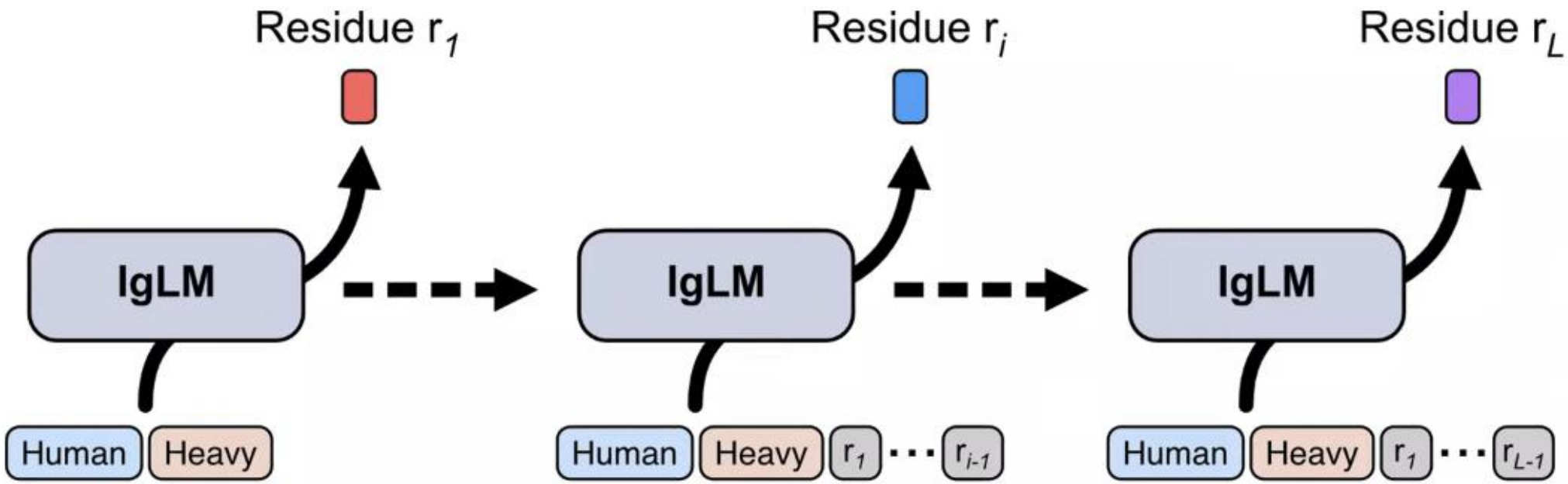
Explore in-context learning

IgLM: next residue prediction

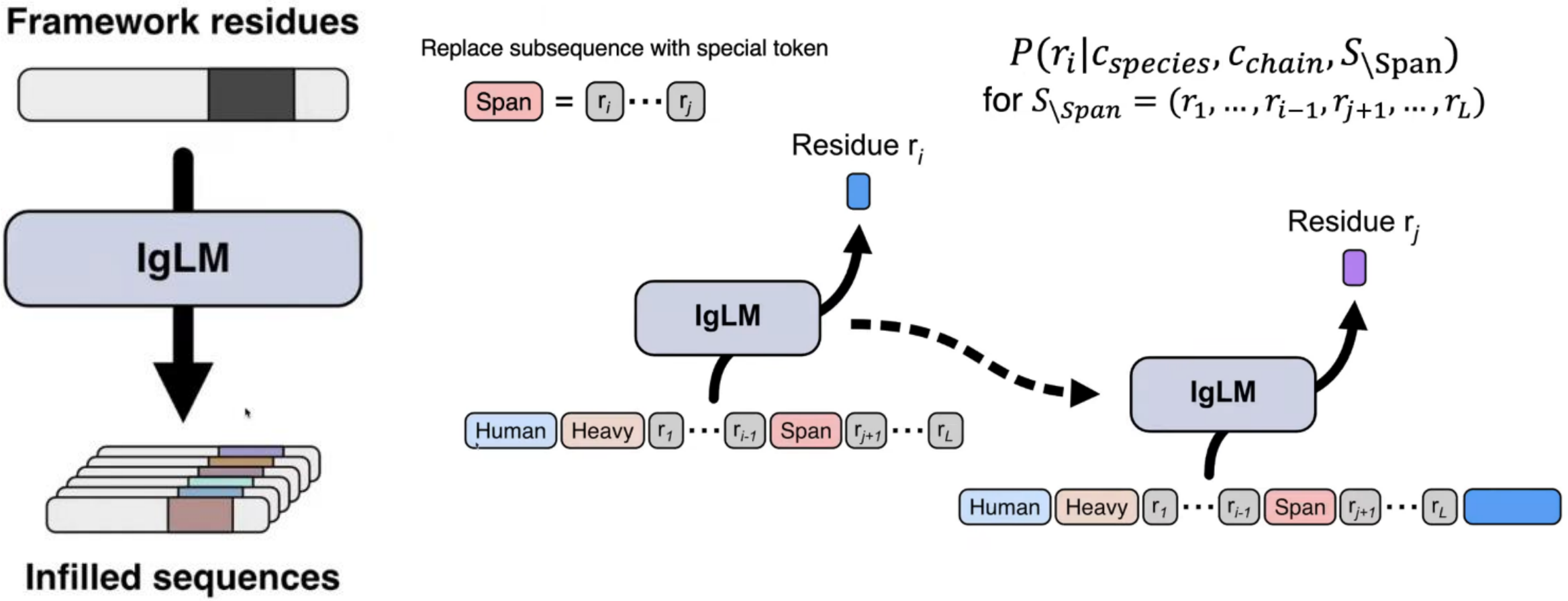


Species and chain ID as conditions

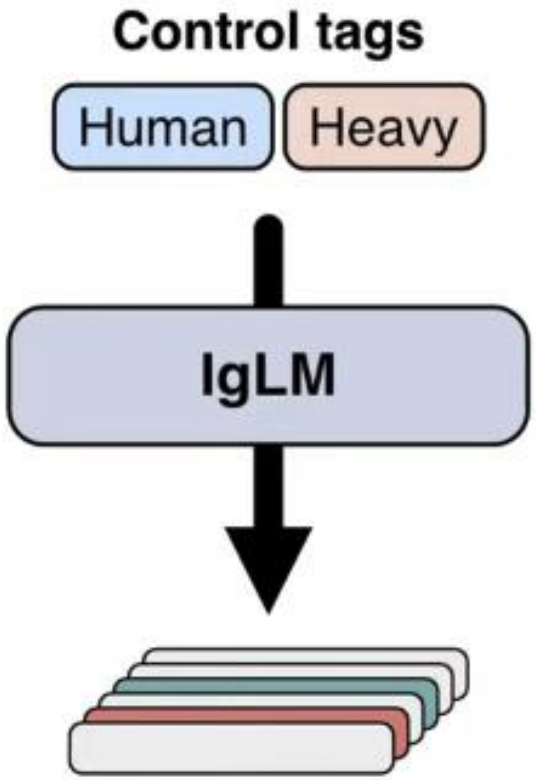
$$P(r_i|c_{species}, c_{chain}, r_1, \ldots r_{i-1})$$



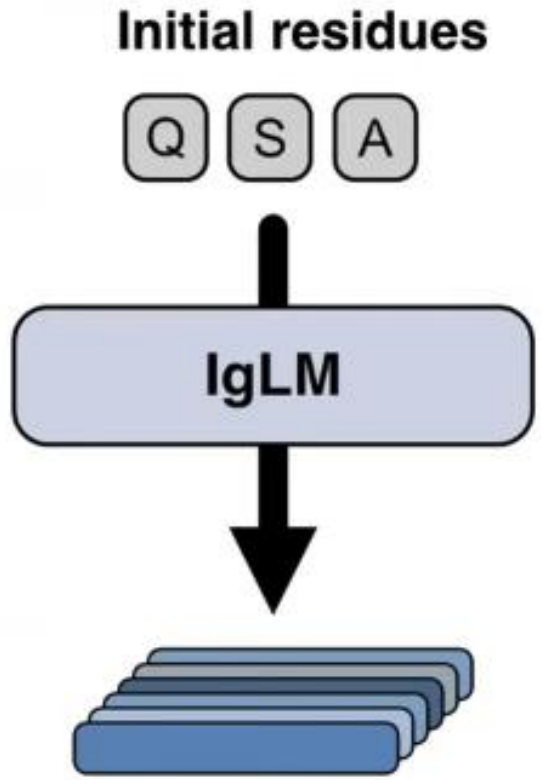
Generating CDR regions: infilling



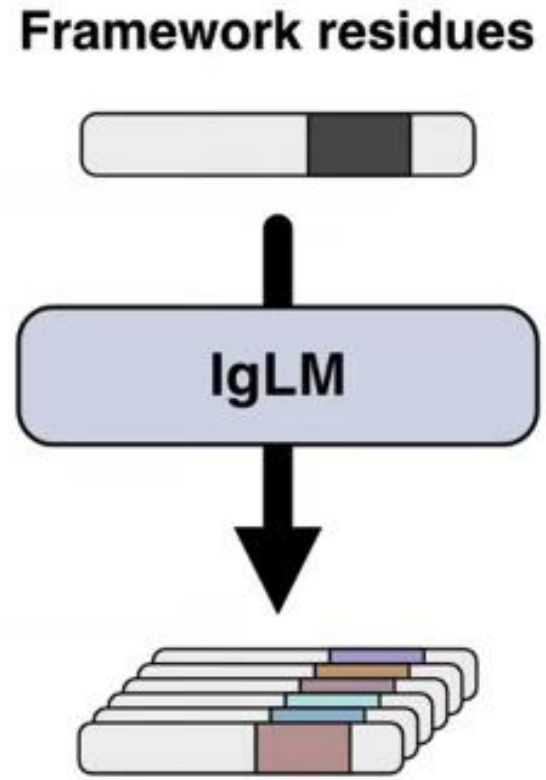
Conditional generation of antibody sequences



Prompt-guided sequence generation



Targeted **CDR loop diversification**



Protein Language Models (pLM)



BERT (Encoder-only)

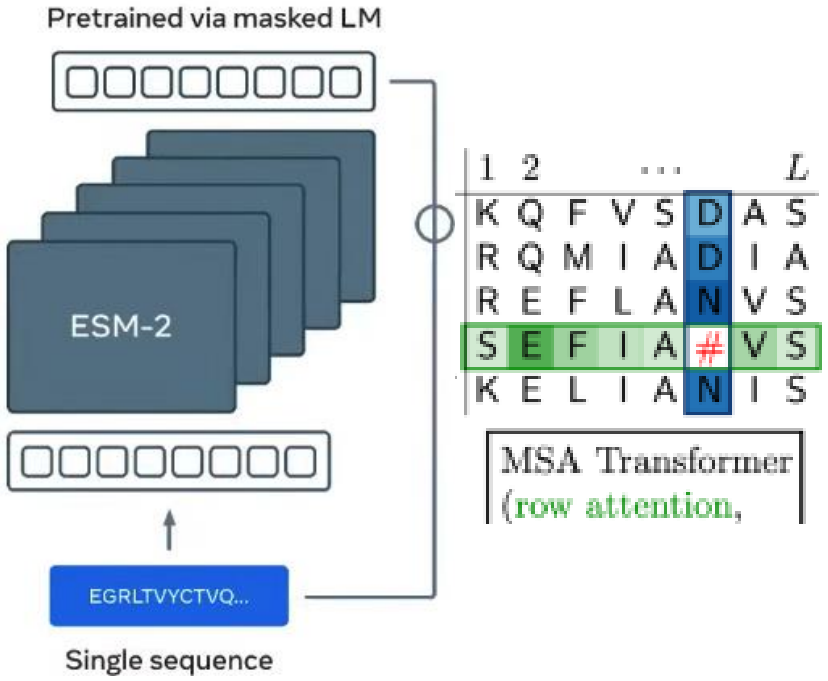
T5

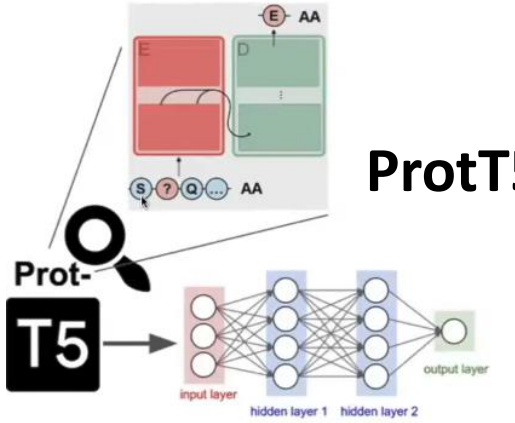
T5 (Encoder-decoder)

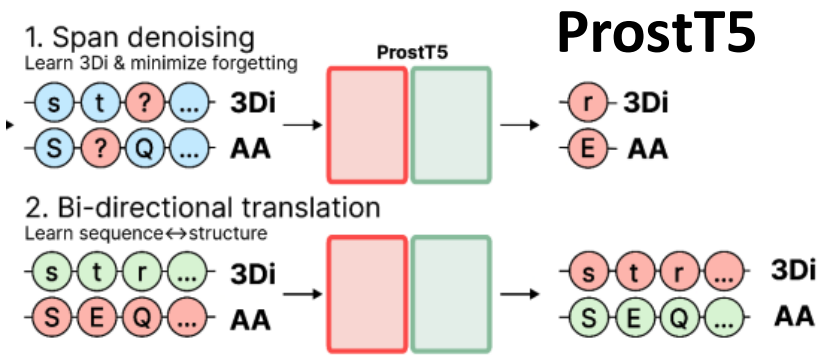
OpenAI

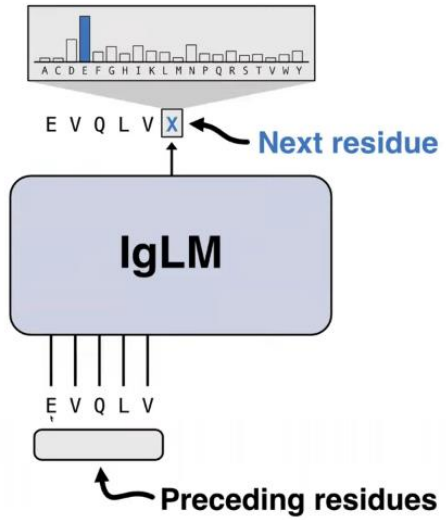
GPT - 4

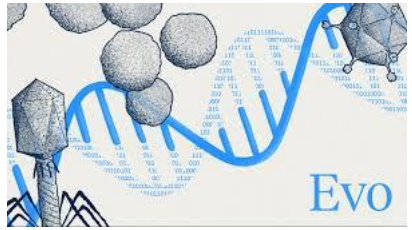
Decoder-only











Masked token prediction

Denoising span-mask prediction

Next token prediction